

Internet Appendix for Consumption in Asset Returns

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Abstract

The Internet Appendix provides additional tables, figures, and results supporting the main text.

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A Simulation design

We assume the same data-generating process of the log consumption growth as in Bansal and Yaron (2004), with the only exception being that we introduce a square-root process for the variance, as in Hansen, Heaton, Lee, and Roussanov (2007); that is:

$$\begin{aligned}\Delta c_{t,t+1} &= \mu + x_t + \sigma_t \eta_{t+1}, \\ x_{t+1} &= \rho x_t + \phi_e \sigma_t e_{t+1}, \text{ and} \\ \sigma_{t+1}^2 &= \sigma^2(1 - v_1) + v_1 \sigma_t^2 + \sigma_w \sigma_t \omega_{t+1},\end{aligned}$$

where $\eta_{t+1}, e_t, \omega_t \stackrel{\text{iid}}{\sim} \mathcal{N}(0, 1)$. The calibrated monthly parameter values are $\mu = 0.0015$, $\rho = 0.979$, $\phi_e = 0.044$, $\sigma = 0.0078$, $v_1 = 0.987$, and $\sigma_w = 0.00029487$.

We use this particular long-run risk (LRR) calibration because it ensures the nonnegativity of the volatility process and, most importantly, it gives the best chance of detecting the predictability of consumption using the simple ARIMA selection methods commonly used in empirical work: The contribution of the conditional mean to the variance of the consumption growth in this calibration is about 12% at the quarterly frequency, the largest share among leading LRR calibrations.

When also simulating return data, we further assume that log excess returns ($\mathbf{r}_{t+1}^e$) follow

$$\underset{N \times 1}{\mathbf{r}_{t+1}^e} = \underset{N \times 1}{\boldsymbol{\mu}_r} + \underset{N \times 1}{\boldsymbol{\rho}^r} e_{t+1} + \underset{N \times 1}{\mathbf{w}_{t+1}^r},$$

where $\boldsymbol{\mu}_r$ is a vector of average monthly log excess returns, $\boldsymbol{\rho}^r$ is a vector of returns' loadings on the contemporaneous shock, e_{t+1} , which drives the conditional mean of $\Delta c_{t,t+1}$, and $\mathbf{w}_{t+1}^r \stackrel{\text{iid}}{\sim} \mathcal{N}(0, \boldsymbol{\Sigma}_r)$. For simplicity, we assume that $\boldsymbol{\Sigma}_r$ is diagonal and constant over time in the simulations. Values of $\boldsymbol{\mu}_r$, $\boldsymbol{\rho}^r$, and $\boldsymbol{\Sigma}_r$ are selected to be their sample estimates in the main text.² After simulating monthly sequences of $(\Delta c_{t,t+1}, \mathbf{r}_{t+1}^e)$, we aggregate them into quarterly sequences by summing all three monthly observations within a quarter. We simulate 1,000 sample paths with 214 quarterly observations. Because we approximate the log consumption growth using an MA(14) process, there are 200 effective quarterly observations.

²More precisely, we obtain the estimates of $\boldsymbol{\mu}_r$, $\boldsymbol{\rho}^r$, and $\boldsymbol{\Sigma}_r$ using quarterly data. In the simulation of monthly returns, we divide all three parameters above by 3.

B Additional methodological details

B.1 State-space estimation with constant volatilities

We can rewrite the dynamic model in equations (2) and (3) in state-space form, assuming Gaussian innovations with constant volatility (the general SV version is presented in Appendix B.2), as

$$\mathbf{z}_t = \mathbf{F}\mathbf{z}_{t-1} + \mathbf{v}_t, \quad \mathbf{v}_t \sim \mathcal{N}(\mathbf{0}_{\bar{S}+K}; \Psi), \quad \text{and} \quad (\text{IA.1})$$

$$\mathbf{y}_t = \boldsymbol{\mu} + \mathbf{H}\mathbf{z}_t + \mathbf{w}_t, \quad \mathbf{w}_t \sim \mathcal{N}(\mathbf{0}_{N+1}, \Sigma), \quad (\text{IA.2})$$

where $\mathbf{y}_t := [\Delta c_t, \mathbf{r}_t^{e'}]$, $\mathbf{z}_t := [f_t, \dots, f_{t-\bar{S}}, \mathbf{g}_t']'$, $\boldsymbol{\mu} := [\mu_c, \boldsymbol{\mu}_r']'$, $\mathbf{v}_t := [f_t, \mathbf{0}_{\bar{S}}', \mathbf{g}_t']'$, $\mathbf{w}_t := [w_t^c, \mathbf{w}_t^{r'}]'$,

$$\Psi := \underbrace{\begin{bmatrix} 1 & \mathbf{0}_{\bar{S}}' & \mathbf{0}_{K-1}' \\ \mathbf{0}_{\bar{S}} & \mathbf{0}_{\bar{S} \times \bar{S}} & \mathbf{0}_{\bar{S} \times (K-1)} \\ \mathbf{0}_{K-1} & \mathbf{0}_{(K-1) \times \bar{S}} & \mathbf{I}_{K-1} \end{bmatrix}}_{(S+K) \times (S+K)}, \quad \mathbf{F} := \underbrace{\begin{bmatrix} \mathbf{0}_{\bar{S}}' & \mathbf{0}_K' \\ \mathbf{I}_{\bar{S}} & \mathbf{0}_{\bar{S} \times K} \\ \mathbf{0}_{(K-1) \times \bar{S}} & \mathbf{0}_{(K-1) \times K} \end{bmatrix}}_{(S+K) \times (S+K)}, \quad \Sigma := \underbrace{\begin{bmatrix} \sigma_c^2 & \mathbf{0}_N' \\ \mathbf{0}_N & \Sigma_r \end{bmatrix}}_{(N+1) \times (N+1)}, \quad \mathbf{H} := \underbrace{\begin{bmatrix} \rho_0 & \rho_1 & \dots & \rho_{\bar{S}} & \mathbf{0}_{K-1}' \\ \boldsymbol{\rho}_f^r & \mathbf{0}_N & \dots & \mathbf{0}_N & \boldsymbol{\rho}_g^r \end{bmatrix}}_{(N+1) \times (S+K)}, \quad (\text{IA.3})$$

and $\mathbf{I}_{\bar{S}}$ and $\mathbf{0}_{\bar{S} \times \bar{S}}$ denote an identity matrix and a matrix of zeros of dimension $\bar{S}$.

Similarly, the dynamic system in equations (5)–(6) can be represented in the state-space form (IA.1)–(IA.2) with $\mathbf{z}_t := [f_t, \dots, f_{t-\bar{S}}, b_t, \dots, b_{t-\bar{S}}]'$; $\mathbf{v}_t := [f_t, \mathbf{0}_{\bar{S}}', b_t, \mathbf{0}_{\bar{S}}']' \sim \mathcal{N}(\mathbf{0}; \Psi)$; Ψ and $\mathbf{F}$ being block diagonal with blocks repeated twice and given by the first two matrices in equation (IA.3) (when $K = 1$); and space equation matrix of coefficients given by

$$\mathbf{H} := \underbrace{\begin{bmatrix} \rho_0 & \dots & \dots & \rho_{\bar{S}} & \theta_0 & \dots & \dots & \theta_{\bar{S}} \\ \rho_1^r & 0 & \dots & 0 & \theta_1^b & 0 & \dots & 0 \\ \dots & \dots & \dots & \dots & \dots & \dots & \dots & \dots \\ \rho_{N_b}^r & 0 & \dots & 0 & \theta_{N_b}^b & 0 & \dots & 0 \\ \dots & 0 & \dots & 0 & 0 & 0 & \dots & 0 \\ \rho_N^r & 0 & \dots & 0 & 0 & 0 & \dots & 0 \end{bmatrix}}_{(N+1) \times 2(\bar{S}+1)}. \quad (\text{IA.4})$$

The state-space model above implies the following conditional likelihood for the data,

$$\mathbf{y}_t | \mathcal{I}_{t-1}, \boldsymbol{\mu}, \mathbf{H}, \Psi, \Sigma, \mathbf{z}_t \sim \mathcal{N}(\boldsymbol{\mu} + \mathbf{H}\mathbf{z}_t; \Sigma), \quad (\text{IA.5})$$

where $\mathcal{I}_{t-1}$ denotes the history of the state variables until time $t - 1$. Hence, under a diffuse (Jeffreys') prior, conditional on the history of $\mathbf{z}_t$ and $\mathbf{y}_t$, and given the diagonal structure of Σ , we have the standard Normal-Inverse-Gamma posterior distribution for the parameters of the model (see, e.g., Bauwens, Lubrano, and Richard (1999)). Moreover, the posterior

of the unobservable $\mathbf{z}_t$, conditional on the data and the parameters, is constructed using a standard Kalman filter and smoother approach (see, e.g., Primiceri (2005)).

Let $\mathbf{\Pi}' := [\boldsymbol{\mu}, \mathbf{H}]$ and $\mathbf{x}'_t := [1, \mathbf{z}'_t]$. Under a diffuse prior, the likelihood of the data in equation (IA.5) implies the posterior distribution

$$\mathbf{\Pi}' | \boldsymbol{\Sigma}, \{\mathbf{z}_t\}_{t=1}^T, \{\mathbf{y}_t\}_{t=1}^T \sim \mathcal{N} \left(\hat{\mathbf{\Pi}}'_{OLS}; \boldsymbol{\Sigma} \otimes (\mathbf{x}' \mathbf{x})^{-1} \right),$$

where $\mathbf{x}$ contains the stacked regressors, and the posterior distribution of each element on the main diagonal of $\boldsymbol{\Sigma}$ is given by³

$$\sigma_j^2 | \{\mathbf{z}_t\}_{t=1}^T, \{\mathbf{y}_t\}_{t=1}^T \sim \text{Inv-}\Gamma \left((T - m_j - 1) / 2, T \hat{\sigma}_{j,OLS}^2 / 2 \right),$$

where m_j is the number of estimated coefficients in the j -th equation. That is, the conditional posterior has a Normal-Inverse-Gamma structure. Moreover, $\mathbf{F}$ and $\boldsymbol{\Psi}$ have a Dirac posterior distribution at the points defined in equation (IA.3). Therefore, the missing part necessary for taking draws via MCMC using a Gibbs sampler is the conditional distributions of $\mathbf{z}_t$. Because

$$\begin{bmatrix} \mathbf{y}_t \\ \mathbf{z}_t \end{bmatrix} \Big| \mathcal{I}_{t-1}, \mathbf{H}, \boldsymbol{\Psi}, \boldsymbol{\Sigma} \sim \mathcal{N} \left(\begin{bmatrix} \boldsymbol{\mu} \\ \mathbf{F} \mathbf{z}_{t-1} \end{bmatrix}; \begin{bmatrix} \boldsymbol{\Omega} & \mathbf{H} \\ \mathbf{H}' & \boldsymbol{\Psi} \end{bmatrix} \right),$$

where $\boldsymbol{\Omega} := \text{Var}_{t-1}(\mathbf{y}_t) = \mathbf{H} \boldsymbol{\Psi} \mathbf{H}' + \boldsymbol{\Sigma}$, $\mathbf{z}_t$ can be drawn using a standard Kalman filter and smoother approach.

B.1.1 Kalman filter algorithm

Let $\mathbf{z}_{t|\tau} := E[\mathbf{z}_t | \mathbf{y}^\tau, \mathbf{H}, \boldsymbol{\Psi}, \boldsymbol{\Sigma}]$ and $\mathbf{V}_{t|\tau} := \text{Var}(\mathbf{z}_t | \mathbf{y}^\tau, \mathbf{H}, \boldsymbol{\Psi}, \boldsymbol{\Sigma})$, where $\mathbf{y}^\tau$ denotes the history of $\mathbf{y}_t$ until τ . Then, given $\mathbf{z}_{0|0}$ and $\mathbf{V}_{0|0}$, the Kalman filter delivers: $\mathbf{z}_{t|t-1} = \mathbf{F} \mathbf{z}'_{t-1|t-1}$; $\mathbf{V}_{t|t-1} = \mathbf{F} \mathbf{V}_{t-1|t-1} \mathbf{F}' + \boldsymbol{\Psi}$; $\mathbf{K}_t = \mathbf{V}_{t|t-1} \mathbf{H}' (\mathbf{H} \mathbf{V}_{t|t-1} \mathbf{H}' + \boldsymbol{\Sigma})^{-1}$; $\mathbf{z}_{t|t} = \mathbf{z}_{t|t-1} + \mathbf{K}_t (\mathbf{y}_t - \boldsymbol{\mu} - \mathbf{H} \mathbf{z}_{t|t-1})$; and $\mathbf{V}_{t|t} = \mathbf{V}_{t|t-1} - \mathbf{K}_t \mathbf{H} \mathbf{V}_{t|t-1}$. The last elements of the recursion, $\mathbf{z}_{T|T}$ and $\mathbf{V}_{T|T}$, are the mean and variance of the normal distribution used to draw $\mathbf{z}_T$. The draw of $\mathbf{z}_T$ and the output of the filter can then be used for the first step of the backward recursion, which delivers the $\mathbf{z}_{T-1|T}$ and $\mathbf{V}_{T-1|T}$ values necessary to make a draw for $\mathbf{z}_{T-1}$ from a Gaussian distribution. The backward recursion can be continued until time zero, drawing each value of $\mathbf{z}_t$ in the process, with the following updating formulas for the following generic time- t recursion: $\mathbf{z}_{t|t+1} = \mathbf{z}_{t|t} + \mathbf{V}_{t|t} \mathbf{F}' \mathbf{V}_{t+1|t}^{-1} (\mathbf{z}_{t+1} - \mathbf{F} \mathbf{z}_{t|t})$ and $\mathbf{V}_{t|t+1} = \mathbf{V}_{t|t} - \mathbf{V}_{t|t} \mathbf{F}' \mathbf{V}_{t+1|t}^{-1} \mathbf{F} \mathbf{V}_{t|t}$.

Hence, parameters and states can be drawn via the Gibbs sampler, as follows:

1. Begin with a guess of $\tilde{\mathbf{\Pi}}'$ and $\tilde{\boldsymbol{\Sigma}}^{-1}$ (e.g., the frequentist maximum likelihood estimates)

³By relaxing the diagonality assumption, the posterior distribution of $\boldsymbol{\Sigma}^{-1}$ becomes a Wishart centered at the OLS estimates.

and use it to construct initial draws for $\boldsymbol{\mu}$ and $\mathbf{H}$. Using $\mathbf{F}$ and $\boldsymbol{\Psi}$, draw the $\mathbf{z}_t$ history using the Kalman recursion above with (Kalman step): $\mathbf{z}_t \sim \mathcal{N}(\mathbf{z}_{t|t+1}; \mathbf{V}_{t|t+1})$.

2. Conditioning on $\{\mathbf{z}_t\}_{t=1}^T$ (drawn in the previous step) and $\{\mathbf{y}_t\}_{t=1}^T$, run *OLS* imposing the zero restrictions and get $\hat{\boldsymbol{\Pi}}'_{OLS}$ and $\hat{\boldsymbol{\Sigma}}_{OLS}$, and draw $\tilde{\boldsymbol{\Pi}}'$ and $\tilde{\boldsymbol{\Sigma}}^{-1}$ from the Normal-Inverse-Gamma (N-i- Γ step). Use these draws as the initial guess for the previous point of the algorithm, and repeat.

B.2 A generalized model with stochastic volatilities

To complete the specification in equations (2) and (16)–(19), we need to formalize the autoregressive volatility processes and the prior formulations. We do so in what follows, and we also provide the sampling algorithm for the generalized state-space model for consumption, returns, and their volatilities.

B.2.1 Stochastic volatility of w_t^c

Let $y_{ct}^* = \log((w_t^c)^2) = h_{ct} + \log(\epsilon_{ct}^2)$ and $h_{ct} = \psi_c + \delta_c(h_{c,t-1} - \psi_c) + \sigma_{\eta c}\eta_{ct}$, where $\epsilon_{ct} \stackrel{\text{iid}}{\sim} \mathcal{N}(0, 1)$ and $\eta_{ct} \stackrel{\text{iid}}{\sim} \mathcal{N}(0, 1)$, and they are independent. Let $e_{ct} = \log(\epsilon_{ct}^2) \sim \log(\chi^2(1))$. We estimate the model following Kim, Shephard, and Chib (1998). More specifically, we approximate e_{ct} using a mixture of Gaussian distributions, $e_{ct} = \sum_{j=1}^7 q_j \mathcal{N}(m_j - 1.2704, v_j^2)$, or, equivalently, $e_{ct} | S_{ct} = j \sim \mathcal{N}(m_j - 1.2704, v_j^2)$, where $\sum_{j=1}^7 q_j = 1$, and values of $\{m_j\}_{j=1}^7$ and $\{v_j\}_{j=1}^7$ can be found in Table 4 of Kim, Shephard, and Chib (1998).

Following is the algorithm used to estimate the unknown parameters $\{h_{ct}, S_{ct}, \psi_c, \delta_c, \sigma_{\eta c}\}$:

Step 1: Initialize $\{S_{ct}\}_{t=1}^T, \psi_c, \delta_c, \sigma_{\eta c}$.

Step 2: Sample $\{h_{ct}\}_{t=1}^T$ from $h_c | y_c^*, S_c, \psi_c, \delta_c, \sigma_{\eta c}$ (Kalman Smoother step).

Step 3: Sample S_{ct} from $p(S_{ct} | y_{ct}^*, h_{ct})$, where

$$\begin{aligned} p(S_{ct} = j | y_{ct}^*, h_{ct}) &\propto q_j \times p(y_{ct}^* | h_{ct}, S_{ct} = j) \\ &\propto q_j \times f_N(y_{ct}^* | h_{ct} + m_j - 1.2704, v_j^2). \end{aligned}$$

Step 4: Update $\psi_c, \delta_c, \sigma_{\eta c}$. An inverse-gamma prior for $\sigma_{\eta c}^2$, that is, $\sigma_{\eta c}^2 \sim \Gamma^{-1}(\frac{\sigma_0}{2}, \frac{s_\sigma}{2})$, yields

$$\begin{aligned} p(\sigma_{\eta c}^2 | y_c^*, S_c, \psi_c, \delta_c, h_c) &\propto p(h_c | \psi_c, \delta_c, \sigma_{\eta c}^2) \pi(\sigma_{\eta c}^2) \propto p(h_{c,1} | \psi_c, \delta_c, \sigma_{\eta c}^2) \prod_{t=0}^{T-1} p(h_{c,t+1} | \psi_c, \delta_c, \sigma_{\eta c}^2) \pi(\sigma_{\eta c}^2) \\ &\propto (\sigma_{\eta c}^2)^{-\frac{T}{2}} \exp \left\{ -\frac{\sum_{t=1}^{T-1} [h_{c,t+1} - \psi_c - \delta_c(h_{c,t} - \psi_c)]^2 + (h_{c,1} - \psi_c)^2 (1 - \delta_c^2)}{2\sigma_{\eta c}^2} \right\} (\sigma_{\eta c}^2)^{-\frac{\sigma_0}{2}-1} \exp \left\{ -\frac{s_\sigma}{2\sigma_{\eta c}^2} \right\}. \end{aligned}$$

Therefore, the conditional posterior is

$$\sigma_{\eta c}^2 \mid y_c^*, S_c, \psi_c, \delta_c, h_c \sim \Gamma^{-1} \left(\frac{T + \sigma_0}{2}, \frac{s_\sigma + \sum_{t=1}^{T-1} [h_{c,t+1} - \psi_c - \delta_c(h_{ct} - \psi_c)]^2 + (h_{c,1} - \psi_c)^2(1 - \delta_c^2)}{2} \right).$$

Let $\delta_c = 2\phi - 1$, and $\phi \sim \text{Beta}(\phi^{(1)}, \phi^{(2)})$. Therefore, the prior distribution of δ_c is $\pi(\delta_c) \propto (1 + \delta_c)^{\phi^{(1)}-1}(1 - \delta_c)^{\phi^{(2)}-1}$. The posterior distribution of δ_c is

$$p(\delta_c \mid \psi_c, h_c, \sigma_{\eta c}^2) \propto \pi(\delta_c)p(h_c \mid \psi_c, \delta_c, \sigma_{\eta c}^2),$$

$$p(h_c \mid \psi_c, \delta_c, \sigma_{\eta c}^2) \propto (1 - \delta_c^2)^{\frac{1}{2}} \exp \left\{ -\frac{\sum_{t=1}^{T-1} [h_{c,t+1} - \psi_c - \delta_c(h_{ct} - \psi_c)]^2 + (h_{c,1} - \psi_c)^2(1 - \delta_c^2)}{2\sigma_{\eta c}^2} \right\}, \text{ and}$$

$$\log p(h_c \mid \psi_c, \delta_c, \sigma_{\eta c}^2) \propto -\frac{\sum_{t=1}^{T-1} [h_{c,t+1} - \psi_c - \delta_c(h_{ct} - \psi_c)]^2 + (h_{c,1} - \psi_c)^2(1 - \delta_c^2)}{2\sigma_{\eta c}^2} + \frac{1}{2} \log(1 - \delta_c^2).$$

The function above is concave in δ_c for all values of $\phi^{(1)}$ and $\phi^{(2)}$. Hence, δ_c can be sampled using a reject-accept algorithm. Let

$$\hat{\delta}_c = \frac{\sum_{t=1}^{T-1} [h_{c,t+1} - \psi_c][h_{c,t} - \psi_c]}{\sum_{t=1}^{T-1} [h_{c,t} - \psi_c]^2} \text{ and } V_{\delta_c} = \frac{\sigma_{\eta c}^2}{\sum_{t=1}^{T-1} [h_{c,t} - \psi_c]^2}.$$

We first sample a proposal δ_c^* from a normal distribution $\mathcal{N}(\hat{\delta}_c, V_{\delta_c})$ and accept the new value δ_c^* with probability $\min\{1, \exp(g(\delta_c^*) - g(\delta_c^{i-1}))\}$, where

$$g(\delta_c) = \log(\pi(\delta_c)) - \frac{(h_{c,1} - \psi_c)^2(1 - \delta_c^2)}{2\sigma_{\eta c}^2} + \frac{1}{2} \log(1 - \delta_c^2).$$

Finally, we assign a diffuse prior for ψ_c , as follows:

$$p(\psi_c \mid h_c, \delta_c, \sigma_{\eta c}^2) \propto \exp \left\{ -\frac{\sum_{t=1}^{T-1} [h_{c,t+1} - \psi_c - \delta_c(h_{ct} - \psi_c)]^2 + (h_{c,1} - \psi_c)^2(1 - \delta_c^2)}{2\sigma_{\eta c}^2} \right\}.$$

Therefore, we obtain the posterior distribution of ψ_c : $\psi_c \mid h_c, \delta_c, \sigma_{\eta c}^2 \sim \mathcal{N}(\hat{\psi}_c, \sigma_\psi^2)$, where $\sigma_\psi^2 = \frac{\sigma_{\eta c}^2}{[(T-1)(1-\delta_c)^2 + (1-\delta_c^2)]}$ and $\hat{\psi}_c = \sigma_\psi^2 \left\{ \frac{(1-\delta_c^2)h_{c,1}}{\sigma_{\eta c}^2} + \frac{(1-\delta_c)\sum_{t=1}^{T-1} [h_{c,t+1} - \delta_c h_{ct}]}{\sigma_{\eta c}^2} \right\}$.

Step 5: Return to Step 2 until convergence.

B.2.2 Stochastic volatility of f_t

The shock f_t follows a normal distribution with the stochastic volatility process given by

$$y_{ft}^* = \log(f_t^2) = h_{ft} + \log(\epsilon_{ft}^2), \quad h_{ft} = \delta_f h_{f,t-1} + \sqrt{1 - \delta_f^2} \eta_{ft},$$

where $\epsilon_{ft} \stackrel{\text{iid}}{\sim} \mathcal{N}(0, 1)$ and $\eta_{ft} \stackrel{\text{iid}}{\sim} \mathcal{N}(0, 1)$, and they are independent. Let $e_{ft} = \log(\epsilon_{ft}^2) \sim \log(\chi^2(1))$. The process above for h_{ft} is simply an AR(1) normalized to have unconditional zero mean and unit variance. As previously, we approximate e_{ft} using a mixture of Gaussians; that is, $e_{ft} = \sum_{j=1}^7 q_j \mathcal{N}(m_j - 1.2704, v_j^2)$. The sampling algorithm is given below.

Step 1: Initialize $\{S_{ft}\}_{t=1}^T, \delta_f$.

Step 2: Sample $\{h_{ft}\}_{t=1}^T$ from $h_f \mid y_f^*, S_f, \delta_f$ (Kalman smoother step).

Step 3: Sample S_{ft} from $p(S_{ft} \mid y_{ft}^*, h_{ft})$.

Step 4: Update δ_f using the Metropolis algorithm. Similar to the prior distribution of δ_c , we assume that $\delta_f = 2\phi - 1$, and $\phi \sim \text{Beta}(\phi^{(1)}, \phi^{(2)})$. The posterior distribution of δ_f is

$$\begin{aligned} p(\delta_f \mid h_f, \psi_f, \gamma_f = 0) &\propto \pi(\delta_f) p(h_f \mid \psi_f, \delta_f, \gamma_f = 0) \\ &\propto (1 + \delta_f)^{\phi^{(1)}-1} (1 - \delta_f)^{\phi^{(2)}-1} (1 - \delta_f^2)^{-\frac{T-1}{2}} \exp \left\{ -\frac{\sum_{t=2}^T (h_{ft} - \delta_f h_{f,t-1})^2}{2(1 - \delta_f^2)} - \frac{h_{f1}^2}{2} \right\}. \end{aligned}$$

We draw δ_f using the Metropolis algorithm: (M1) initialize δ_f^0 ; (M2) draw δ_f^* from a normal distribution $\mathcal{N}(\delta_f^{(i-1)}, c_{mh}^2)$ ⁴; (M3) calculate $\rho(\delta_f^*, \delta_f^{(i-1)}) = \min\{1, \frac{p(\delta_f^* \mid h_f)}{p(\delta_f^{(i-1)} \mid h_f)}\}$; (M4) set $\delta_f^{(i)} = \delta_f^{(i-1)}$ with probability $1 - \rho(\delta_f^*, \delta_f^{(i-1)})$ and $\delta_f^{(i)} = \delta_f^*$ with probability $\rho(\delta_f^*, \delta_f^{(i-1)})$.

Step 5: Return to Step 2 until convergence.

B.2.3 Stochastic volatility of asset returns

Let $\mathbf{y}_{rt}^* = \boldsymbol{\kappa}_0 + \boldsymbol{\kappa}_1 h_{rt} + \mathbf{e}_{rt}$ and $h_{rt} = \delta_r h_{r,t-1} + \sqrt{1 - \delta_r^2} \eta_{rt}$, where

$$\mathbf{y}_{rt}^* = \begin{pmatrix} \log(w_{1t}^r)^2 \\ \vdots \\ \log(w_{Nt}^r)^2 \end{pmatrix}, \quad \mathbf{e}_{rt} = \begin{pmatrix} e_{r,1,t} \\ \vdots \\ e_{r,N,t} \end{pmatrix} = \begin{pmatrix} \log(\epsilon_{1t}^2) \\ \vdots \\ \log(\epsilon_{Nt}^2) \end{pmatrix},$$

and the ϵ_{it} shocks are independent across different assets. In the model above, we assume that one hidden state, h_{rt} , drives the common component of asset-specific stochastic volatilities. In order to identify the model, we normalize h_{rt} to have zero mean and unit variance. In

⁴ c_{mh} determines the step size in the Metropolis algorithm. We aim to choose c_{mh} such that the frequency of accepting a new δ_f is about 50%.

order to simplify the estimation, we further exclude κ_0 by demeaning $\mathbf{y}_{rt}^*$ to have the same sample mean as $\mathbf{e}_{rt}$. Therefore, the model is simplified as

$$\kappa_0 = \mathbf{y}_{rt}^* - \bar{\mathbf{y}}_{rt}^*, \quad \bar{\mathbf{y}}_{rt}^* = \kappa_1 h_{rt} + \mathbf{e}_{rt}, \quad h_{rt} = \delta_r h_{r,t-1} + \sqrt{1 - \delta_r^2} \eta_{rt},$$

for $i = 1, 2, \dots, N$, $e_{i,rt} = \sum_{j=1}^7 q_j \mathcal{N}(m_j - 1.2704, v_j^2)$, or equivalently, $e_{i,rt} \mid S_{i,rt} = j \sim \mathcal{N}(m_j - 1.2704, v_j^2)$. Therefore, $\bar{\mathbf{y}}_{i,rt}^* \mid \kappa_1^{(i)}, h_{rt}, S_{i,rt} = j \stackrel{\text{iid}}{\sim} \mathcal{N}(\kappa_1^{(i)} h_{rt} + m_j - 1.2704, v_j^2)$. Let

$$\bar{\mathbf{Y}}_{\mathbf{r}}^{*,(i)} = \begin{pmatrix} \bar{y}_{r1}^{*,(i)} - m_{i1} + 1.2704 \\ \vdots \\ \bar{y}_{rT}^{*,(i)} - m_{iT} + 1.2704 \end{pmatrix}, \mathbf{V}_{\mathbf{r}}^{(i)} = \begin{pmatrix} v_{i1}^2 & & \\ & \ddots & \\ & & v_{iT}^2 \end{pmatrix}, \mathbf{H}_{\mathbf{r}} = \begin{pmatrix} h_{r1} \\ \vdots \\ h_{rT} \end{pmatrix},$$

where $m_{it} = m_j$ and $v_{it}^2 = v_j^2$ if $S_{it} = j$. Assuming a diffuse prior for $\kappa_1^{(i)}$, we have

$$\begin{aligned} p(\kappa_1^{(i)} \mid \bar{\mathbf{Y}}_{\mathbf{r}}^{*,(i)}, \mathbf{H}_{\mathbf{r}}, \{S_{i,rt}\}_{t=1}^T) &\propto p(\bar{\mathbf{Y}}_{\mathbf{r}}^{*,(i)} \mid \kappa_1^{(i)}, \mathbf{H}_{\mathbf{r}}, \{S_{i,rt}\}_{t=1}^T) \\ &\propto \exp\left\{-\frac{1}{2}(\bar{\mathbf{Y}}_{\mathbf{r}}^{*,(i)} - \mathbf{H}_{\mathbf{r}}\kappa_1^{(i)})^\top (\mathbf{V}_{\mathbf{r}}^{(i)})^{-1} (\bar{\mathbf{Y}}_{\mathbf{r}}^{*,(i)} - \mathbf{H}_{\mathbf{r}}\kappa_1^{(i)})\right\}, \text{ and} \\ \kappa_1^{(i)} \mid \bar{\mathbf{Y}}_{\mathbf{r}}^{*,(i)}, \mathbf{H}_{\mathbf{r}}, \{S_{i,rt}\}_{t=1}^T &\sim \mathcal{N}(\hat{\kappa}_1^{(i)}, [\mathbf{H}_{\mathbf{r}}^\top (\mathbf{V}_{\mathbf{r}}^{(i)})^{-1} \mathbf{H}_{\mathbf{r}}]^{-1}), \end{aligned}$$

where $\hat{\kappa}_1^{(i)} = [\mathbf{H}_{\mathbf{r}}^\top (\mathbf{V}_{\mathbf{r}}^{(i)})^{-1} \mathbf{H}_{\mathbf{r}}]^{-1} \mathbf{H}_{\mathbf{r}}^\top (\mathbf{V}_{\mathbf{r}}^{(i)})^{-1} \bar{\mathbf{Y}}_{\mathbf{r}}^{*,(i)}$. Finally, we update $\{S_{i,rt}\}_{t=1}^T$ as follows:

$$\begin{aligned} p(S_{i,rt} = j \mid \bar{y}_{rt}^{*,(i)}, h_{rt}, \kappa_1^{(i)}) &\propto q_j \times p(\bar{y}_{rt}^{*,(i)} \mid h_{rt}, S_{i,rt} = j, \kappa_1^{(i)}) \\ &\propto q_j \times f_N(\bar{y}_{rt}^{*,(i)} \mid \kappa_1^{(i)} h_{rt} + m_j - 1.2704, v_j^2). \end{aligned}$$

The posterior draws of h_{rt} and δ_r are then obtained in a similar way as previously.

B.2.4 The consumption growth excess return mean-equation coefficients

Because shocks in equations (2) and (16) are uncorrelated, we can sample the unknown parameters equation by equation. We introduce the following notations:

$$\begin{aligned} \boldsymbol{\rho}^c &= \begin{pmatrix} \mu_c \\ \rho_0 \\ \vdots \\ \rho_{\bar{S}} \end{pmatrix}, \quad \boldsymbol{\Delta C} = \begin{pmatrix} \Delta C_{0,1} \\ \Delta C_{1,2} \\ \vdots \\ \Delta C_{T-1,T} \end{pmatrix}, \quad \boldsymbol{\Sigma}_w^c = \begin{pmatrix} \sigma_{c,0}^2 & 0 & \dots & 0 \\ 0 & \sigma_{c,1}^2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \sigma_{c,T-1}^2 \end{pmatrix}, \quad \mathbf{X}^c = \begin{pmatrix} 1 & f_1 & \dots & f_{1-\bar{S}} \\ 1 & f_2 & \dots & f_{2-\bar{S}} \\ \vdots & \vdots & \ddots & \vdots \\ 1 & f_T & \dots & f_{T-\bar{S}} \end{pmatrix}, \\ \boldsymbol{\rho}_i^r &= \begin{pmatrix} \mu_{ri} \\ \rho_i^r \\ \beta_{ci} \\ \beta_{fi} \\ \beta_{ri} \end{pmatrix}, \quad \mathbf{X}_i^r = \begin{pmatrix} 1 & f_1 & \sigma_{c,0}^2 & \sigma_{f,0}^2 & \sigma_{r,0}^2 \\ 1 & f_2 & \sigma_{c,1}^2 & \sigma_{f,1}^2 & \sigma_{r,1}^2 \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ 1 & f_T & \sigma_{c,T-1}^2 & \sigma_{f,T-1}^2 & \sigma_{r,T-1}^2 \end{pmatrix}, \text{ and } \boldsymbol{\Sigma}_{w,r,i} = \begin{pmatrix} \sigma_{r,i,0}^2 & 0 & \dots & 0 \\ 0 & \sigma_{r,i,1}^2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \sigma_{r,i,T-1}^2 \end{pmatrix}. \end{aligned}$$

Then the posterior distribution of ρ^c under a flat prior is as follows:⁵

$$\rho^c \mid \Delta C, X^c, \Sigma_w^c \sim \mathcal{N}(\hat{\rho}_{gls}, [(\mathbf{X}^c)^\top (\Sigma_w^c)^{-1} \mathbf{X}^c]^{-1}), \quad \hat{\rho}_{gls} = [(\mathbf{X}^c)^\top (\Sigma_w^c)^{-1} \mathbf{X}^c]^{-1} (\mathbf{X}^c)^\top (\Sigma_w^c)^{-1} \Delta C.$$

The conditional posterior for the return equation coefficients under a flat prior is then

$$\rho_i^r \mid \mathbf{R}_i, \mathbf{X}_i^r, \{\sigma_{ft}^2\}_{t=0}^{T-1}, \{\sigma_{r,i,t}^2\}_{t=0}^{T-1} \sim \mathcal{N}(\hat{\rho}_{i,gls}^r, [(\mathbf{X}_i^r)^\top \Sigma_{wr,i}^{-1} \mathbf{X}_i^r]^{-1}), \text{ and}$$

$$\hat{\rho}_{i,gls}^r = [(\mathbf{X}_i^r)^\top \Sigma_{wr,i}^{-1} \mathbf{X}_i^r]^{-1} (\mathbf{X}_i^r)^\top \Sigma_{wr,i}^{-1} \mathbf{R}_i.$$

B.2.5 Drawing the conditional consumption mean shocks

To sample $\{f_t\}_{t=1}^T$, let

$$\mathbf{y}_t = \begin{pmatrix} \Delta c_{t-1,t} \\ \mathbf{r}_t^e \end{pmatrix}, \quad \tilde{\boldsymbol{\mu}}_t = \begin{pmatrix} \mu_c \\ \boldsymbol{\mu}_r + \beta_c \sigma_{c,t-1}^2 + \beta_f \sigma_{f,t-1}^2 + \beta_r \sigma_{r,t-1}^2 \end{pmatrix}, \quad \mathbf{H} = \begin{pmatrix} \rho_0 & \rho_1 & \dots & \rho_{\bar{S}} \\ \boldsymbol{\rho}^r & \mathbf{0}_N & \dots & \mathbf{0}_N \end{pmatrix},$$

$$\mathbf{w}_t = \begin{pmatrix} w_t^c \\ \mathbf{w}_t^r \end{pmatrix} \sim \mathcal{N}(\mathbf{0}_{N+1}, \Sigma_t), \quad \mathbf{z}_t = \begin{pmatrix} f_t \\ f_{t-1} \\ \vdots \\ f_{t-\bar{S}} \end{pmatrix}, \quad \text{and } \Sigma_t = \begin{pmatrix} \sigma_{c,t-1}^2 & 0 & \dots & 0 \\ 0 & \sigma_{r,1,t-1}^2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \sigma_{r,N,t-1}^2 \end{pmatrix}.$$

Hence, the joint distribution of observables and f shocks is

$$\begin{pmatrix} \mathbf{y}_t \\ \mathbf{z}_t \end{pmatrix} \mid \mathcal{I}_{t-1}, \tilde{\boldsymbol{\mu}}_t, \mathbf{H}, \Sigma_t, \sigma_{f,t-1}^2 \sim \mathcal{N} \left(\begin{pmatrix} \tilde{\boldsymbol{\mu}}_t \\ \mathbf{F} \mathbf{z}_{t-1} \end{pmatrix}, \begin{pmatrix} \boldsymbol{\Omega}_t & \mathbf{H} \\ \mathbf{H}^\top & \boldsymbol{\Psi}_t \end{pmatrix} \right), \quad (\text{IA.6})$$

where $\boldsymbol{\Psi}_t = \begin{pmatrix} \sigma_{f,t-1}^2 & \mathbf{0}_{\bar{S}}^\top \\ \mathbf{0}_{\bar{S}} & \mathbf{0}_{\bar{S} \times \bar{S}} \end{pmatrix}$, $\mathbf{F} := \begin{bmatrix} \mathbf{0}_{\bar{S}}' & 0 \\ \mathbf{I}_{\bar{S}} & \mathbf{0}_{\bar{S}} \end{bmatrix}$, and $\boldsymbol{\Omega}_t = \mathbf{H} \boldsymbol{\Psi}_t \mathbf{H}^\top + \Sigma_t$. We then use the Kalman smoother to draw f_t , following the same procedure as in Internet Appendix B.1.1.

B.3 Model comparison

The model comparison for the restricted and unrestricted specification in Table IA.I of Section I is performed using Bayes factors (i.e., the the marginal likelihoods of the various

⁵ The likelihood function of the data is $\Delta C \mid \rho^c, X^c, \Sigma_w^c \sim \mathcal{N}(X^c \rho^c, \Sigma_w^c)$. By applying the diffuse prior for ρ^c , the posterior distribution of ρ^c becomes

$$p(\rho^c \mid \Delta C, X^c, \Sigma_w^c) \propto p(\Delta C \mid \rho^c, X^c, \Sigma_w^c) \propto \exp\left\{-\frac{1}{2}(\rho^c - \hat{\rho}_{gls})^\top (\mathbf{X}^c)^\top (\Sigma_w^c)^{-1} \mathbf{X}^c (\rho^c - \hat{\rho}_{gls})\right\}.$$

models) and posterior probabilities. Based on equation (IA.6), the likelihood function of the observed data $\{\mathbf{y}_t\}_{t=1}^T$ (after integrating out f_t) is

$$\begin{aligned} p(\mathbf{Y} \mid \tilde{\boldsymbol{\mu}}_t, \mathbf{H}, \{\boldsymbol{\Sigma}_t\}_{t=1}^T, \{\sigma_{ft}^2\}_{t=1}^T) &= \prod_{t=1}^T f_N(\mathbf{y}_t \mid \tilde{\boldsymbol{\mu}}_t, \boldsymbol{\Omega}_t) \\ &= (2\pi)^{-\frac{T(N+1)}{2}} \prod_{t=1}^T |\boldsymbol{\Omega}_t|^{-\frac{1}{2}} \exp\left\{-\frac{1}{2}(\mathbf{y}_t - \tilde{\boldsymbol{\mu}}_t)^\top \boldsymbol{\Omega}_t^{-1}(\mathbf{y}_t - \tilde{\boldsymbol{\mu}}_t)\right\} \\ &= (2\pi)^{-\frac{T(N+1)}{2}} \prod_{t=1}^T |\mathbf{H}\boldsymbol{\Psi}_t\mathbf{H}^\top + \boldsymbol{\Sigma}_t|^{-\frac{1}{2}} \exp\left\{-\frac{1}{2}(\mathbf{y}_t - \tilde{\boldsymbol{\mu}}_t)^\top (\mathbf{H}\boldsymbol{\Psi}_t\mathbf{H}^\top + \boldsymbol{\Sigma}_t)^{-1}(\mathbf{y}_t - \tilde{\boldsymbol{\mu}}_t)\right\}. \end{aligned}$$

A full Bayesian analysis requires us to specify a proper prior for $\boldsymbol{\mu}$, $\mathbf{H}$ and the parameters underlying stochastic volatility processes. The difficulty is that there is no closed-form solution for the marginal likelihood of data. Furthermore, a flat prior for $(\boldsymbol{\mu}, \mathbf{H}, \boldsymbol{\beta}_f, \boldsymbol{\beta}_r, \boldsymbol{\beta}_c)$ is improper, hence, the marginal likelihood of the data is unnormalized and there would be an undetermined constant term in model comparison. And even if we were to assign a proper prior, the numerical integration would be imprecise due to the high dimensionality of the parameter and hidden state spaces. Therefore, we follow the literature and approximate the marginal likelihood of the data using the Schwartz criterion (i.e., a Laplace, or particular second-order approximation of the marginal likelihood), as follows

$$\log(BF_{1,2}) \approx \log p(\mathbf{Y} \mid \hat{\theta}(\mathbb{M}_1)) - \log p(\mathbf{Y} \mid \hat{\theta}(\mathbb{M}_2)) - \frac{d_1 - d_2}{2} \log(T), \quad (\text{IA.7})$$

where $\mathbf{Y}$ is the observed data, T is the sample size, $\mathbb{M}_1$ and $\mathbb{M}_2$ represent Models 1 and 2, $\hat{\theta}(\mathbb{M}_1)$ and $\hat{\theta}(\mathbb{M}_2)$ are the posterior mean of parameter θ under Models 1 and 2, and d_1 and d_2 are model dimensions.⁶ The Bayes factor in equation (IA.7) ignores the prior distribution; hence, we do not need to change our current improper prior. Note that the model selection based on the above is analogous to likelihood ratio testing (the LR statistic is proportional to the first two terms in the equation (IA.7)) or the BIC-based model selection.

Posterior model probabilities are then computed using the approximation above of the Bayes factor and equal prior probability for all specifications; for example, the posterior probability of Model 1 is computed as $\frac{BF_{1,i}}{\sum_j BF_{j,i}}$, where the identity of the reference Model i is irrelevant.

⁶Note that the vector of parameters encompasses both the frequentist parameters $(\mu_c, \boldsymbol{\mu}_r, \boldsymbol{\beta}_f, \boldsymbol{\beta}_r, \boldsymbol{\beta}_c, \boldsymbol{\rho}^c, \boldsymbol{\rho}^r, \psi_c, \delta_c, \sigma_{\eta c}, \delta_f, \boldsymbol{\kappa}_0, \boldsymbol{\kappa}_0, \delta_r)$ and the latent states $(\{f_t, \sigma_{ct}^2, \sigma_{ft}^2, \sigma_{rt}^2\}_{t=1}^T)$.

C Data description

Bond holding returns are calculated on a quarterly basis using the zero coupon yield data constructed by Gurkaynak, Sack, and Wright (2007),⁷ which fit the Nelson-Siegel-Svensson curves daily since June 1961, and excess returns are computed by subtracting the return on a three-month Treasury bill. We consider the following maturities: six months, one, two, three, four, five, six, seven, and 10 years, which gives us a set of nine bond portfolios.

We consider several portfolios of stock returns. In addition to the bond portfolios, the baseline specification relies on the 25 size and book-to-market Fama-French portfolios (Fama and French (1992)) and 12 industry portfolios. In robustness checks, we consider four additional sets of characteristic-sorted portfolios, including (1) 32 size-profitability-investment-sorted portfolios, (2) 32 size-value-investment-sorted portfolios, (3) 32 size-value-profitability-sorted portfolios, and (4) 74 decile 1 and 10 portfolios from Kozak, Nagel, and Santosh (2018). These portfolios are sorted by 37 firm characteristics that have data since July 1963.⁸ The stock portfolio data of the first three cross-sections are available from the Kenneth French data library, while we obtain the data of 74 decile 1 and 10 portfolios from Serhiy Kozak’s website. We consider monthly returns from July 1963 to December 2019 and accumulate them to form quarterly returns, matching the frequency of consumption data. Excess returns are then formed by subtracting the corresponding return on the three-month Treasury bill.

Consumption flow is measured as the real (chain-weighted) expenditure on nondurable goods per capita available from the National Income and Product Accounts (NIPA). As in, for example, Parker and Julliard (2005), we do not include services in our baseline definition of consumption because these are likely to mechanically bias both the persistence of the consumption proxy (due to, e.g., utilities and health care) and the comovement with market returns (due to, e.g., financial services and insurance). Furthermore, nondurable consumption seems less affected by interpolation and other measurement issues (see, e.g., Savov (2011) and Kroencke (2017)). Our results are robust to the usage of alternative measures and refinements of the consumption proxy (e.g., the exclusion of shoes and clothing,

⁷The data are regularly updated and available at <http://www.federalreserve.gov/pubs/feds/2006/200628/200628abs.html>.

⁸The author provides the data of portfolios sorted by more than 50 firm characteristics. However, some characteristic-sorted portfolios do not have available data before the 1970s; hence, we use 37 characteristics to ensure that the time-series sample is consistent with the baseline analysis.

as in Lettau and Ludvigson (2001, 2001b), due to their semi-durable nature). We use the end-of-period timing convention and assume that all of the expenditure occurs at the end of the period between t and $t + 1$. We make this (common) choice because, under this convention, the entire period covered by time t consumption is part of the information set of the representative agent before time $t + 1$ returns are realized. All the returns are made real using the corresponding consumption deflator.

In Section III.1.2, we conduct horse races with the canonical predictors to explore whether the conditional consumption mean process uncovered by our state-space formulation indeed captures the consumption predictability. In particular, the real GDP growth comes from FRED St. Louis. The SPF forecasts are downloaded from the Federal Reserve Bank of Philadelphia. The news indices in Liu and Matthies (2022) are obtained from the authors' websites. Shareholders' and nonshareholders' consumption growth are downloaded from Tobias Moskowitz's website.

In some analyses, we include the market's price-dividend (P/D) ratio and the Chicago Fed National Activity Index (CFNAI) to predict stock and bond portfolio returns, respectively. We download the CFNAI data from the Federal Reserve Bank of Chicago, and the sample begins in January 1967.

We estimate the P/D ratio by comparing the gross return with the ex-dividends return. Suppose we have a cross-section of test assets, $i = 1, 2, \dots, N$, and D_{it} and P_{it} are the dividends and market prices of portfolio i between time $t - 1$ and t . The total and ex-dividend portfolio returns are defined as follows:

$$R_{it} = \frac{P_{it} + D_{it} - P_{i,t-1}}{P_{i,t-1}}, \quad R_{it,-d} = \frac{P_{it} - P_{i,t-1}}{P_{i,t-1}}, \implies \frac{D_{it}}{P_{i,t-1}} = R_{it} - R_{it,-d}, \text{ and } P_{i,t-1} = \frac{P_{it}}{1 + R_{it,-d}}.$$

We define the market's P/D ratio as follows:

$$PD_t = \frac{\sum_{i=1}^N P_{it}}{\sum_{s=0}^3 \sum_{i=1}^N D_{i,t-s}},$$

which is a smooth version P/D ratio used in past research, such as Welch and Goyal (2008) and Campbell and Thompson (2008).

Finally, we use several uncertainty measures commonly studied in the previous literature. Specifically, the VIX volatility index is downloaded from CRSP (and has been publicly available since 1986:Q1). The financial/real/macro uncertainty measures from Jurado, Ludvigson, and Ng (2015) and Ludvigson, Ma, and Ng (2021) are downloaded from the authors'

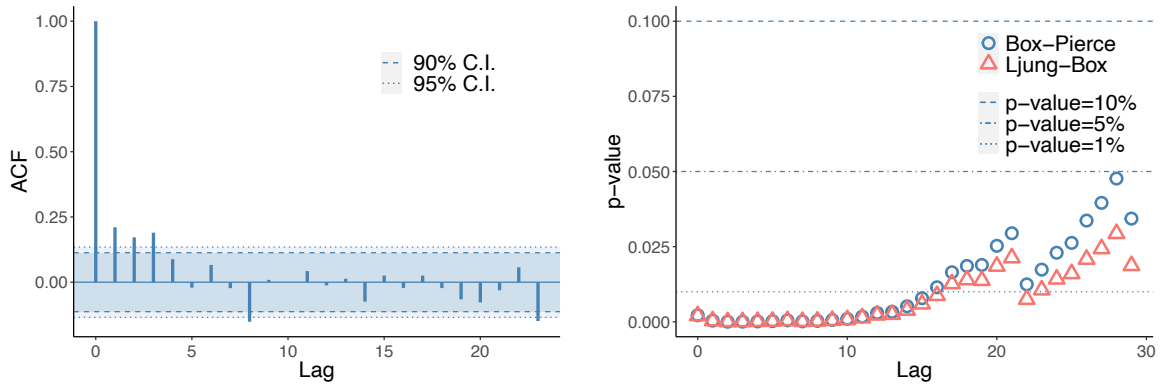
websites.

D Preliminary empirical evidence

To motivate the structure of the state-space model for consumption and asset returns, we first establish a set of empirical facts via model-free reduced-form approaches. We document that *a)* consumption growth is autocorrelated and *b)* not only asset returns predict future levels of consumption growth, but they also do it better than the past values of consumption itself. A detailed description of the data is reported in Appendix C.

First, Figure IA.1 plots the autocorrelation function (left panel), and the p -values (right panel) of the Ljung and Box (1978) and Box and Pierce (1970) tests of joint significance of the autocorrelations, of the one quarter log consumption growth ($\Delta c_{t,t+1}$). The figure shows that the autocorrelations are individually statistically significant up to the one-year horizon (left panel) and jointly statistically significant (right panel) at the 1% level, even after 14 quarters (and significant at lower confidence levels at even longer horizons). That is, there is substantial persistence in the time-series of consumption growth.⁹

Figure IA.1: Autocorrelation structure of consumption growth.



Left panel: Autocorrelation function of consumption growth ($\Delta c_{t,t+1}$) with 90% and 95% confidence bands. Right panel: p -values of Ljung and Box (1978) (triangles) and Box and Pierce (1970) (circles) tests.

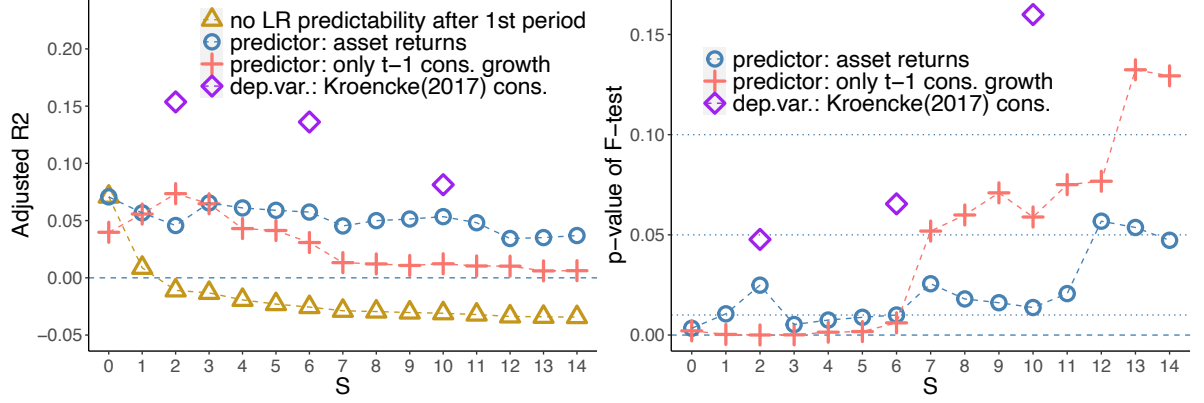
Second, we run multivariate linear predictive regressions of cumulated log consumption growth $\Delta c_{t,t+1+S}$ (for several values of S) on the first eight principal components of time t asset returns.¹⁰ Figure IA.2 depicts summary statistics for these predictive regressions at

⁹Note that, even in the seminal examination of the random walk hypothesis of Hall (1978), the presence of predictability in consumption growth could not be rejected.

¹⁰We use the first eight principal components of the 25 size and book-to-market Fama-French portfolios, 12

different horizons (S). In particular, the left panel plots the time-series adjusted R^2 of these regressions, and the right panel plots the p -value of the F -test of joint significance of the regressors for this as well as other additional specifications.

Figure IA.2: Predictive regressions of $\Delta c_{t,t+1+S}$ on time t asset returns and consumption.



The figure shows the predictive regressions of $\Delta c_{t,t+1+S}$ on the first eight principal components of asset returns between time $t - 1$ and t for different values of S . Left panel: Adjusted R^2 (blue line with circles) and theoretical adjusted R^2 (yellow dashed line with triangles) if all the predictability was driven by the first period. The red dashed line with pluses stands for the adjusted R^2 when only $t - 1$ consumption growth is used as a predictor. The dotted line with rhombi corresponds to using asset returns to predict the unfiltered consumption growth of Kroencke (2017). Right panel: p -value of the F -test of joint significance of the covariates, as well as the 10%, 5%, and 1% significance thresholds.

Several observations are to be mentioned. At $S = 0$, the time-series adjusted R^2 is quite large, being about 6.3%. Moreover, the regressors are jointly statistically significant (the p -value of the F -test is less than 1%). For $S > 0$, because $\Delta c_{t,t+1+S} \equiv \Delta c_{t,t+1} + \Delta c_{t+1,t+1+S}$, if asset returns did not predict the autocorrelated component of future consumption growth, the adjusted R^2 should actually decrease monotonically in S , as depicted by the yellow dashed lines with triangles in the left panel of Figure IA.2. Instead, for $S > 0$, the figure shows no such decrease in the data (blue dashed line with circles). In fact, predictability increases at intermediate horizons. Moreover, the regressors are jointly statistically significant for any horizon up to 12 quarters following the returns.

Could one achieve the same level of predictability by using only consumption data, either due to a persistent component (independent of returns) propagating through the actual consumption growth (as, e.g., an AR(1)), or through accumulated non-classical measurement errors that display a certain degree of persistence? This is unlikely. The red crosses in Figure

industry portfolios, and nine bond portfolios, because they explain about 95% of the asset returns variance. Using fewer, or more, principal components, or even directly the asset returns series, we have obtained very similar results to those reported.

IA.2 depict the degree of predictability obtained using only lagged consumption growth, $\Delta c_{t-1,t}$. Although highly significant at horizons up to six quarters, using lagged consumption as a predictor is inferior to extracting information from asset returns: Not only does this variable fail to capture the long range of true predictability, but even at the short horizon, it is almost always underperforming stock and bond returns.

Measurement errors in consumption are unlikely to yield such a persistent level of predictability either. Although non-classical errors could possibly contribute to a wide range of statistical artefacts, most of their impact should either disappear within a horizon of about one year (if it is related to seasonal smoothing) or be much smaller in magnitude. In order to test this conjecture, we repeat the same predictive exercise with the unfiltered consumption data of Kroenke (2017)¹¹ (purple diamonds in Figure IA.2). If the predictability result is an accidental by-product of a countercyclical measurement error due to smoothing, it must go away when using the unfiltered data. If anything, as the figure shows, the power of asset returns to forecast consumption becomes even more apparent. Unfortunately, because only yearly data are available for unfiltered consumption, the sample is naturally shorter, which increases standard errors and leads to the feasible use of only three predictive horizons within our time window. However, even taking these limitations into account, asset returns still remain significant predictors of future consumption.

The results above highlight that not only is there substantial predictability in consumption growth, but it is also best captured by asset returns.

E Dew-Becker (2017) bounds

Dew-Becker (2017) provides bounds for the long-run standard deviation (LRSD) of consumption growth. Within our framework, this can be expressed as

$$LRSD \equiv std(\Delta \mathbb{E}_t \sum_{j=0}^{\infty} \Delta c_{t+j}) = std((\sum_{j=0}^{\bar{S}} \rho_j) f_t + w_t^c) = \sqrt{(\sum_{j=0}^{\bar{S}} \rho_j)^2 + \sigma_c^2}.$$

In our baseline analysis using nondurable consumption growth, we find that the posterior mean of (annualized) LRSD is about 2.3%, with [1.6%, 3.1%] as 95% CI. Using instead nondurable plus service consumption, Dew-Becker (2017) estimates that the LRSD of con-

¹¹We are grateful to Tim Kroenke for making the data available on his website.

sumption growth in the postwar sample is 2.5% per year, with an upper bound being the 95% confidence interval of 4.9%. Using the same consumption measure, we estimate a posterior mean of 2.2% for the LRSD, with [1.6%, 2.9%] as 95% CI. Hence, the estimates of the LRSD implied by our state-space method are close to the one in Dew-Becker (2017) and within his bounds.

F Estimation with monthly consumption data

In this section, we consider the consumption dynamics at the monthly frequency. We download the real nondurable consumption data from Table 2.8.3 of BEA and the monthly US population from FRED St. Louis to calculate the real monthly nondurable consumption growth per capita.¹²

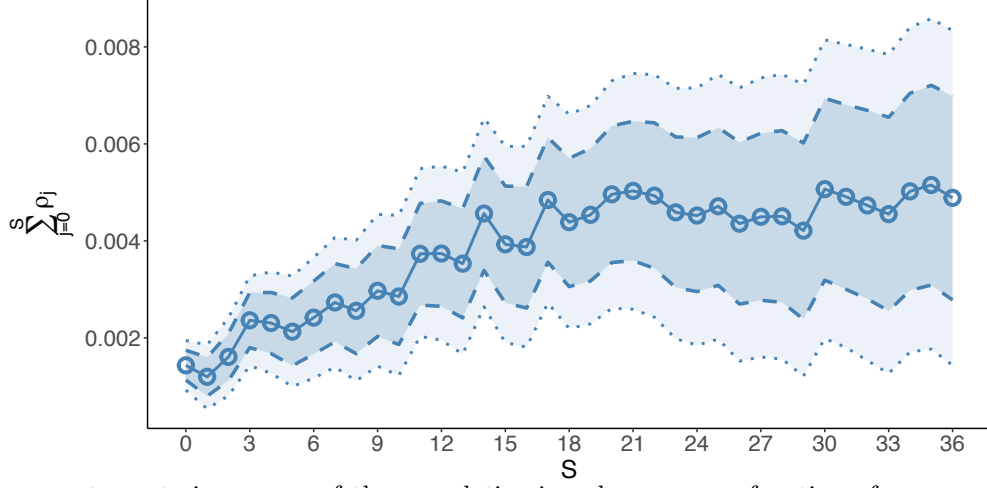
We first consider the estimation without stochastic volatility. Figure IA.3 plots the cumulative response function of monthly consumption growth to a one-standard-deviation f_t shock. Even at the monthly frequency without the concern of time-aggregation bias, we find that consumption growth slowly adjusts to the asset return shocks. The cumulative effects are substantial: They keep increasing even after 15 months, and they jointly explain about 11% of the time-series variation of (one-period) monthly consumption growth, as can be seen in Figure IA.4. Moreover, monthly consumption growth shows significantly negative autocorrelations (see the left panel of Figure IA.5), which may potentially originate from the mean-reverting measurement error in the monthly consumption data. Consequently, the null hypothesis of uncorrelated forecast errors in consumption growth is rejected in the data (see the right panel of Figure IA.5).

F.1 Stochastic Volatility at the monthly frequency

To study the stochastic volatilities of consumption at the monthly frequency, we need to explicitly model the mean-reverting measurement error; otherwise, we would obtain spurious volatility clustering (see Figure IA.5). We consider a simple IID measurement error in

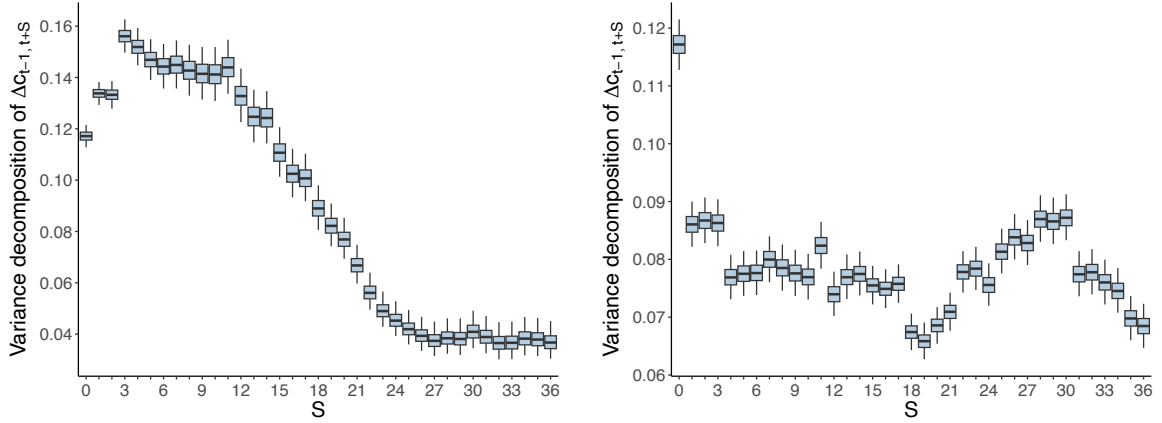
¹² The monthly consumption data are used by previous literature, such as Hansen and Singleton (1983) and Heaton (1995). However, Breeden, Gibbons, and Litzenberger (1989) show several reasons for using quarterly consumption rather than monthly data, such as sampling error and infrequent reporting of consumption.

Figure IA.3: Cumulative response function of consumption growth to a one-standard-deviation shock spanned by asset returns: Monthly frequency.



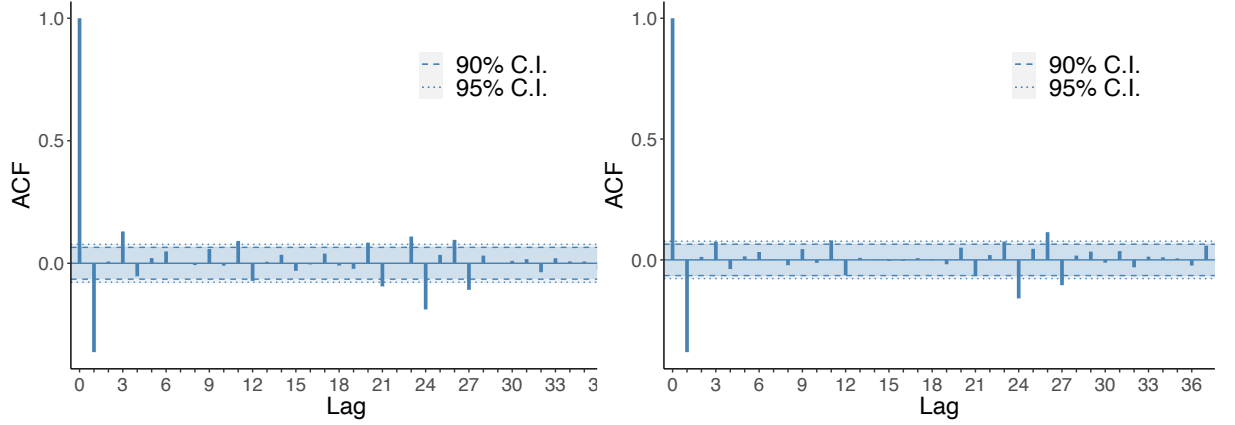
The graph presents posterior means of the cumulative impulse response function of consumption growth (solid line with circles), along with the centered posterior 90% (dotted lines) and 68% (dashed lines) coverage regions. These shocks account for 11% of monthly consumption growth time-series variation (monthly data, 07/1963–12/2019).

Figure IA.4: Share of consumption growth variance driven by its moving average component.



The figure presents box plots (posterior 95% coverage area) of the percentage of time-series variances of consumption growth explained by the MA component. These plots report the unadjusted R-squared. Left panel: Cumulated consumption growth $\Delta c_{t-1,t+S}$. Right panel: One-period consumption growth $\Delta c_{t-1+j,t+j}$. We study a single-factor model of asset returns, with $\bar{S} = 36$. The cross-section of test assets includes 25 size- and value-sorted portfolios, 12 industry portfolios, and nine bond portfolios (monthly data, 07/1963–12/2019).

Figure IA.5: Autocorrelation structure of consumption growth & its forecast errors.



Autocorrelation function of $\Delta c_{t,t+1}$ (left panel) and its forecast error (right panel) with 95% and 99% confidence bands (monthly data, 07/1963–12/2019).

monthly log consumption level (as in Schorfheide, Song, and Yaron (2018)),

$$c_t = c_t^* + \sigma_{\epsilon c} \cdot \epsilon_{c,t}, \quad \text{where } \epsilon_{c,t} \stackrel{\text{iid}}{\sim} \mathcal{N}(0, 1),$$

where c_t^* is the latent true log consumption level without measurement errors, and this implies the following consumption growth dynamics,

$$\Delta c_t = \mu_c + \underbrace{\sum_{j=0}^{\bar{S}} \rho_j f_{t-j}}_{MA(\bar{S})} + \underbrace{\sigma_{\epsilon c} \cdot (\epsilon_{c,t} - \epsilon_{c,t-1})}_{\text{measurement error}} + w_t^c, \quad w_t^c \sim \mathcal{N}\left(0, \exp(h_{ct})\right), \quad \sigma_{c,t-1}^2 \quad (\text{IA.8})$$

where we assume that the measurement error is uncorrelated with the innovation to expected consumption growth, f_t , and the short-run consumption shock, w_t^c . The measurement error in monthly consumption growth, $\sigma_{\epsilon c} \cdot (\epsilon_{c,t} - \epsilon_{c,t-1})$, can explain the significantly negative one-lag autocorrelation coefficient in Figure IA.5. The return dynamics is the same as in equation (16).

To estimate the system with an additional measurement error, we need to expand the space of latent states. Specifically, we write down the following state-space model:

$$\mathbf{z}_t = \mathbf{F}\mathbf{z}_{t-1} + \mathbf{v}_t, \quad \mathbf{v}_t \sim \mathcal{N}(\mathbf{0}_{\bar{S}+3}, \mathbf{\Psi}_t), \quad \text{and} \quad (\text{IA.9})$$

$$\mathbf{y}_t = \boldsymbol{\mu}_t + \mathbf{H}\mathbf{z}_t + \mathbf{w}_t, \quad \mathbf{w}_t \sim \mathcal{N}(\mathbf{0}_{N+1}, \boldsymbol{\Sigma}_t), \quad (\text{IA.10})$$

where $\mathbf{y}_t := [\Delta c_{t-1,t}, \mathbf{r}_t^e]'$, $\mathbf{v}_t := [f_t, \mathbf{0}'_{\bar{S}}, \epsilon_{c,t}, 0]'$, $\mathbf{w}_t := [\tilde{w}_t^c, \mathbf{w}_t^{r'}]'$,

$$z_t = \begin{bmatrix} f_t \\ \vdots \\ f_{t-\bar{S}} \\ \epsilon_{c,t} \\ \epsilon_{c,t-1} \end{bmatrix}, \quad \mathbf{F} = \begin{bmatrix} \mathbf{0}_{\bar{S}}^\top & 0 & 0 & 0 \\ \mathbf{I}_{\bar{S}} & \mathbf{0}_{\bar{S}} & \mathbf{0}_{\bar{S}} & \mathbf{0}_{\bar{S}} \\ \mathbf{0}_{\bar{S}}^\top & 0 & 0 & 0 \\ \mathbf{0}_{\bar{S}}^\top & 0 & 1 & 0 \end{bmatrix}, \quad \mathbf{\Psi}_t = \begin{bmatrix} \exp(h_{ft}) & \mathbf{0}_{\bar{S}}^\top & 0 & 0 \\ \mathbf{0}_{\bar{S}} & \mathbf{0}_{\bar{S} \times \bar{S}} & \mathbf{0}_{\bar{S}} & \mathbf{0}_{\bar{S}} \\ 0 & \mathbf{0}_{\bar{S}}^\top & 1 & 0 \\ 0 & \mathbf{0}_{\bar{S}}^\top & 0 & 0 \end{bmatrix},$$

$$\boldsymbol{\mu}_t = \begin{bmatrix} \mu_c \\ \boldsymbol{\mu}_r + \beta_c \sigma_{c,t-1}^2 + \beta_f \sigma_{f,t-1}^2 + \beta_r \sigma_{r,t-1}^2 \end{bmatrix}, \quad \mathbf{H} = \begin{bmatrix} \rho_0 & \dots & \rho_{\bar{S}} & \sigma_{ec} & -\sigma_{ec} \\ \boldsymbol{\rho}^r & \dots & \mathbf{0}_N & \mathbf{0}_N & \mathbf{0}_N \end{bmatrix}, \quad \boldsymbol{\Sigma}_t = \begin{bmatrix} \exp(h_{ct}) & \\ & \boldsymbol{\Sigma}_{r,t-1} \end{bmatrix}.$$

In the monthly data, the measurement error, quantified by σ_{ec} , is sizable. The posterior mean of σ_{ec} is 0.0036. Note that the volatility of the monthly consumption growth is 0.0072; hence, ignoring measurement errors would induce mechanical volatility clustering.

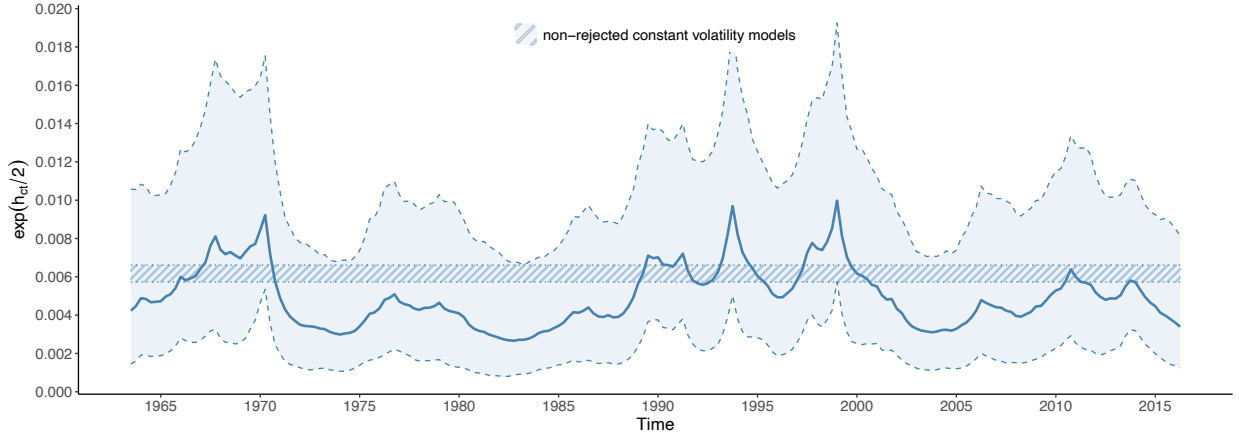
Results are reported in Figures IA.6 and IA.7. First, the short-run consumption volatility is still hard to detect. As Panel A of Figure IA.6 shows, there is an entire range of constant volatility levels for $\sigma_{c,t}^2$. Second, the long-run consumption volatility, $\sigma_{f,t}^2$, is sharply identified using the monthly data. Third, the common market volatility, $\sigma_{r,t}^2$, becomes much spikier at the monthly than quarterly frequency, especially in the early 2000s. Finally, neither short- nor long-run consumption volatility ($\sigma_{c,t}^2$ or $\sigma_{f,t}^2$) significantly predicts excess returns (see Panels A and B of Figure IA.7). In contrast, the common market volatility negatively predicts the next-month excess returns, although most of the coefficients are not significant.

G Mixed-frequency state-space estimation

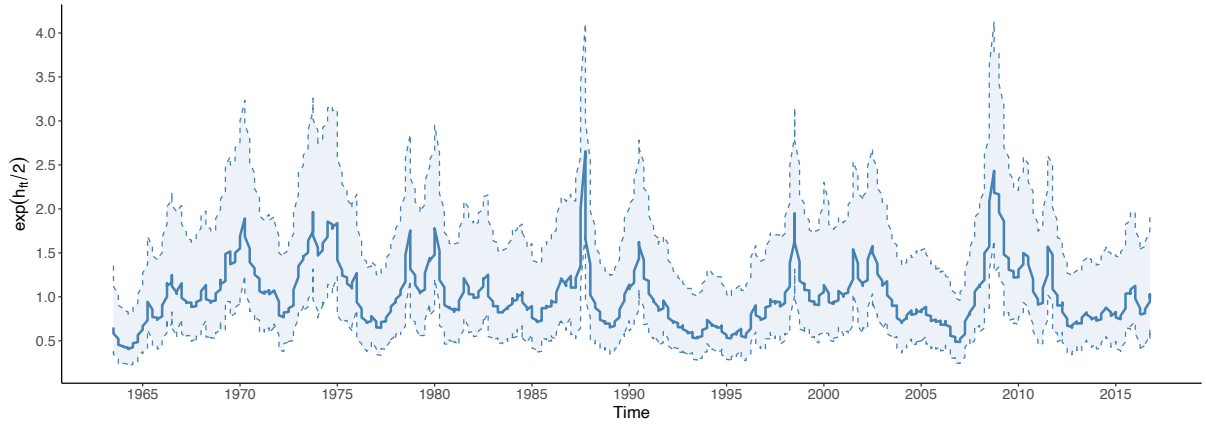
In the main analysis of the quarterly consumption growth, we ignore the fact that consumption is measured over a time interval, whereas asset prices are observed at a higher frequency (e.g., monthly). In this section, we derive a mixed-frequency state-space model to handle the time-aggregation bias. Specifically, we assume that we observe only the quarterly consumption growth in the final month of a quarter but monthly portfolio returns. However, the estimation in this section does not rely on monthly consumption growth. The estimation based on monthly consumption data is reported in Section F.

Throughout this section, we use t to denote the month t . For instance, $\mathbf{r}_t^e$ denote the excess returns in month t . Consumption data, C_t , are observed at the quarterly frequency, which equals the sum of the monthly consumption within the quarter: $C_t = C_t^* + C_{t-1}^* + C_{t-2}^*$, where C_t^* denotes the unobserved monthly consumption. Using the log-linearization (see,

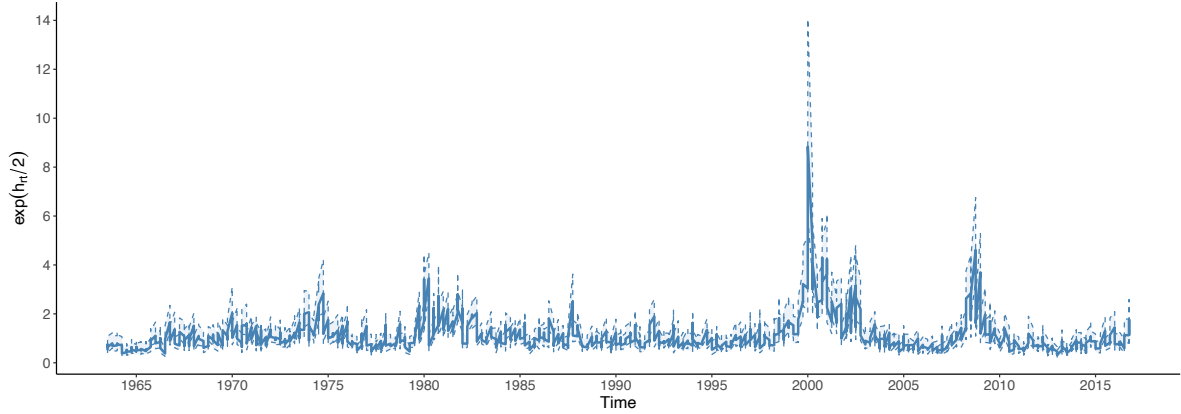
Figure IA.6: Filtered stochastic volatilities of *monthly* consumption and returns.



Panel A: Log volatility of the short-run consumption shock (w_t^c)



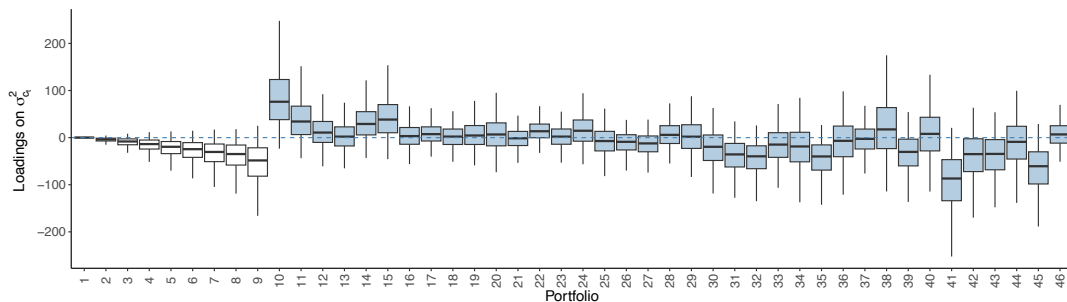
Panel B: Log volatility of the shock to the conditional mean of consumption growth (f_t).



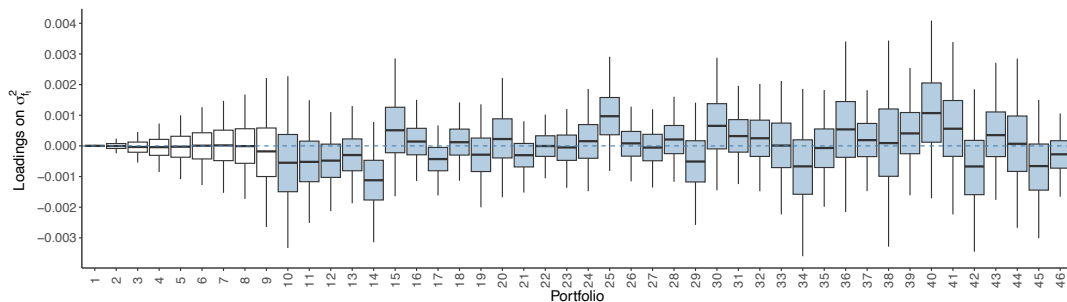
Panel C: Common log volatility of asset return (h_{rt} of equation (19)).

The figure shows the estimated stochastic volatilities of the model in equations (16) and (IA.8). Solid blue lines depict the posterior median of the log volatility, whereas dotted red lines denote 2.5% and 97.5% credible intervals. Shaded (patterned) areas reflect constant volatility levels that would not be rejected given the credible intervals. Both consumption and asset returns are observed at the monthly frequency.

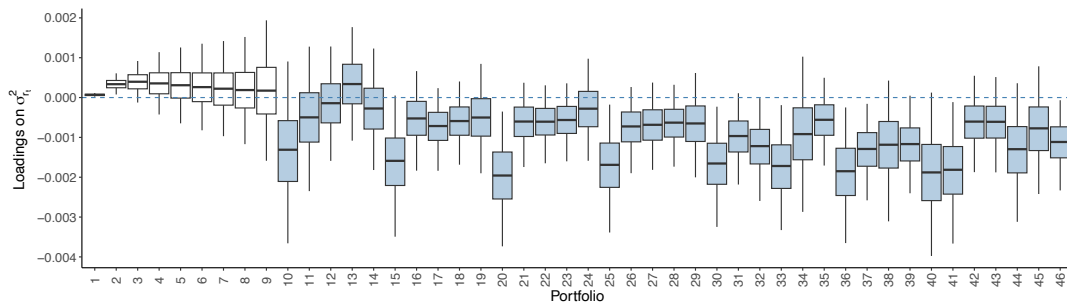
Figure IA.7: Loadings of excess returns on *monthly* consumption and returns volatilities.



Panel A: Posterior distribution of excess return loadings (β_c) on the variance of short-run consumption shocks ($\sigma_{c,t-1}^2$) in equation (16).



Panel B: Posterior distribution of excess return loadings (β_f) on the variance of shocks to the conditional consumption growth mean ($\sigma_{f,t-1}^2$) in equation (16).



Panel C: Posterior distributions of excess return loadings (β_r) on the common financial return variance ($\sigma_{r,t-1}^2$) in equation (16).

The figure shows the box plots of the posterior distributions of the loadings of portfolio excess returns on the variance of shocks to the conditional consumption growth and the common financial returns variance. Portfolios are ordered with bonds first (1–9), Fama-French 25 size and book-to-market second (10–34), and industry portfolios last. Both consumption and asset returns are observed at the monthly frequency.

for example, Schorfheide, Song, and Yaron (2018)), we obtain $\log C_t = \frac{1}{3}(\log C_t^* + \log C_{t-1}^* + \log C_{t-2}^*)$. Hence, the quarterly consumption growth, for all t , is

$$\Delta c_{t-3,t} = \log C_t - \log C_{t-3} = \frac{1}{3}\Delta c_t^* + \frac{2}{3}\Delta c_{t-1}^* + \Delta c_{t-2}^* + \frac{2}{3}\Delta c_{t-3}^* + \frac{1}{3}\Delta c_{t-4}^*, \quad (\text{IA.11})$$

where $\Delta c_t^* = \log C_t^* - \log C_{t-1}^*$ denotes the unobserved monthly consumption growth.

We assume the same data-generating process of asset returns as before,

$$\mathbf{r}_t^e = \boldsymbol{\mu}_r + \boldsymbol{\rho}_f^r f_t + \mathbf{w}_t^r, \quad (\text{IA.12})$$

where the only difference is that we now observe $\mathbf{r}_t^e$ at the monthly frequency. Moreover, monthly consumption growth can react slowly to the f_t shocks, as follows:

$$\Delta c_t^* = \mu_c^* + \underbrace{\sum_{j=0}^{\bar{S}} \rho_j f_{t-j}}_{MA(\bar{S})} + w_t^{c*}, \quad (\text{IA.13})$$

which implies the following MA representation of quarterly consumption growth $\Delta c_{t-3,t}$:

$$\Delta c_{t-3,t} = \mu_c + \sum_{j=0}^{\bar{S}} \rho_j \left(\frac{1}{3} f_{t-j} + \frac{2}{3} f_{t-j-1} + f_{t-j-2} + \frac{2}{3} f_{t-j-3} + \frac{1}{3} f_{t-j-4} \right) + w_t^c, \quad \text{and} \quad (\text{IA.14})$$

$$\mu_c = 3\mu_c^*, \quad w_t^c = \frac{1}{3}w_t^{c*} + \frac{2}{3}w_{t-1}^{c*} + w_{t-2}^{c*} + \frac{2}{3}w_{t-3}^{c*} + \frac{1}{3}w_{t-4}^{c*}.$$

We take an agnostic view on the dynamics of w_t^{c*} ; hence, we do not impose a MA(4) process for w_t^c . It is worth noting that the distribution of w_t^c does not affect the consistency of μ_c and ρ^c , as long as it is a zero-mean process and uncorrelated with f_t .

We introduce the following matrix notations:

$$\mathbf{z}_t = \begin{bmatrix} f_t \\ \vdots \\ f_{t-\bar{S}} \\ f_{t-\bar{S}-1} \\ f_{t-\bar{S}-2} \\ f_{t-\bar{S}-3} \\ f_{t-\bar{S}-4} \end{bmatrix}, \quad \boldsymbol{\rho}_0^c = \begin{bmatrix} \rho_0 \\ \vdots \\ \rho_{\bar{S}} \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \quad \boldsymbol{\rho}_1^c = \begin{bmatrix} 0 \\ \vdots \\ \rho_{\bar{S}-1} \\ \rho_{\bar{S}} \\ 0 \\ 0 \\ 0 \end{bmatrix}, \quad \boldsymbol{\rho}_2^c = \begin{bmatrix} 0 \\ \vdots \\ \rho_{\bar{S}-2} \\ \rho_{\bar{S}-1} \\ \rho_{\bar{S}} \\ 0 \\ 0 \end{bmatrix}, \quad \boldsymbol{\rho}_3^c = \begin{bmatrix} 0 \\ \vdots \\ \rho_{\bar{S}-3} \\ \rho_{\bar{S}-2} \\ \rho_{\bar{S}-1} \\ \rho_{\bar{S}} \\ 0 \end{bmatrix}, \quad \boldsymbol{\rho}_4^c = \begin{bmatrix} 0 \\ \vdots \\ \rho_{\bar{S}-4} \\ \rho_{\bar{S}-3} \\ \rho_{\bar{S}-2} \\ \rho_{\bar{S}-1} \\ \rho_{\bar{S}} \end{bmatrix},$$

which imply the following dynamics for quarterly consumption growth:

$$\Delta c_{t-3,t} = \mu_c + (\boldsymbol{\rho}^c)^\top \mathbf{z}_t + w_t^c, \quad \text{where: } \boldsymbol{\rho}^c = \frac{1}{3}\boldsymbol{\rho}_0^c + \frac{2}{3}\boldsymbol{\rho}_1^c + \boldsymbol{\rho}_2^c + \frac{2}{3}\boldsymbol{\rho}_3^c + \frac{1}{3}\boldsymbol{\rho}_4^c. \quad (\text{IA.15})$$

Note that we observe $\Delta c_{t-3,t}$ only once per quarter. To handle the missing data issue,

we follow Mariano and Murasawa (2003) and define $\Delta c_{t-3,t}^+$ as follows:

$$\Delta c_{t-3,t}^+ = \begin{cases} \Delta c_{t-3,t}, & t \text{ is the last month of the quarter,} \\ d_t, & \text{otherwise,} \end{cases}$$

where d_t is a random draw from a distribution that does not depend on the model parameters (denoted by $\boldsymbol{\theta}$). As in Mariano and Murasawa (2003), we assume $d_t \sim \mathcal{N}(0, 1)$, yielding the likelihood

$$p(\{\Delta c_{t-3,t}^+, \mathbf{r}_t^e\}_{t=1}^T | \boldsymbol{\theta}) = p(\{\Delta c_{t-3,t}, \mathbf{r}_t^e\}_{t=1}^T | \boldsymbol{\theta}) \prod_{t \in A} p(d_t),$$

where $\Delta c_{t-3,t}$ is missing for all $t \in A$. Therefore, the inference for $\boldsymbol{\theta}$ does not depend on d_t . For simplicity, we follow Mariano and Murasawa (2003) and set $d_t = 0$ for all t .

We can now write down a mixed-frequency state-space system,

$$\mathbf{z}_t = \mathbf{F}\mathbf{z}_{t-1} + \mathbf{v}_t, \quad \mathbf{v}_t \sim \mathcal{N}(\mathbf{0}_{\bar{S}+5}, \boldsymbol{\Psi}), \quad \text{and} \quad (\text{IA.16})$$

$$\mathbf{y}_t = \boldsymbol{\mu}_t + \mathbf{H}_t \mathbf{z}_t + \mathbf{w}_t, \quad \mathbf{w}_t \sim \mathcal{N}(\mathbf{0}_{N+1}, \boldsymbol{\Sigma}_t) \quad (\text{IA.17})$$

where $\mathbf{y}_t := [\Delta c_{t-3,t}^+, \mathbf{r}_t^e]'$, $\boldsymbol{\mu}_t := [\mu_{c,t}, \boldsymbol{\mu}_r']'$, $\mathbf{v}_t := [f_t, \mathbf{0}_{\bar{S}+4}]'$, $\mathbf{w}_t := [\tilde{w}_t^c, \mathbf{w}_t^{rr}]'$,

$$\boldsymbol{\Psi} := \underbrace{\begin{bmatrix} 1 & \mathbf{0}_{\bar{S}+4}' \\ \mathbf{0}_{\bar{S}+4} & \mathbf{0}_{(\bar{S}+4) \times (\bar{S}+4)} \end{bmatrix}}_{(\bar{S}+5) \times (\bar{S}+5)}, \quad \mathbf{F} := \underbrace{\begin{bmatrix} \mathbf{0}_{\bar{S}+4}' & 0 \\ \mathbf{I}_{\bar{S}+4} & \mathbf{0}_{\bar{S}+4} \end{bmatrix}}_{(\bar{S}+5) \times (\bar{S}+5)}, \quad \boldsymbol{\Sigma}_t := \underbrace{\begin{bmatrix} \sigma_{c,t}^2 & \mathbf{0}_N' \\ \mathbf{0}_N & \boldsymbol{\Sigma}_r \end{bmatrix}}_{(N+1) \times (N+1)}, \quad \mathbf{H}_t := \underbrace{\begin{bmatrix} \mathbf{H}_{1,t} \\ \boldsymbol{\rho}_f^r & \mathbf{0}_N & \dots & \mathbf{0}_N \end{bmatrix}}_{(N+1) \times (\bar{S}+5)},$$

$$\mu_{c,t} = \begin{cases} \mu_c, & t \text{ is the last month of the quarter} \\ 0, & \text{otherwise} \end{cases} \quad \mathbf{H}_{1,t} = \begin{cases} (\boldsymbol{\rho}^c)^\top, & t \text{ is the last month of the quarter} \\ \mathbf{0}_{\bar{S}+5}^\top, & \text{otherwise} \end{cases}$$

$$\tilde{w}_{c,t} = \begin{cases} w_{c,t}, & t \text{ is the last month of the quarter} \\ d_t, & \text{otherwise} \end{cases} \quad \sigma_{c,t}^2 = \begin{cases} \sigma_c^2, & t \text{ is the last month of the quarter} \\ 1, & \text{otherwise} \end{cases}.$$

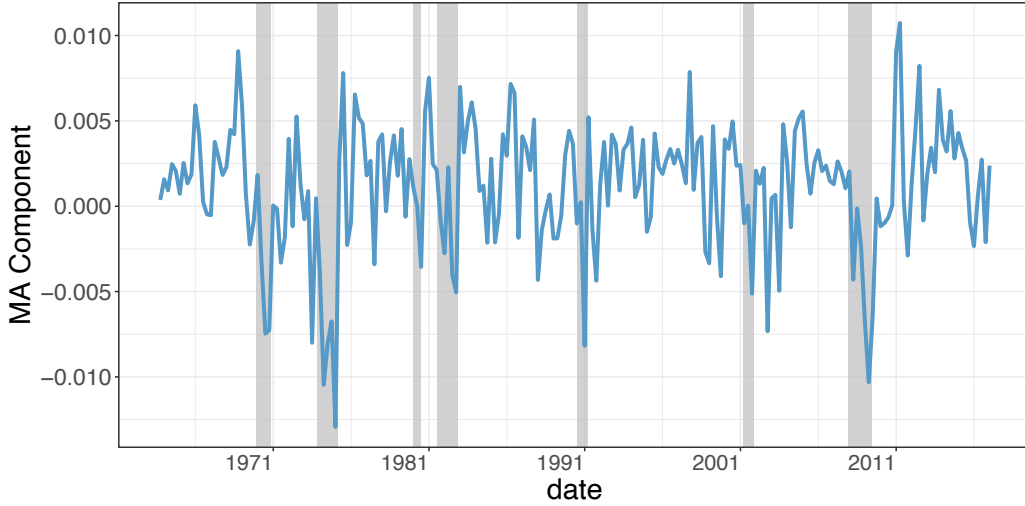
There are two major differences from the benchmark analysis based on the time-aggregated quarterly data. First, there is a measurement error, denoted by d_t , in the consumption growth equation when $\Delta c_{t-3,t}$ is missing (that is, when t is not the last month of the quarter). Second, we infer the common factor f_t using monthly asset returns and quarterly observed consumption growth. However, when $\Delta c_{t-3,t}$ is missing, it does not play a role in estimating

f_t . To see the last point, let's consider the standard Kalman filter,

$$\begin{aligned} \mathbf{z}_{t|t-1} &= \mathbf{F}' \mathbf{z}_{t-1|t-1}; \quad \mathbf{V}_{t|t-1} = \mathbf{F} \mathbf{V}_{t-1|t-1} \mathbf{F}' + \Psi; \quad \mathbf{K}_t = \mathbf{V}_{t|t-1} \mathbf{H}_t' (\mathbf{H}_t \mathbf{V}_{t|t-1} \mathbf{H}_t' + \Sigma_t)^{-1} \\ \mathbf{z}_{t|t} &= \mathbf{z}_{t|t-1} + \mathbf{K}_t (\mathbf{y}_t - \boldsymbol{\mu}_t - \mathbf{H}_t \mathbf{z}_{t|t-1}); \quad \mathbf{V}_{t|t} = \mathbf{V}_{t|t-1} - \mathbf{K}_t \mathbf{H}_t \mathbf{V}_{t|t-1}; \\ \mathbf{z}_{t|\tau} &:= \mathbb{E}[\mathbf{z}_t | \mathbf{y}^\tau, \mathbf{H}_t, \Psi, \Sigma_t]; \quad \text{and } \mathbf{V}_{t|\tau} := \text{Cov}(\mathbf{z}_t | \mathbf{y}^\tau, \mathbf{H}_t, \Psi, \Sigma_t). \end{aligned}$$

When t is not the final month of a quarter, the first row of $\mathbf{H}_t$ consists of zeros, so the Kalman gain $\mathbf{K}_t$ depends only on the factor loadings of asset returns ($\boldsymbol{\rho}_f^r$). Moreover, the first element of $\mathbf{y}_t - \boldsymbol{\mu}_t - \mathbf{H}_t \mathbf{z}_{t|t-1}$ is zero when $\Delta c_{t-3,t}$ is not observed. This observation echoes our previous argument that the assumption of d_t 's data-generating process is not essential as long as it does not rely on $\boldsymbol{\theta}$.

Figure IA.8: MA component of consumption growth: Mixed-frequency estimation.



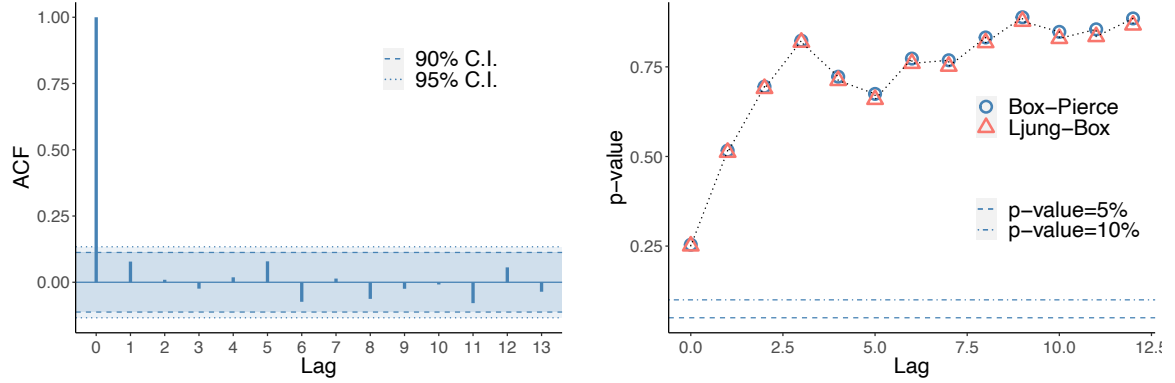
The figure shows the posterior mean of the moving average f_t component of quarterly consumption growth. Grey areas denote NBER recessions. We study a mixed-frequency single-factor model in equations (3) and (IA.14), with $\bar{S} = 36$ (months). The cross-section of test assets includes 25 size- and value-sorted portfolios, 12 industry portfolios, and nine bond portfolios.

G.1 Incorporating stochastic volatility at mixed frequencies

We consider the following return dynamics at the monthly frequency:

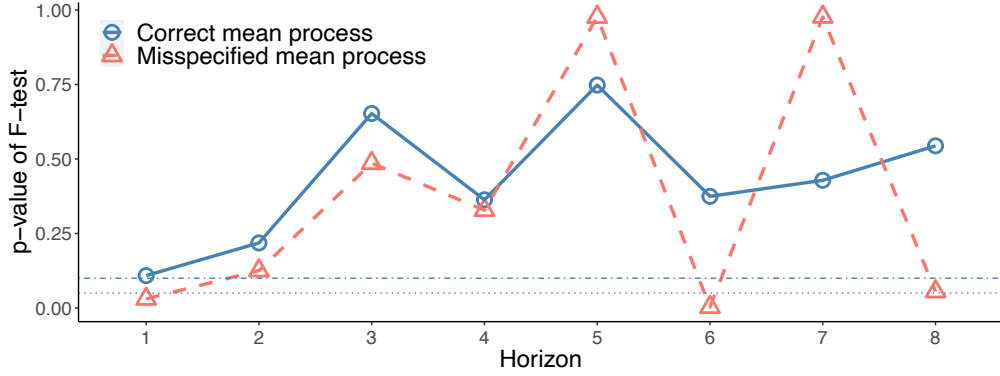
$$\mathbf{r}_t^e = \underbrace{\boldsymbol{\mu}_r}_{N \times 1} + \underbrace{\boldsymbol{\rho}^r}_{N \times 1} f_t + \underbrace{\boldsymbol{\beta}_f}_{N \times 1} \sigma_{f,t-1}^2 + \underbrace{\boldsymbol{\beta}_r}_{N \times 1} \sigma_{r,t-1}^2 + \underbrace{\mathbf{w}_t^r}_{N \times 1}, \quad f_t \sim \mathcal{N} \left(0, \exp(h_{f_t}) \right), \quad (\text{IA.18})$$

Figure IA.9: Autocorrelation structure of consumption growth squared forecast errors: Quarterly frequency.



Left panel: Autocorrelation function of $\widehat{Var}_t(\Delta c_{t,t+1})$ with 95% and 99% confidence bands. Right panel: p -values of Ljung and Box (1978) (red triangles) and Box and Pierce (1970) (blue circles) tests (quarterly data, 1963:Q3–2019:Q4). The model is based on the mixed-frequency state-space estimation.

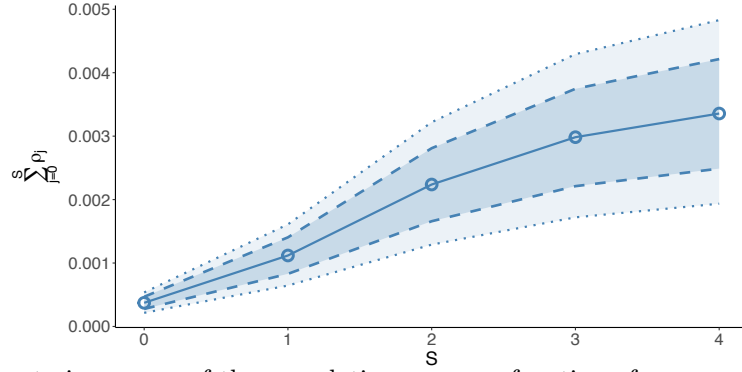
Figure IA.10: Predictability of consumption squared forecast errors.



Predictive regressions of $\widehat{Var}_{t+h}(\Delta c_{t+h,t+h+1})$ on the time- t first eight principal components of quarterly asset returns at several horizons h . We report the p -value of the F -test of joint significance of the covariates as well as the 10% and 5% significance thresholds (respectively, horizontal dot-dashed and dotted lines). The solid blue line with circles denotes statistics for the correctly specified conditional mean for the consumption growth process, while the dashed red line with triangles corresponds to the assumption of a constant conditional mean. In the correct mean process, we estimate a mixed-frequency state-space model of quarterly consumption growth and monthly asset returns.

where $\sigma_{r,t-1}^2$ is a common market volatility process that affects the volatility of all excess return shocks ($\mathbf{w}^r$), and $\sigma_{f,t-1}^2$ is the long-run consumption volatility (in equation (IA.15)). The key distinction from the analysis in the main text is that we consider monthly asset returns in equation (IA.18) and, hence, monthly stochastic volatility of f_t and $\mathbf{w}_t^r$; meanwhile, we still estimate the stochastic volatility process of the quarterly consumption growth shock (w_t^c). At the mixed frequency, we do not include $\sigma_{c,t-1}^2$ into equation (IA.18). In Section F.1, we consider monthly consumption data and further incorporate the short-run stochastic volatility of consumption, $\sigma_{c,t-1}^2$, into the return dynamics.

Figure IA.11: Cumulative response function of consumption growth to a one-standard-deviation shock spanned by asset returns: Mixed-frequency estimation with $\bar{S} = 0$.



The figure presents posterior means of the cumulative response function of consumption growth (solid line with circles), along with the centered posterior 90% (dotted lines) and 68% (dashed lines) coverage regions. We estimate a mixed-frequency state-space model of quarterly consumption growth and monthly asset returns, with $\bar{S} = 0$. These monthly f_t shocks account for **6.4%** of quarterly consumption growth time-series variation (quarterly data, 1963:Q3–2019:Q4).

We next consider the consumption dynamics, as follows:

$$\Delta c_{t-3,t} = \mu_c + \sum_{j=0}^{\bar{S}} \rho_j \left(\frac{1}{3} f_{t-j} + \frac{2}{3} f_{t-j-1} + f_{t-j-2} + \frac{2}{3} f_{t-j-3} + \frac{1}{3} f_{t-j-4} \right) + w_t^c, \quad w_t^c \sim \mathcal{N} \left(0, \exp(h_{ct}) \right). \quad (\text{IA.19})$$

We can now write a mixed-frequency state-space system with three separate stochastic volatility processes,

$$\mathbf{z}_t = \mathbf{F} \mathbf{z}_{t-1} + \mathbf{v}_t, \quad \mathbf{v}_t \sim \mathcal{N}(\mathbf{0}_{\bar{S}+5}, \mathbf{\Psi}_t), \quad \text{and} \quad (\text{IA.20})$$

$$\mathbf{y}_t = \boldsymbol{\mu}_t + \mathbf{H}_t \mathbf{z}_t + \mathbf{w}_t, \quad \mathbf{w}_t \sim \mathcal{N}(\mathbf{0}_{N+1}, \boldsymbol{\Sigma}_t) \quad (\text{IA.21})$$

where $\mathbf{y}_t := [\Delta c_{t-3,t}^+, \mathbf{r}_t^{ef}]$, $\boldsymbol{\mu}_t := [\mu_{c,t}, \boldsymbol{\mu}_{r,t}']'$, $\mathbf{v}_t := [f_t, \mathbf{0}_{\bar{S}+4}']'$, $\mathbf{w}_t := [\tilde{w}_t^c, \mathbf{w}_t^{r'}]'$,

$$\mathbf{\Psi}_t := \underbrace{\begin{bmatrix} \exp(h_{ft}) & \mathbf{0}_{\bar{S}+4}' \\ \mathbf{0}_{\bar{S}+4} & \mathbf{0}_{(\bar{S}+4) \times (\bar{S}+4)} \end{bmatrix}}_{(\bar{S}+5) \times (\bar{S}+5)}, \quad \mathbf{F} := \underbrace{\begin{bmatrix} \mathbf{0}_{\bar{S}+4}' & 0 \\ \mathbf{I}_{\bar{S}+4} & \mathbf{0}_{\bar{S}+4} \end{bmatrix}}_{(\bar{S}+5) \times (\bar{S}+5)}, \quad \boldsymbol{\Sigma}_t := \underbrace{\begin{bmatrix} \tilde{\sigma}_{c,t}^2 & \mathbf{0}_N' \\ \mathbf{0}_N & \boldsymbol{\Sigma}_{r,t-1} \end{bmatrix}}_{(N+1) \times (N+1)}, \quad \mathbf{H}_t := \underbrace{\begin{bmatrix} \mathbf{H}_{1,t} \\ \boldsymbol{\rho}_f^r & \mathbf{0}_N & \dots & \mathbf{0}_N \end{bmatrix}}_{(N+1) \times (\bar{S}+5)},$$

$$\mu_{c,t} = \begin{cases} \mu_c, & t \text{ is the last month of the quarter} \\ 0, & \text{otherwise} \end{cases}, \quad \mathbf{H}_{1,t} = \begin{cases} (\boldsymbol{\rho}^c)^\top, & t \text{ is the last month of the quarter} \\ \mathbf{0}_{\bar{S}+5}^\top, & \text{otherwise} \end{cases},$$

$$\tilde{w}_{c,t} = \begin{cases} w_{c,t}, & t \text{ is the last month of the quarter} \\ d_t, & \text{otherwise} \end{cases}, \quad \tilde{\sigma}_{c,t}^2 = \begin{cases} \exp(h_{ct}), & t \text{ is the last month of the quarter} \\ 1, & \text{otherwise} \end{cases}.$$

Finally, we apply the Kalman smoother to extract the latent state f_t . The estimation method

of other parameters is the same as in the previous analysis.

H Benchmarking

The covariance between consumption growth and asset returns can be biased by the benchmarking process used by the US Bureau of Economic Analysis (BEA). The benchmarking of NIPA data have been discussed in the past literature (see, e.g., Wilcox (1992) and Triplett (1997)). Simply speaking, the Monthly Retail Trade Survey (MRTS), which is used to construct monthly and quarterly consumption estimates, is of lower quality than the Annual Retail Trade Survey (ARTS). Hence, the four quarters of consumption will never precisely equal the corresponding annual measure; consequently, the quarterly estimates are ex post revised to benchmark the annual ones. As explained in Chapter 4, estimating methods, of the NIPA handbook,¹³ *“for the periods for which annual estimates are available and the quarterly estimates must be forced to average to these annual totals”*.

Without observing the unfiltered data, we cannot directly incorporate the benchmarking equation into our state-space model. Nevertheless, we can explore the mechanical effect originating from benchmarking through simulation. However, the exact benchmarking procedure (e.g., how BEA smoothes the consumption data) is unknown. Therefore, in this section, we consider several simulation settings and aim to explore when the benchmarking process will (and will not) distort the impulse responses of quarterly consumption growth to asset return shocks.

We simulate monthly consumption in log units; that is, $c_t^* = c_{t-1}^* + \mu_c + \rho_0 f_t + w_t^c$, $w_t^c \stackrel{\text{iid}}{\sim} \mathcal{N}(0, \sigma_c^2)$, and $f_t \stackrel{\text{iid}}{\sim} \mathcal{N}(0, 1)$. Note that the true monthly consumption growth is IID and correlates with only the contemporaneous f_t in simulations. However, we do not observe c_t^* but only its noisy proxy based on the MRTS,

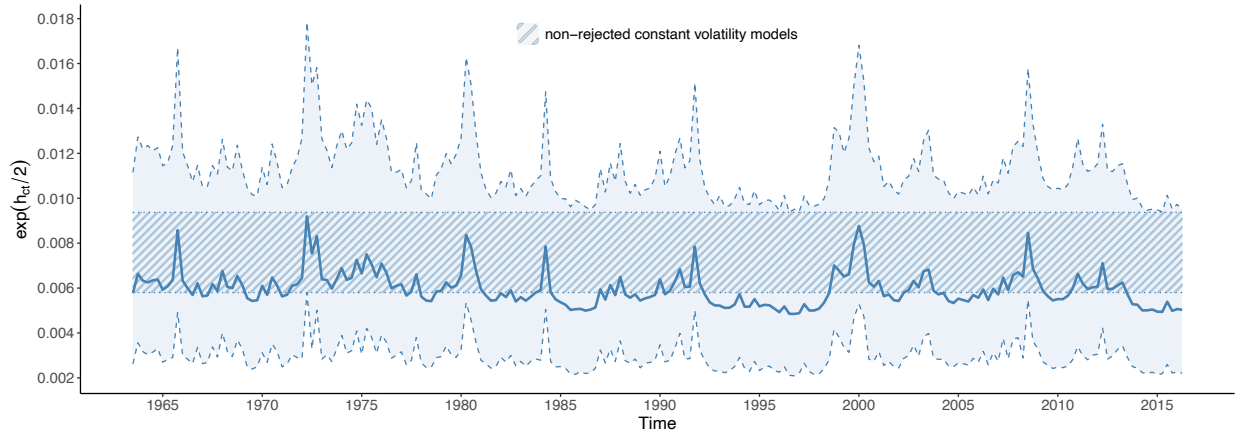
$$c_t = \beta c_t^* + (1 - \beta)\mu_c t + \epsilon_{c,t}, \quad \epsilon_{c,t} \stackrel{\text{iid}}{\sim} \mathcal{N}(0, \sigma_{\epsilon c}^2), \quad (\text{IA.22})$$

where $\beta \leq 1$, and $\epsilon_{c,t}$ is the measurement error. Equation (IA.22) implies the following dynamics of monthly consumption growth:

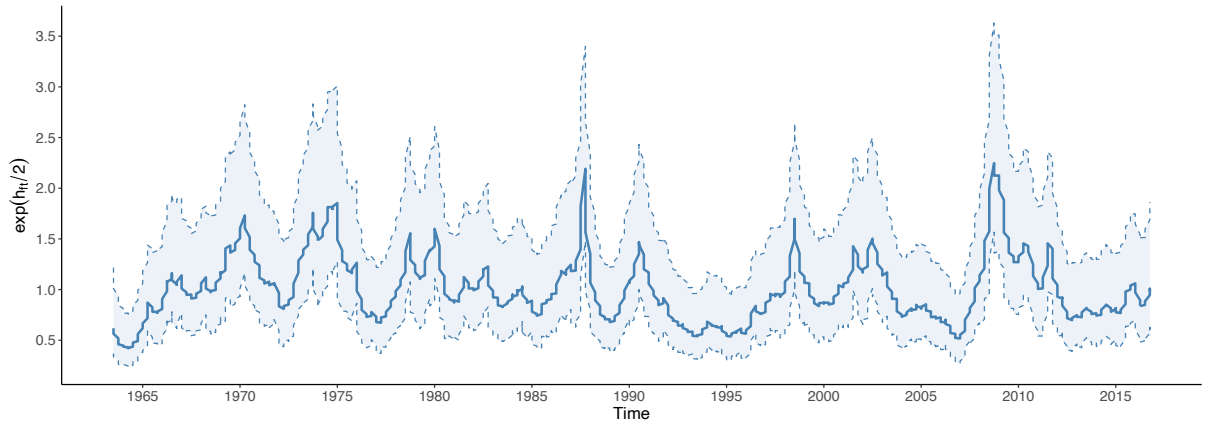
$$\Delta c_t = \beta \Delta c_t^* + (1 - \beta)\mu_c + (\epsilon_{c,t} - \epsilon_{c,t-1}). \quad (\text{IA.23})$$

¹³See <https://www.bea.gov/resources/methodologies/nipa-handbook/pdf/chapter-04.pdf>.

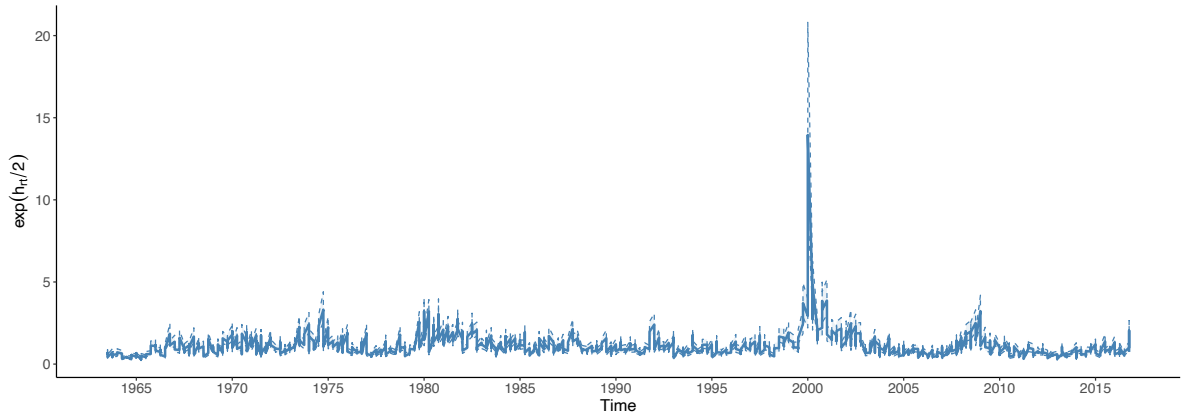
Figure IA.12: Filtered stochastic volatilities of consumption and returns: Mixed-frequency state-space model.



Panel A: Log volatility of the short-run consumption shock (w_t^c) of equation (IA.19)



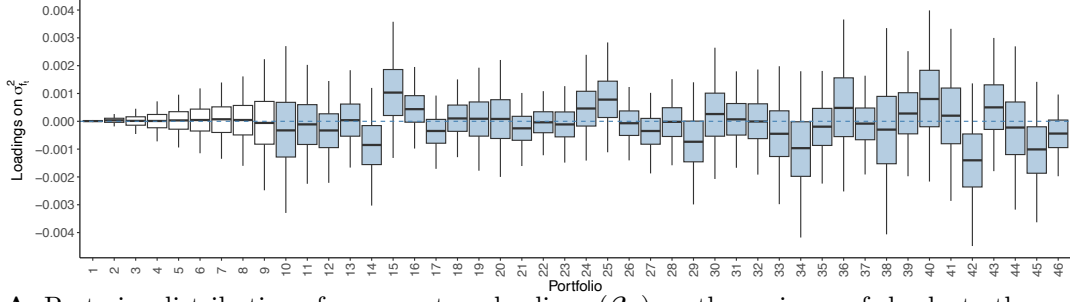
Panel B: Log volatility of the shock to the conditional mean of consumption growth (f_t).



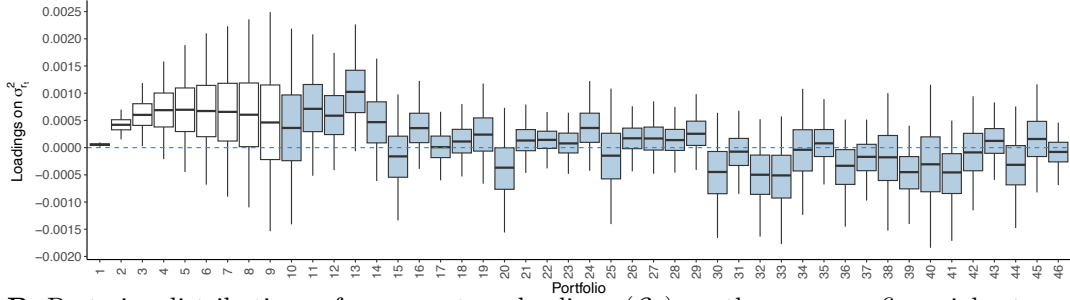
Panel C: Common log volatility of asset return (h_{rt}).

The figure shows the estimated stochastic volatilities of the mixed-frequency model in equations (IA.18)–(IA.19). Solid blue lines depict the posterior median of the log volatility, whereas dotted red lines denote 2.5% and 97.5% credible intervals. Shaded (patterned) areas reflect constant volatility levels that would not be rejected given the credible intervals.

Figure IA.13: Loadings of excess returns on consumption and returns volatilities: Mixed-frequency state-space model.



Panel A: Posterior distribution of excess return loadings (β_f) on the variance of shocks to the conditional consumption growth mean ($\sigma_{f,t-1}^2$) in equation (IA.18).



Panel B: Posterior distributions of excess return loadings (β_r) on the common financial return variance ($\sigma_{r,t-1}^2$) in equation (IA.18).

This figure shows the box plots of the posterior distributions of the loadings of portfolio excess returns on the variance of shocks to the conditional consumption growth and the common financial returns variance. Portfolios are ordered with bonds first (1–9), Fama-French 25 size and book-to-market second (10–34), and industry portfolios last.

That is, monthly consumption growth from MRTS (Δc_t) has the same unconditional mean μ_c as the pseudo-true process (Δc_t^*) that we do not observe.

We further observe the annual consumption data from the ARTS, which is considerably more comprehensive than the MRTS. Suppose that t is the final month of the year. The annual consumption flow is denoted by $C_{t,t+12}^a$, which equals $\sum_{j=1}^{12} \exp(c_{t+j}^*)$. Since $C_{t,t+12}^a$ never matches the sum of monthly consumption flows obtained from the MRTS ($\sum_{j=1}^{12} \exp(c_{t+j})$), benchmarking implies the following revision of the monthly estimates:

$$\exp(\tilde{c}_{t+j}) = \exp(c_{t+j}) \times \frac{\sum_{j=1}^{12} \exp(c_{t+j}^*)}{\sum_{j=1}^{12} \exp(c_{t+j})}, \quad (\text{IA.24})$$

where $\tilde{c}_{t+j}$ denotes the revised estimates of monthly consumption in log units. Using $\tilde{c}_{t+j}$, we compute the quarterly consumption growth (denoted by $\tilde{c}_{t-3,t}^q$) and estimate its impulse responses to quarterly asset return shocks, defined as $f_{t-3,t}^q = (f_t + f_{t-1} + f_{t-2})/\sqrt{3}$.

There are two representative simulation settings to consider: (1) $\beta = 1$ and (2) $\beta < 1$ in equation (IA.22). First, when $\beta = 1$, the benchmarking process in equation (IA.24) does not distort (in population) the impulse responses of quarterly consumption growth to f_t shocks. To see this intuitively, we can rewrite the adjustment factor in equation (IA.24) as follows:

$$\frac{\sum_{j=1}^{12} \exp(c_{t+j}^*)}{\sum_{j=1}^{12} \exp(c_{t+j})} = \frac{\sum_{j=1}^{12} \exp(c_{t+j}^*)}{\sum_{j=1}^{12} \exp(c_{t+j}^* + \epsilon_{c,t+j})}.$$

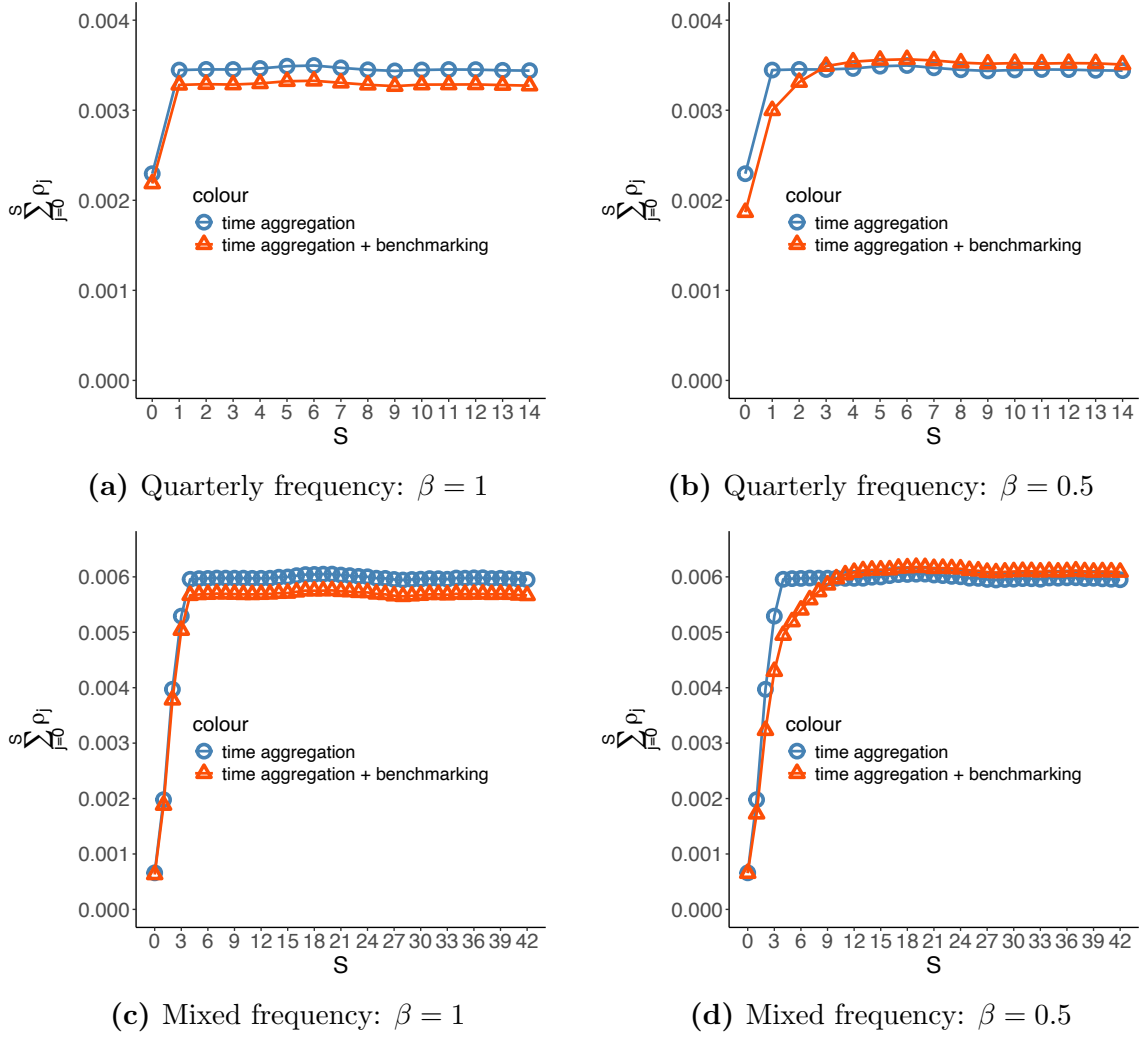
Smoothing the monthly estimates based on the MRTS to match the annual ones, on average, reallocates only measurement errors over different periods and, hence, does not change the impulse responses to f_t . Note that assuming $\beta = 1$ is a common way of modeling consumption measurement error in the literature (see, e.g., Schorfheide, Song, and Yaron (2018)).

Second, instead, when $\beta < 1$, benchmarking introduces a mechanical autocorrelation between f_t shocks and observed consumption growth. The intuition is that, in this case, benchmarking smooths both measurement errors and f_t shocks over 12 months, leading to the slow adjustment of consumption growth to f_t shocks.

To illustrate the above points, we simulate consumption data following equations (IA.22)–(IA.23). In the simulations, the contemporaneous correlation between the true monthly consumption growth (Δc_t^*) and f_t is 0.40, whereas the measurement error is calibrated such that the observed monthly consumption growth (Δc_t) has a correlation of 0.30 with f_t . After obtaining the monthly growth rates, we aggregate the simulated data to quarterly consumption growth, which is then normalized to have the same standard deviation (0.0072) as in the true data.

Panel (a) of Figure IA.14 shows the simulated impulse responses based on $\beta = 1$ at the quarterly frequency. Due to the time-aggregation bias (the blue line with circles), the cumulative impulse responses increase from 0.002 to about 0.0035 at $S = 1$ and stay flat after one quarter. The benchmarking process does not introduce additional autocorrelations between consumption growth and f_t (see orange line with triangles). Panel (b) displays the case with $\beta = 0.5$ at the quarterly frequency. Unlike Panel (a), quarterly consumption growth responds to both the one-quarter lagged f_t shocks and the lagged f_t of the preceding four quarters. The cumulative impulse responses become flat after four quarters. Panels (c) and (d) of Figure IA.14 repeat similar simulations with mixed-frequency estimation; that is, it displays how quarterly consumption growth responds to a one-standard-deviation *monthly*

Figure IA.14: Simulated cumulative response function of quarterly consumption growth.



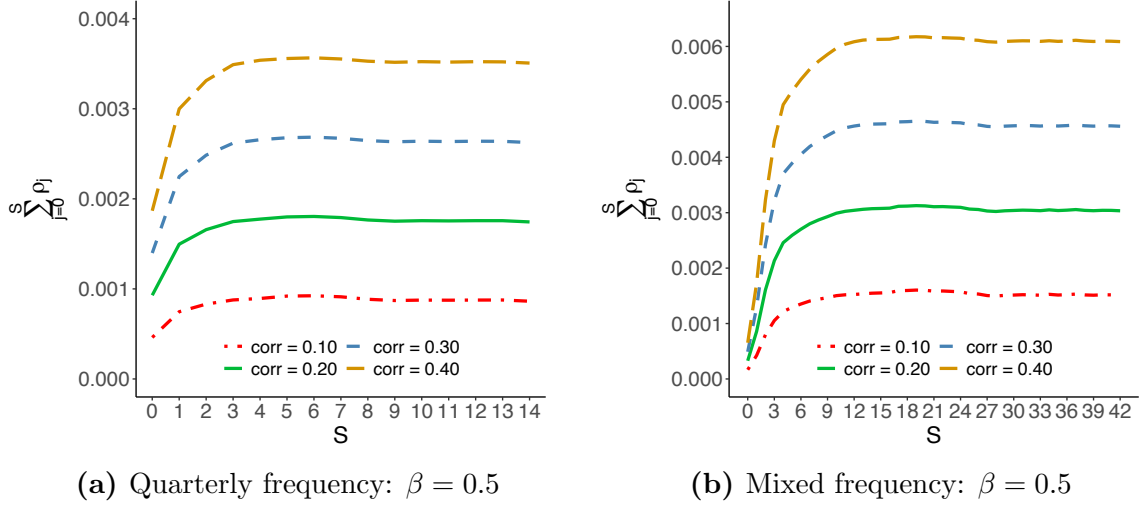
The figure plots the cumulative impulse response function (CIRF) of quarterly consumption growth. The blue lines with circles show the impulses responses with the time aggregation bias, whereas the orange lines with triangulars display those with both the time aggregation bias and benchmarking. Panels (a) and (b), which assume $\beta = 1$ and $\beta = 0.5$ in equation (IA.23) respectively, consider the impulse responses at the quarterly frequency, i.e., how quarterly consumption growth responds to quarterly f_t shocks. Panels (c) and (d), instead, study the mixed-frequency CIRFs of quarterly consumption growth to a one-standard-deviation monthly f_t shock. We simulate 100,000 years of data. Quarterly consumption growth is standardized to have the same standard deviation (0.0072) as in the data.

f_t shock. With only the time-aggregation bias, quarterly consumption growth reacts to both contemporaneous and four lagged monthly f_t shocks. However, as we further incorporate benchmarking of consumption data, the CIRFs continue to increase until 12 months after the impulse and stay flat afterwards.¹⁴

We further explore the impulse responses with time-aggregation bias and benchmarking

¹⁴ Note that the quarterly CIRFs roughly equal the mixed-frequency estimates divided by $\sqrt{3}$.

Figure IA.15: Simulated cumulative response function of quarterly consumption growth: Different coefficients of $\text{corr}(\Delta c_t^*, f_t)$.



The figure plots the cumulative impulse response function (CIRF) of quarterly consumption growth. We consider both time-aggregation bias and benchmarking ($\beta = 0.5$ in equation (IA.23)). The contemporaneous correlation between true monthly consumption growth and f_t , $\text{corr}(\Delta c_t^*, f_t)$, ranges from 0.10 to 0.40. Panel (a) considers the impulse responses at the quarterly frequency, i.e., how quarterly consumption growth responds to quarterly f_t shocks. Panel (b), instead, presents the mixed-frequency CIRFs of quarterly consumption growth to a one-standard-deviation monthly f_t shock. We simulate 100,000 years of data. Quarterly consumption growth is standardized to have the same standard deviation (0.0072) as in the data.

of consumption data, assuming different contemporaneous correlations between true monthly consumption growth and f_t . Figure IA.15 plots these CIRFs for correlations ranging from 0.10 to 0.40. In the monthly data, this contemporaneous correlation coefficient is estimated to be about 0.17; hence, the assumption of a correlation of 0.20 (the solid green line) represents the most realistic case.

A crucial observation is that the benchmarking process cannot fully explain the patterns that we detected in the real data. First, in Figures 3 and 6, the CIRFs continue to increase until the seventh quarter (or 21 months), whereas the benchmarking process in our simulations leads to slowly increasing CIRFs within only the first four quarters. Second, the CIRFs of the IID consumption data with benchmarking are less than 0.002 at the quarterly frequency (0.003 at the mixed frequency) under the assumption of 0.2 correlation (in the most realistic calibration), as shown in green solid lines. Instead, we estimate much more sizable CIRFs of about 0.01 at the quarterly frequency in Figure 3 (0.015 in the mixed-frequency estimation in Figure 6).

I Model selection: Consumption mean and volatility risks in asset returns

Commonly, time-varying risk premia in consumption-based asset pricing models are induced by the time variation in the volatility of consumption shocks; that is, the models (a) assume the existence of time-varying volatility and (b) postulate that returns are linear in this volatility. Our state-space formulation allows us to formally test these common theoretical assumptions because it nests, as particular cases, the most popular structural models.

Table IA.I, Panel A, reports Bayes factors and posterior probabilities for several restricted and unrestricted versions of the specification in equations (2) and (16)–(19), and Section B.2. In particular, we test both commonalities in the various volatility processes (for returns and consumption mean shocks) and their impact on excess returns. Note that the posterior model probabilities (and Bayes factors) are particularly appropriate for this type of test because they yield valid model selection even over the space of misspecified models. That is, they select the model that has the highest probability of being the true data-generating process (not solely the model with the highest likelihood). See, for example, Schervish (1996).

The data favor a specification in which *i*) volatilities do *not* affect excess returns (Models I–V) and *ii*) returns and consumption have *distinct* volatility processes—the posterior probability of such a formulation (Model I) is almost 100%.¹⁵

Panel B of Table IA.I shows that the share of variance of asset returns explained by the shock spanned by consumption (f_t) is stable across specification, even when we include the stochastic volatilities in the return equation (Models VI–X). This stresses the robustness of our identification approach to recover the conditional mean shocks to consumption. Furthermore, Panel C emphasizes that even when allowing for stochastic volatility in all the latent shocks, the correlation between the variances of consumption and asset returns remains considerably small.

One may wonder whether our model selection findings are largely driven by the likelihood function for the joint dynamics of returns and consumption being dominated by the former due to their much higher volatility and, hence, reflect only a model for returns, rather than their joint dynamics. Consequently, Table IA.X repeats the same model selection exercise after scaling returns to have unit variance and consumption to have variance equal to the

¹⁵As shown in Table IA.II, these results are stable if we exclude bond returns from the state-space model.

Table IA.I: Model comparison using log Bayes factors: Baseline stock and bond portfolios

	Models									
	I	II	III	IV	V	VI	VII	VIII	IX	X
Panel A: Model probabilities and Bayes factors										
Log of Bayes factor:	0	-27	-37	-51	-85	-105	-161	-223	-236	-332
Posterior probability:.	100%	0%	0%	0%	0%	0%	0%	0%	0%	0%
Panel B: Variance Decomposition of Returns										
f_t	62.3%	62.4%	62.2%	62.3%	62.2%	57.6%	57.8%	57.7%	58.4%	57.8%
$\sigma_{c,t-1}^2$										0.2%
$\sigma_{f,t-1}^2$								0.3%	0.3%	0.3%
$\sigma_{r,t-1}^2$						5.1%	4.8%	4.9%	4.5%	4.9%
Panel C: Correlations among stochastic volatility processes										
$cor(\sigma_{ct}^2, \sigma_{ft}^2)$	0.06	0.12				0.14		0.08		0.07
	[-0.09, 0.29]	[-0.03, 0.32]				[-0.02, 0.35]		[-0.08, 0.31]		[-0.09, 0.30]
$cor(\sigma_{ct}^2, \sigma_{rt}^2)$	0.12	0.12	0.12			0.14		0.15		0.15
	[-0.03, 0.32]	[-0.03, 0.32]	[-0.03, 0.32]			[-0.02, 0.35]		[-0.02, 0.36]		[-0.02, 0.36]
$cor(\sigma_{ft}^2, \sigma_{rt}^2)$	0.23	1		0.23		1		0.22	0.22	0.22
	[0.07, 0.42]	[1, 1]		[0.07, 0.43]		[1, 1]		[0.06, 0.41]	[0.06, 0.41]	[0.06, 0.41]

Model I: w_t^c , f_t and w_t^r follow SV processes, and $r_t^e = \mu_r + \rho^r f_t + w_t^r$.

Model II: w_t^c , f_t and w_t^r follow SV processes, $r_t^e = \mu_r + \rho^r f_t + w_t^r$, and $h_{ft} = h_{rt}$.

Model III: w_t^c and w_t^r follow SV processes, $f_t \stackrel{iid}{\sim} \mathcal{N}(0, 1)$, and $r_t^e = \mu_r + \rho^r f_t + w_t^r$.

Model IV: f_t and w_t^r follow SV processes, $w_t^c \stackrel{iid}{\sim} \mathcal{N}(0, \sigma_c^2)$, and $r_t^e = \mu_r + \rho^r f_t + w_t^r$.

Model V: w_t^r follows SV process, $w_t^c \stackrel{iid}{\sim} \mathcal{N}(0, \sigma_c^2)$, $f_t \stackrel{iid}{\sim} \mathcal{N}(0, 1)$, and $r_t^e = \mu_r + \rho^r f_t + w_t^r$.

Model VI: w_t^c , f_t and w_t^r follow SV processes, $r_t^e = \mu_r + \rho^r f_t + \beta_f \sigma_{f,t-1}^2 + w_t^r$, and $h_{ft} = h_{rt}$.

Model VII: w_t^r follows SV process, $w_t^c \stackrel{iid}{\sim} \mathcal{N}(0, \sigma_c^2)$, $f_t \stackrel{iid}{\sim} \mathcal{N}(0, 1)$, and $r_t^e = \mu_r + \rho^r f_t + \beta_r \sigma_{r,t-1}^2 + w_t^r$.

Model VIII: w_t^c , f_t and w_t^r follow SV processes, and $r_t^e = \mu_r + \rho^r f_t + \beta_f \sigma_{f,t-1}^2 + \beta_r \sigma_{r,t-1}^2 + w_t^r$.

Model IX: f_t and w_t^r follow SV processes, $w_t^c \stackrel{iid}{\sim} \mathcal{N}(0, \sigma_c^2)$, and $r_t^e = \mu_r + \rho^r f_t + \beta_f \sigma_{f,t-1}^2 + \beta_r \sigma_{r,t-1}^2 + w_t^r$.

Model X: w_t^c , f_t and w_t^r follow SV processes, and $r_t^e = \mu_r + \rho^r f_t + \beta_c \sigma_{c,t-1}^2 + \beta_f \sigma_{f,t-1}^2 + \beta_r \sigma_{r,t-1}^2 + w_t^r$.

The table summarizes the model comparison for restricted and unrestricted versions of the specification in equations (2) and (16)–(19), and Section B.2. Panel A reports log Bayes factors and posterior probabilities. We approximate the Bayes factor using the Schwartz criterion. We use Model I as a benchmark and calculate the (log) odds of each model compared to Model I. A negative number implies that the chosen model is less likely than Model I conditional on the observed data. The model posterior probabilities are computed under the prior of the specifications being all equally likely. Panel B reports the variance decomposition of asset returns for the model-specific sources of time variation. Panel C reports the correlations among the stochastic volatility processes.

number of assets. Again, we find that there is no support for stochastic volatilities driving time-varying risk premia or for the volatility of consumption and returns being proportional to each other.

J Stochastic volatility with leverage

In Section III.6, we consider stochastic volatility models without the leverage effect; that is, shocks to asset returns are independent of the shocks to their stochastic volatility processes.

Table IA.II: Model comparison using log Bayes factors with only equity portfolios

	Models									
	I	II	III	IV	V	VI	VII	VIII	IX	X
Panel A: Model probabilities and Bayes factors										
Log of Bayes factor:	0	-26	-39	-48	-87	-103	-146	-185	-201	-278
Posterior probability:.	100%	0%	0%	0%	0%	0%	0%	0%	0%	0%
Panel B: Variance Decomposition of Returns										
f_t	77.3%	77.3%	77.0%	77.2%	77.0%	73.2%	71.4%	71.5%	71.8%	71.2%
$\sigma_{c,t-1}^2$										0.2%
$\sigma_{f,t-1}^2$								0.2%	0.2%	0.2%
$\sigma_{r,t-1}^2$						3.4%	5.2%	5.4%	4.9%	5.2%
Panel C: Correlations among stochastic volatility processes										
$cor(\sigma_{ct}^2, \sigma_{ft}^2)$	0.07	0.12				0.12		0.07		0.07
	[-0.09, 0.29]	[-0.03, 0.34]				[-0.03, 0.34]		[-0.09, 0.30]		[-0.09, 0.30]
$cor(\sigma_{ct}^2, \sigma_{rt}^2)$	0.12	0.12	0.12			0.12		0.13		0.14
	[-0.03, 0.34]	[-0.03, 0.34]	[-0.03, 0.33]			[-0.03, 0.34]		[-0.02, 0.36]		[-0.02, 0.38]
$cor(\sigma_{ft}^2, \sigma_{rt}^2)$	0.25	1		0.25		1		0.22	0.21	0.23
	[0.07, 0.45]	[1, 1]		[0.08, 0.46]		[1, 1]		[0.05, 0.45]	[0.05, 0.44]	[0.06, 0.45]

Model I: w_t^c , f_t and w_t^r follow SV processes, and $r_t^e = \mu_r + \rho^r f_t + w_t^r$.

Model II: w_t^c , f_t and w_t^r follow SV processes, $r_t^e = \mu_r + \rho^r f_t + w_t^r$, and $h_{ft} = h_{rt}$.

Model III: w_t^c and w_t^r follow SV processes, $f_t \stackrel{iid}{\sim} \mathcal{N}(0, 1)$, and $r_t^e = \mu_r + \rho^r f_t + w_t^r$.

Model IV: f_t and w_t^r follow SV processes, $w_t^c \stackrel{iid}{\sim} \mathcal{N}(0, \sigma_c^2)$, and $r_t^e = \mu_r + \rho^r f_t + w_t^r$.

Model V: w_t^r follows SV process, $w_t^c \stackrel{iid}{\sim} \mathcal{N}(0, \sigma_c^2)$, $f_t \stackrel{iid}{\sim} \mathcal{N}(0, 1)$, and $r_t^e = \mu_r + \rho^r f_t + w_t^r$.

Model VI: w_t^c , f_t and w_t^r follow SV processes, $r_t^e = \mu_r + \rho^r f_t + \beta_f \sigma_{f,t-1}^2 + w_t^r$, and $h_{ft} = h_{rt}$.

Model VII: w_t^r follows SV process, $w_t^c \stackrel{iid}{\sim} \mathcal{N}(0, \sigma_c^2)$, $f_t \stackrel{iid}{\sim} \mathcal{N}(0, 1)$, and $r_t^e = \mu_r + \rho^r f_t + \beta_r \sigma_{r,t-1}^2 + w_t^r$.

Model VIII: w_t^c , f_t and w_t^r follow SV processes, and $r_t^e = \mu_r + \rho^r f_t + \beta_f \sigma_{f,t-1}^2 + \beta_r \sigma_{r,t-1}^2 + w_t^r$.

Model IX: f_t and w_t^r follow SV processes, $w_t^c \stackrel{iid}{\sim} \mathcal{N}(0, \sigma_c^2)$, and $r_t^e = \mu_r + \rho^r f_t + \beta_f \sigma_{f,t-1}^2 + \beta_r \sigma_{r,t-1}^2 + w_t^r$.

Model X: w_t^c , f_t and w_t^r follow SV processes, and $r_t^e = \mu_r + \rho^r f_t + \beta_c \sigma_{c,t-1}^2 + \beta_f \sigma_{f,t-1}^2 + \beta_r \sigma_{r,t-1}^2 + w_t^r$.

The table summarizes the model comparison for restricted and unrestricted versions of the specification in equations (2) and (16)–(19), and Section B.2. Panel A reports log Bayes factors and posterior probabilities. We approximate the Bayes factor using the Schwartz criterion. We use Model I as a benchmark and calculate the (log) odds of each model compared to Model I. A negative number implies that the chosen model is less likely than Model I conditional on the observed data. The model posterior probabilities are computed under the prior of the specifications being all equally likely. Panel B reports the variance decomposition of asset returns for the model-specific sources of time variation. Panel C reports the correlations among the stochastic volatility processes.

In this Appendix, we further consider the leverage effect in the stochastic volatility of shocks to the conditional consumption mean, f_t . Specifically, f_t follows a normal distribution with the stochastic volatility process given by

$$f_t = \exp\left(\frac{h_{ft}}{2}\right)\epsilon_{ft}, \quad y_{ft}^* = \log(f_t^2) = h_{ft} + \log(\epsilon_{ft}^2), \quad h_{ft} = \delta_f h_{f,t-1} + \sqrt{1 - \delta_f^2} \eta_{f,t-1}, \quad (\text{IA.25})$$

where $\epsilon_{ft} \stackrel{\text{iid}}{\sim} \mathcal{N}(0, 1)$ and $\eta_{ft} \stackrel{\text{iid}}{\sim} \mathcal{N}(0, 1)$. Although macro-finance literature normally assumes that ϵ_{ft} and η_{ft} are independent, many earlier papers (e.g., Black (1976), Nelson (1991), and Yu (2005)) document strong leverage effects in asset returns: An increase in volatility often follows a drop in equity returns; that is, ϵ_{ft} and η_{ft} tend to be negatively correlated.

To incorporate the leverage effect, we consider the following distributional assumption:

$$\begin{pmatrix} \epsilon_{ft} \\ \eta_{ft} \end{pmatrix} \stackrel{\text{iid}}{\sim} \mathcal{N} \left(\mathbf{0}, \begin{bmatrix} 1 & \zeta \\ \zeta & 1 \end{bmatrix} \right), \quad (\text{IA.26})$$

where $\zeta < 0$ if the leverage effect exists: A negative shock to y_{ft}^* (negative ϵ_{ft}) leads to a spike in η_{ft} ; therefore, the stochastic volatility of f_{t+1} , $\exp(h_{f,t+1}/2)$, tends to increase.

We follow the solution proposed by Omori, Chib, Shephard, and Nakajima (2007). In particular, they approximate the joint distribution of (e_{ft}, η_{ft}) using a mixture of bivariate Gaussian densities:

$$p(e_{ft}, \eta_{ft} \mid d_t) \approx \sum_{i=1}^K q_i \cdot \mathcal{N}(e_{ft} \mid m_i, v_i^2) \cdot \mathcal{N}(\eta_{ft} \mid d_t \zeta \exp(m_i/2)[a_i + b_i(e_{ft} - m_i)], 1 - \zeta^2), \quad (\text{IA.27})$$

where $\mathcal{N}(x \mid m, v^2)$ means that $x \sim \mathcal{N}(m, v^2)$, $d_t = \text{sign}(f_t)$, $\{(q_i, m_i, v_i^2, a_i, b_i)\}_{i=1}^K$ are fixed real numbers given by Table 1 of Omori, Chib, Shephard, and Nakajima (2007). As they do, we select $K = 10$, so ten different normal distributions are used to approximate the joint distribution of (e_{ft}, η_{ft}) .

The SV model can be expressed as

$$\begin{pmatrix} y_{ft}^* \\ h_{f,t+1} \end{pmatrix} = \begin{pmatrix} h_{f,t} \\ \delta_f h_{f,t} \end{pmatrix} + \begin{pmatrix} e_{f,t} \\ \sqrt{1 - \delta_f^2} \eta_{f,t} \end{pmatrix}.$$

Using the mixture approximation (IA.27) and introducing the mixture component indicator $S_{ft} \in \{1, \dots, K\}$, we have that

$$\begin{pmatrix} e_{f,t} \\ \eta_{f,t} \end{pmatrix} \mid S_{ft} = i, d_t = \begin{pmatrix} m_i + v_i z_t \\ d_t \zeta \exp(m_i/2)[a_i + b_i v_i z_t] + \sqrt{1 - \zeta^2} z_t^* \end{pmatrix}, \quad \text{and} \quad \begin{pmatrix} z_t \\ z_t^* \end{pmatrix} \stackrel{\text{iid}}{\sim} \mathcal{N}(\mathbf{0}, \mathbf{I}_2).$$

Since (e_{ft}, η_{ft}) are not independent of each other, we augment the space of latent states as

follows:

$$y_{ft}^* - m_i = h_{f,t} + v_i z_t, \quad (\text{observation equation}), \quad (\text{IA.28})$$

and we treat both $h_{f,t}$ and z_t as latent states, with the following state equation:

$$\begin{pmatrix} h_{f,t+1} \\ z_{t+1} \\ c_{t+1} \end{pmatrix} = \begin{pmatrix} \delta_f & d_t \zeta \exp(m_i/2) b_i v_i \sqrt{1 - \delta_f^2} & d_t \zeta \exp(m_i/2) a_i \sqrt{1 - \delta_f^2} \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} h_{f,t} \\ z_t \\ c_t \end{pmatrix} + \begin{pmatrix} \sqrt{(1 - \zeta^2)(1 - \delta_f^2)} z_t^* \\ z_{t+1} \\ 0 \end{pmatrix}, \quad (\text{IA.29})$$

where c_t is introduced to remove the mechanical drift term: $c_{t+1} = c_t = \dots = c_0 = 1$. Using the system (IA.28)–(IA.29), we can use the Kalman smoother to estimate $(h_{f,t}, z_t)$.

The inference for S_{ft} and δ_f are the same as before. The posterior distribution of S_{ft} is

$$\begin{aligned} & p(S_{ft} = i \mid e_{f,t}, \eta_{f,t}, \zeta, d_t) \\ & \propto Pr(S_{ft} = i) \cdot v_i^{-1} \exp \left\{ -\frac{(e_{ft} - m_i)^2}{2v_i^2} - \frac{[\eta_{f,t} - d_t \zeta \exp(m_i/2)[a_i + b_i(e_{ft} - m_i)]]^2}{2(1 - \zeta^2)} \right\}. \end{aligned}$$

We update δ_f using a Metropolis algorithm as before.

Finally, we discuss the posterior inference for ζ , the key parameter governing the leverage effect. We rely on the distributional assumption in equation (IA.26),

$$\begin{pmatrix} f_t \cdot \exp(-\frac{h_{ft}}{2}) \\ \frac{h_{f,t+1} - \delta_f h_{f,t}}{\sqrt{1 - \delta_f^2}} \end{pmatrix} \stackrel{\text{iid}}{\sim} \mathcal{N} \left(\mathbf{0}, \begin{bmatrix} 1 & \zeta \\ \zeta & 1 \end{bmatrix} \right),$$

which implies the following posterior distribution of ζ under a flat prior $\pi(\zeta) \propto 1$:

$$\begin{aligned} p(\zeta \mid \mathbf{h}_f, \mathbf{f}, \delta_f) & \propto (1 - \zeta^2)^{-\frac{T-1}{2}} \exp \left\{ -\frac{\text{SSE}(\mathbf{h}_f, \mathbf{f}, \delta_f, \zeta)}{2} \right\}, \text{ and} \\ \text{SSE}(\mathbf{h}_f, \mathbf{f}, \delta_f, \zeta) & = \sum_{t=1}^{T-1} \begin{bmatrix} f_t \cdot \exp(-\frac{h_{ft}}{2}) & \frac{h_{f,t+1} - \delta_f h_{f,t}}{\sqrt{1 - \delta_f^2}} \end{bmatrix} \begin{bmatrix} 1 & \zeta \\ \zeta & 1 \end{bmatrix}^{-1} \begin{pmatrix} f_t \cdot \exp(-\frac{h_{ft}}{2}) \\ \frac{h_{f,t+1} - \delta_f h_{f,t}}{\sqrt{1 - \delta_f^2}} \end{pmatrix}. \end{aligned}$$

We draw ζ using a Metropolis algorithm, as follows:

- a. initialize ζ^0 ;
- b. draw ζ from a normal distribution $\zeta^* \sim \mathcal{N}(\zeta^{(j-1)}, c_{mh}^2)$;
- c. calculate $\rho(\zeta^*, \zeta^{(j-1)}) = \min\{1, \frac{p(\zeta^* | \mathbf{h}_f, -)}{p(\zeta^{(j-1)} | \mathbf{h}_f, -)}\}$;

- d. set $\zeta^{(j)} = \zeta^{(j-1)}$ with probability $1 - \rho(\zeta^*, \zeta^{(j-1)})$ and $\zeta^{(j)} = \zeta^*$ with probability $\rho(\zeta^*, \zeta^{(j-1)})$.

c_{mh} determines the step size in the Metropolis algorithm. We choose c_{mh} such that the frequency of accepting a new ζ is about 50%.

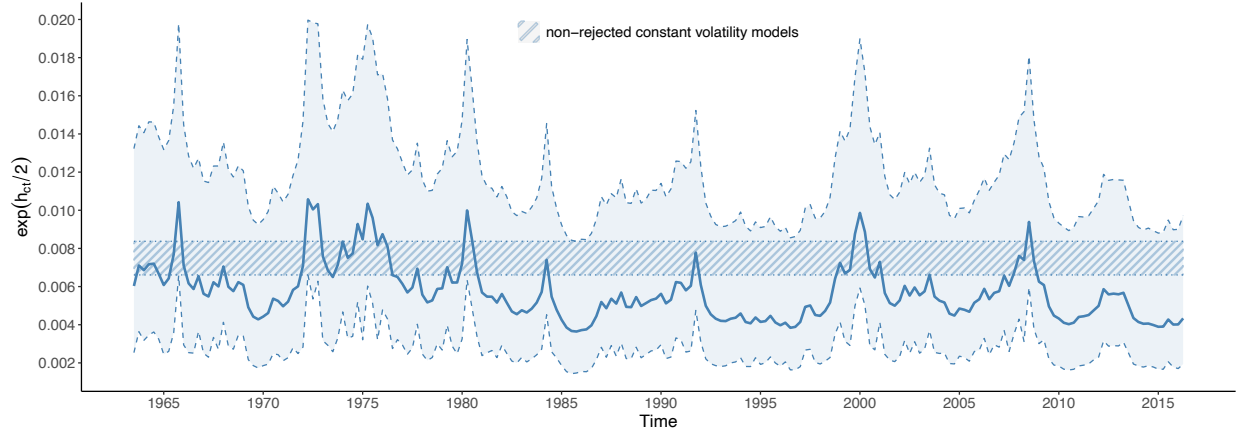
We repeat the analysis in Section III.6 but allow for the leverage effect in f_t . First, we detect significantly negative ζ : The posterior median of ζ is -0.38 , with 90% CIs $[-0.59, -0.08]$. This finding is consistent with the leverage effect. Second, as we show in Figure IA.16, the estimates of long-run consumption volatility, $\sigma_{f_t}^2$, become sharper after introducing the leverage effect. Third, Figure IA.17 suggests that the returns loadings on three stochastic volatility processes are almost unchanged after we introduce the leverage effect to f_t .

K Additional robustness checks

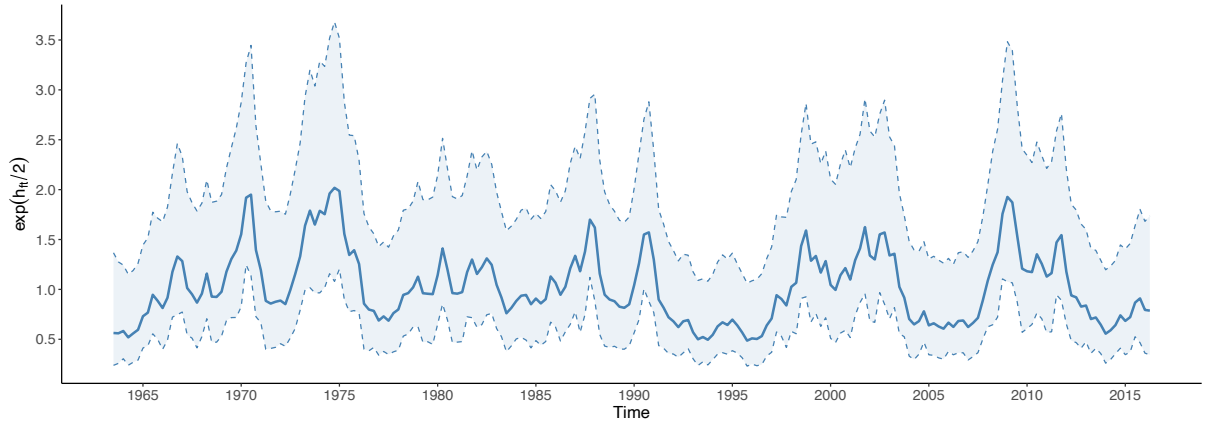
Because the state-space estimation results may depend on the choice of the cross-section of assets used for the analysis, we repeat our analysis using alternative test assets and report the results in Figures IA.31–IA.33. They present all the key empirical results for a five-factor consumption-returns model estimated on 12 industry portfolios, nine bond portfolios, and different cross-sections of stock returns (32 size-profitability-investment-sorted portfolios, 32 size-value-investment-sorted portfolios, and 32 size-value-profitability-sorted portfolios, respectively). Figures IA.31–IA.33 indicate that all our results remain almost unchanged. We also observe the same fanning-out pattern in the term structure of consumption exposure for different cross-sections of test assets, as shown in Figure IA.37.

Throughout all the empirical analyses in the main text, we have used only nondurable consumption growth per capita. However, another popular choice for empirical work in macro-finance is a combination of nondurable consumption growth and services. We reestimate the state-space formulation of our model using this proxy for consumption growth. Because adding services might introduce additional latent dynamics, we present our findings in Figures IA.38 and IA.39 that allow for up to five latent factors. Our results are similar both qualitatively and quantitatively. In particular, the five-factor model, as shown in Figure IA.39, displays the cumulative impulse response function of consumption of about 1% after three years (Panel (a)), and the variance decompositions are rather similar to those previously reported (but with larger confidence bands due to a higher number of param-

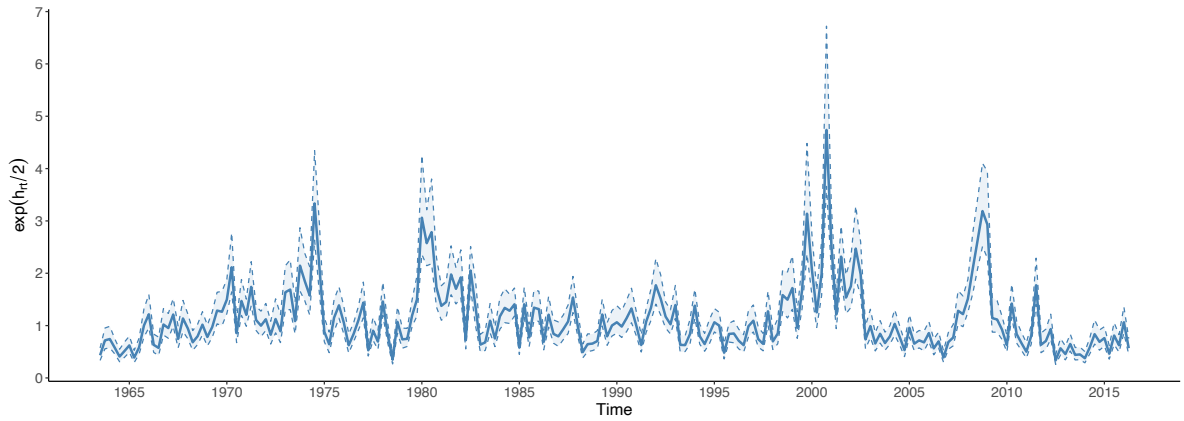
Figure IA.16: Filtered stochastic volatilities of consumption and returns with leverage effect in the stochastic volatility of f_t .



Panel A: Log volatility of the idiosyncratic shock to consumption (w_t^c) of equation (2).



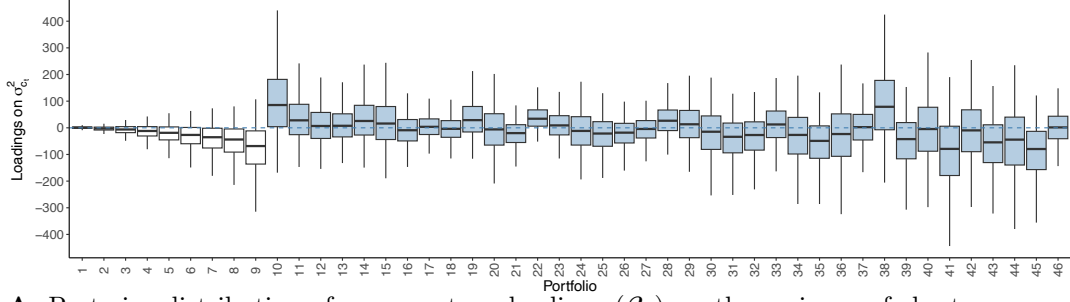
Panel B: Log volatility of the shock to the conditional mean of consumption growth (f_t).



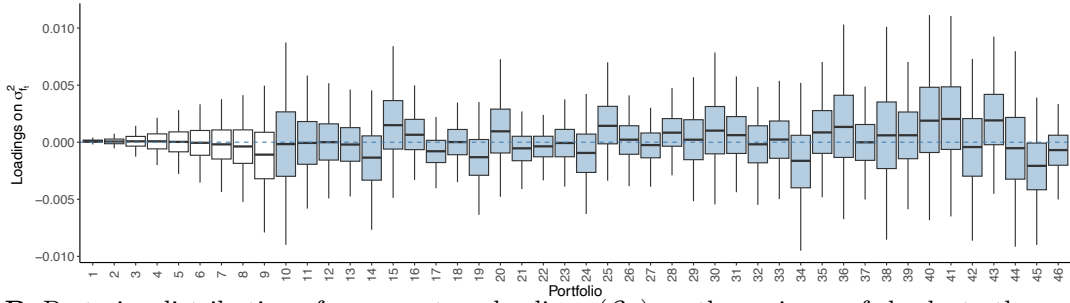
Panel C: Common log volatility of asset return (h_{rt} of equation (19)).

The figure shows the estimated stochastic volatilities of the model in equations (2) and (16)–(19), and Section B.2 under a diffuse prior for the autoregressive volatility coefficients. Solid blue lines depict the posterior median of the log volatility, whereas dotted red lines denote 2.5% and 97.5% credible intervals. Shaded (patterned) areas reflect constant volatility levels that would not be rejected given the credible intervals. f_t follows the stochastic volatility process with the leverage effect (see equations (IA.25)–(IA.26)).

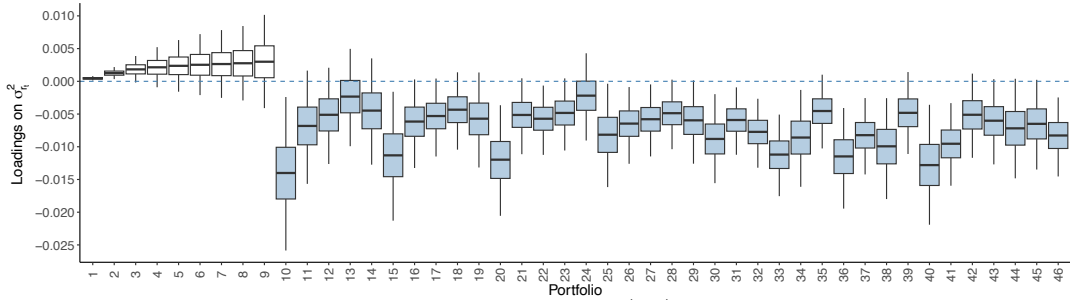
Figure IA.17: Loadings of excess returns on consumption and returns volatilities with leverage effect in the stochastic volatility of f_t .



Panel A: Posterior distribution of excess return loadings (β_c) on the variance of short-run consumption shocks ($\sigma_{c,t-1}^2$) in equation (16).



Panel B: Posterior distribution of excess return loadings (β_f) on the variance of shocks to the conditional consumption growth mean ($\sigma_{f,t-1}^2$) in equation (16).



Panel C: Posterior distributions of excess return loadings (β_r) on the common financial return variance ($\sigma_{r,t-1}^2$) in equation (16).

The figure shows the box plots of the posterior distributions of the loadings of portfolio excess returns on the variance of short-run consumption shocks ($\sigma_{c,t-1}^2$), the variance of shocks to the conditional consumption growth ($\sigma_{f,t-1}^2$), and the common financial returns variance ($\sigma_{r,t-1}^2$). Portfolios are ordered with bonds first (1–9), Fama-French 25 size and book-to-market second (10–34), and industry portfolios last. f_t follows the stochastic volatility process with the leverage effect (see equations (IA.25)–(IA.26)).

ters). One caveat is that the estimation is more demanding in the five-factor model than the single-factor setup due to a much larger parameter space, hence, leading to more disperse distributions of, for instance, R_{adj}^2 .

L VAR estimation of the conditional consumption mean

As a benchmark for comparison of our method, in this Appendix, we employ the Parker (2001) VAR-based estimation of the conditional mean of consumption. Specifically, we estimate a three-variable VAR in market excess return ($r_{r-1,t}^m$), the log consumption level ($\log(C_{t-1,t})$), and the dividend-to-price ratio of the market portfolio (D_t/P_{t-1}) at the quarterly frequency. The number of lags is four, the same as in Parker (2001). Let $\mathbf{y}_t$ denote the three-variable vector in quarter t , yielding the following VAR system:

$$\mathbf{y}_t = \boldsymbol{\mu}_y + \sum_{j=1}^4 \mathbf{A}_j \mathbf{y}_{t-j} + \boldsymbol{\epsilon}_{y,t}. \quad (\text{IA.30})$$

According to Parker (2001, equation (15) therein), the impulse responses of log consumption *level* to innovations in excess return measures the covariance between future cumulative consumption growth and excess returns, as follows:

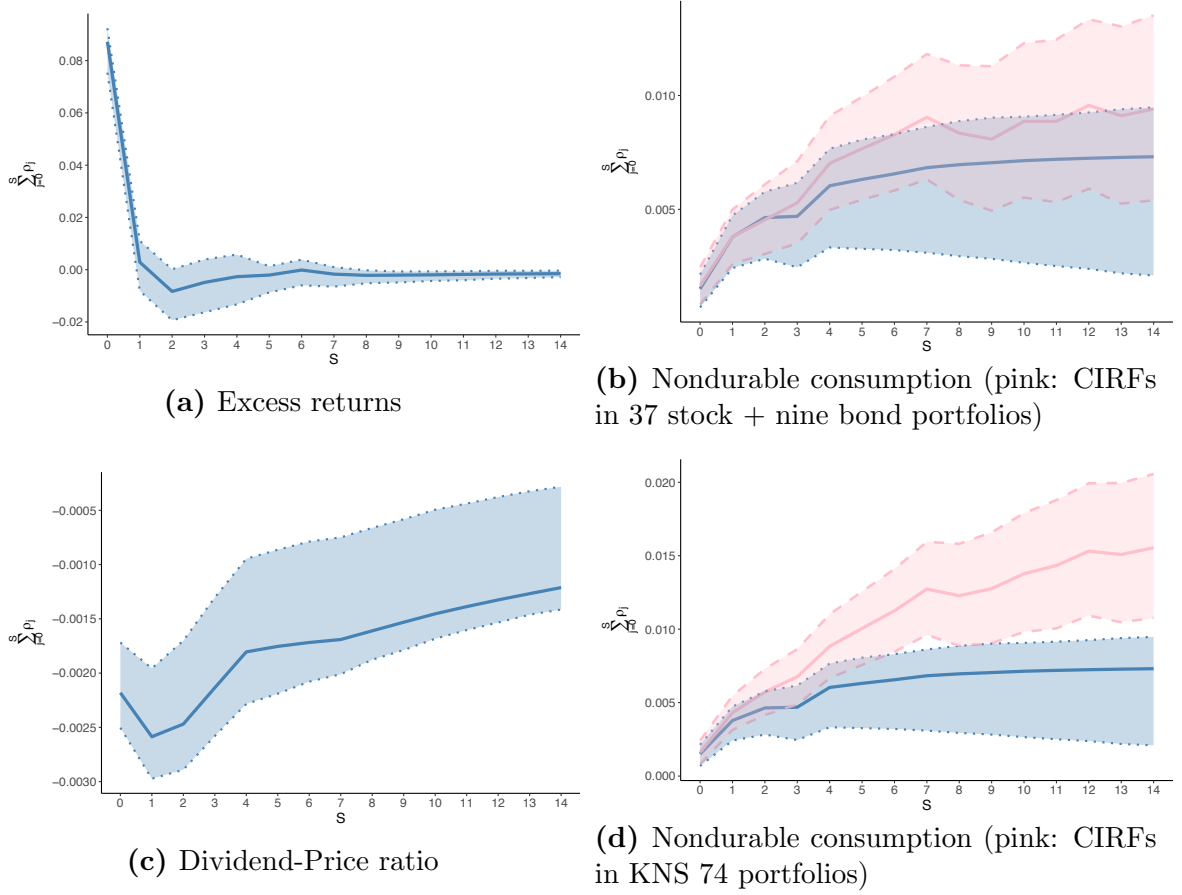
$$\text{cov} \left[\log\left(\frac{C_{1+t+s}}{C_t}\right), r_{t,t+1}^m \right] = \sigma(r_{t,t+1}^m) \cdot IRF_s(\log(C)), \quad (\text{IA.31})$$

where $IRF_s(\log(C))$ represents the impulse response of the log consumption at horizon s to one-standard-deviation asset return shock. We can compare $IRF_s(\log(C))$ with the cumulative impulse responses of log consumption growth implied by our MA model. The estimation of equation (IA.30) attempts to fit the unit root in the log consumption level. We use the Bootstrap to estimate the 90% confidence intervals of $IRF_s(\log(C))$.

Figure IA.18 displays the impulse responses to a one-standard-deviation shock to excess returns. The left two panels report the IRF of excess returns and the dividend-price ratio, while Panels (b) and (d) report (twice) the IRF of the consumption level with overlay the one estimated in our state-space framework using our two cross-sections: our baseline stock and bond portfolios (Panel (b)) and the Kozak, Nagel, and Santosh (2020) anomaly portfolios.

First, we confirm that the patterns and magnitudes of impulse responses in Figure IA.18 are similar to those in Parker (2001). In particular, quarterly consumption slowly adjusts to the market excess return shock, as can be seen in Panel (b). Nevertheless, the VAR approach seems to underestimate the consumption responses in the medium- to long-run compared to the state-space framework. The long-horizon covariance at $S = 14$ quarters implied by our MA model is 28.7% larger than that implied by the VAR(4) in Panel (b), and almost twice

Figure IA.18: VAR impulse responses to a one-standard-deviation shock to excess returns.



The blue lines present the impulse response functions of quarterly market excess return (Panel (a)), log non-durable consumption (Panels (b) and (d)), and dividend-price ratio (Panel (c)) to a one-standard-deviation market excess return shock, along with the centered posterior 90% coverage regions. In Panels (b) and (d), we include the cumulative impulse responses of log nondurable consumption growth estimated based on our MA model (see the pink lines).

as large as the VAR one in Panel (d).

What drives the difference between Parker (2001) VAR and our MA model? The VAR model assumes that the shock in market excess return is the only source of asset return driver in consumption growth. Instead, our MA model, as we show in simulations, can capture an arbitrary linear combination of systematic latent factors, which are identified by a large cross-section of portfolio returns. Hence, when consumption growth reacts to other factors beyond solely the market excess return (as shown in Table 6), we can detect a significant difference between the VAR- and MA-implied IRFs.

But which model—the VAR one or our state-space formulation—does a better job at capturing the conditional consumption mean? To answer this question, we first rewrite the

Table IA.III: Predicting nondurable consumption growth using VAR(4) in Parker (2001)

$s =$	1	2	3	4	5	6	7	8	9	10	11	12
$\mathbb{E}_t^{VAR(4)}[\Delta c_{t+s-1,t+s}]$	0.984	1.028	1.024	0.958	0.853	0.468	0.389	0.307	0.516	0.692	0.531	0.462
s.e. (OLS)	(0.137)	(0.212)	(0.252)	(0.285)	(0.469)	(0.481)	(0.508)	(0.535)	(0.546)	(0.551)	(0.559)	(0.566)
s.e. (NW, lag=12)	(0.172)	(0.228)	(0.256)	(0.298)	(0.654)	(0.636)	(0.656)	(0.760)	(0.773)	(0.792)	(0.785)	(0.780)
Predictive R^2	0.200	0.103	0.074	0.052	0.016	0.005	0.003	0.002	0.004	0.008	0.004	0.003

We regress the future realized growth rates in nondurable consumption on the conditional consumption mean ($\mathbb{E}_t[\Delta c_{t+s-1,t+s}^{VAR(4)}]$) implied by the VAR(4) model in Parker (2001). s ranges from one to 12 quarters. We report (1) the point estimates of the slope coefficients, (2) the OLS and Newey-West (12 lags) standard errors within the parentheses, and (3) the predictive R^2 . Sample: 1963:Q3–2019:Q4.

VAR(4) in companion form; that is,

$$\mathbf{z}_t = \boldsymbol{\mu}_z + \mathbf{B}\mathbf{z}_{t-1} + \boldsymbol{\epsilon}_{z,t}, \quad (\text{IA.32})$$

$$\mathbf{z}_t = \begin{bmatrix} \mathbf{y}_t \\ \mathbf{y}_{t-1} \\ \mathbf{y}_{t-2} \\ \mathbf{y}_{t-3} \end{bmatrix}, \quad \boldsymbol{\mu}_z = \begin{bmatrix} \boldsymbol{\mu}_y \\ \mathbf{0} \\ \mathbf{0} \\ \mathbf{0} \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} \mathbf{A}_1 & \mathbf{A}_2 & \mathbf{A}_3 & \mathbf{A}_4 \\ \mathbf{I}_3 & \mathbf{0} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{I}_3 & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{I}_3 & \mathbf{0} \end{bmatrix}, \quad \text{and } \boldsymbol{\epsilon}_{z,t} = \begin{bmatrix} \boldsymbol{\epsilon}_{y,t} \\ \mathbf{0} \\ \mathbf{0} \\ \mathbf{0} \end{bmatrix}.$$

Using equation (IA.32), we can then predict the future log consumption:

$$\mathbb{E}_t[\mathbf{z}_{t+s}] = \boldsymbol{\mu}_z + \mathbf{B}\mathbb{E}_t[\mathbf{z}_{t+s-1}], \quad s \geq 2, \quad \text{and } \mathbb{E}_t[\mathbf{z}_{t+1}] = \boldsymbol{\mu}_z + \mathbf{B}\mathbf{z}_t. \quad (\text{IA.33})$$

It then follows that the VAR-implied expected consumption growth between horizon $t+s-1$ and $t+s$ is $\mathbb{E}_t[\log(C_{t+s})] - \mathbb{E}_t[\log(C_{t+s-1})]$.

Then, to verify the accuracy of the VAR conditional mean approximation, we can regress $\Delta c_{t+s-1,t+s}$ on $\mathbb{E}_t[\log(C_{t+s})] - \mathbb{E}_t[\log(C_{t+s-1})]$ and test whether the slope coefficients are close to one (as we do for the state-space forecasts in Table 2). The results of the predictive regressions are reported in Table IA.III. Compared with the MA model (see Panel A of Table 2 in the main text), the predictions for one- and two-period ahead consumption growth are similar in VAR(4). However, as the forecast horizon increases, the performance of the VAR(4) prediction tends to deteriorate. Starting from $s \geq 5$, the predictive R^2 is close to zero in the VAR(4), with slope coefficients much smaller than one, although we cannot reject the null hypothesis of the slope coefficients being equal to one.

Finally, under the assumption of additively separable CRRA preferences, we can compute

the coefficient of relative risk aversion implied by the VAR specification,

$$\gamma_s = \frac{\mathbb{E}[r_{t,t+1}^m] + \frac{1}{2}\text{var}(r_{t,t+1}^m)}{\text{cov}\left[\log\left(\frac{C_{1+t+s}}{C_t}\right), r_{t,t+1}^m\right]} = \frac{\mathbb{E}[r_{t,t+1}^m] + \frac{1}{2}\text{var}(r_{t,t+1}^m)}{\sigma(r_{t,t+1}^m) \cdot IRF_s(\log(C))}. \quad (\text{IA.34})$$

M Cross-sectional pricing: Estimation details

We now describe the estimation of f_t 's risk price in Section IV.1 of the main text. The Bayesian framework comes from Bryzgalova, Huang, and Julliard (2024), and details can be found therein.

As in our generalized return dynamics in equation (7), we postulate an approximate factor structure for the cross-section of excess returns, as follows:

$$\mathbf{r}_t^e = \boldsymbol{\mu}_r + \boldsymbol{\rho}^r \mathbf{u}_t + \mathbf{w}_t^r, \quad \mathbf{u}_t \stackrel{\text{iid}}{\sim} \mathcal{N}(\mathbf{0}_K, \mathbf{I}_K), \quad (\boldsymbol{\rho}^r)' \boldsymbol{\rho}^r = \mathbf{I}_K, \quad \mathbf{w}_t^r \stackrel{\text{iid}}{\sim} \mathcal{N}(\mathbf{0}_N, \boldsymbol{\Sigma}_r), \quad (\text{IA.35})$$

where $\mathbf{u}_t$ are K largest latent factors of asset returns. As in equation (2), consumption growth slowly adjusts to asset return shocks $\mathbf{u}_t$, as follows:

$$\Delta c_{t-1,t} = \mu_c + \sum_{j=0}^{\bar{S}} \rho_j \underbrace{\boldsymbol{\eta}^\top \mathbf{u}_{t-j}}_{f_{t-j}} + w_t^c, \quad \boldsymbol{\eta}^\top \boldsymbol{\eta} = 1, \quad \text{var}(f_t) = 1, \quad (\text{IA.36})$$

so f_t is the common driver of consumption growth and asset returns. $\mathbf{g}_t$, the additional $(K-1)$ latent factors in equations (3) and (20) of the main text, are linear combinations of $\mathbf{u}_t$ orthogonal to f_t . It is worth noting that since asset return drivers, $\mathbf{u}_t$, are latent, we cannot identify the exact rotation of $\mathbf{g}_t$. However, f_t is uniquely identified (up to the sign restriction) due to equation (IA.36).

Moreover, to model the cross-sectional dimension, we consider a log linear SDF as follows:

$$m_t = 1 - \boldsymbol{\lambda}_u^\top \mathbf{u}_t = 1 - b_f f_t - \mathbf{b}_g^\top \mathbf{g}_t, \quad (\text{IA.37})$$

where $\boldsymbol{\lambda}_u$ are the risk prices associated with $\mathbf{u}_t$ (also their risk premia since $\text{cov}(\mathbf{u}_t) = \mathbf{I}_K$). The last equality is due to the fact that we can always find a non-singular linear rotation of $\mathbf{u}_t$ such that $(f_t, \mathbf{g}_t^\top)^\top = \mathbf{H} \mathbf{u}_t$ and $(b_f, \mathbf{b}_g^\top) = \boldsymbol{\lambda}_u^\top \mathbf{H}^{-1}$, where $\mathbf{H}$ is a $K \times K$ non-singular matrix.

As standard, we postulate that the exposures to latent factors, $\boldsymbol{\rho}^r$, can partially explain the cross-sectional variation of expected returns:

$$\tilde{\boldsymbol{\mu}}_r = \boldsymbol{\mu}_r + \frac{1}{2}\mathbf{V}_r = \boldsymbol{\rho}^r \boldsymbol{\lambda}_u + \boldsymbol{\alpha}, \quad (\text{IA.38})$$

where we allow for pricing errors $\boldsymbol{\alpha}$ but require them to be orthogonal to the factor loadings $\boldsymbol{\rho}^r$. The latter requirement is standard in the literature (e.g., Giglio and Xiu (2021)) and acts as an identification assumption to recover risk premia estimates. We also include the Jensen correction term, $\frac{1}{2}\mathbf{V}_r$ (a vector of asset returns' variances), into equation (IA.38). Finally, after identifying $\boldsymbol{\lambda}_u$, the risk price of f_t is

$$b_f = -\text{cov}(m_t, f_t) = -\text{cov}(1 - \boldsymbol{\lambda}_u^\top \mathbf{u}_t, \boldsymbol{\eta}^\top \mathbf{u}_t) = \boldsymbol{\eta}^\top \boldsymbol{\lambda}_u, \quad (\text{IA.39})$$

where the first equality comes from the fact that f_t is orthogonal to g_t and has a unit variance.

In the data, asset return drivers, $\mathbf{u}_t$, are unidentified. That is, we can identify only a linear rotation of $\mathbf{u}_t$, denoted by $\tilde{\mathbf{u}}_t = \tilde{\mathbf{H}}\mathbf{u}_t$, and $\boldsymbol{\Sigma}_{\tilde{u}} = \text{cov}(\tilde{\mathbf{u}}_t) = \tilde{\mathbf{H}}\tilde{\mathbf{H}}^\top$. However, the risk price of f_t is point-identified due to the rotation invariance property emphasized in Giglio and Xiu (2021). To show the rotation invariance of b_f , we rewrite the equation system as

$$\begin{aligned} \mathbf{r}_t^e &= \boldsymbol{\alpha} + \underbrace{\boldsymbol{\rho}^r \tilde{\mathbf{H}}^{-1}}_{\tilde{\boldsymbol{\rho}}^r} \underbrace{\tilde{\mathbf{H}}\boldsymbol{\lambda}_u}_{\boldsymbol{\lambda}_{\tilde{u}}} - \frac{1}{2}\mathbf{V}_r + \underbrace{\boldsymbol{\rho}^r \tilde{\mathbf{H}}^{-1}}_{\tilde{\boldsymbol{\rho}}^r} \underbrace{\tilde{\mathbf{H}}\mathbf{u}_t}_{\tilde{\mathbf{u}}_t} + \mathbf{w}_t^r, \\ \Delta c_{t-1,t} &= \mu_c + \sum_{j=0}^{\bar{S}} \rho_j \underbrace{\boldsymbol{\eta}^\top \tilde{\mathbf{H}}^{-1}}_{\tilde{\boldsymbol{\eta}}^\top} \underbrace{\tilde{\mathbf{H}}\mathbf{u}_{t-s}}_{\tilde{\mathbf{u}}_{t-s}} + w_t^c, \text{ and} \\ m_t &= 1 - \boldsymbol{\lambda}_{\tilde{u}}^\top (\tilde{\mathbf{H}}^{-1})^\top \tilde{\mathbf{H}}^{-1} \tilde{\mathbf{u}}_t = 1 - \underbrace{\boldsymbol{\lambda}_{\tilde{u}}^\top \boldsymbol{\Sigma}_{\tilde{u}}^{-1}}_{\tilde{\boldsymbol{\eta}}} \tilde{\mathbf{u}}_t, \quad b_f = \underbrace{\boldsymbol{\eta}^\top \tilde{\mathbf{H}}^{-1}}_{\tilde{\boldsymbol{\eta}}} \underbrace{\tilde{\mathbf{H}}\boldsymbol{\lambda}_u}_{\boldsymbol{\lambda}_{\tilde{u}}}. \end{aligned} \quad (\text{IA.40})$$

The posterior sampler for this estimation framework, as well as its finite-sample performance based on simulation studies, is provided in Bryzgalova, Huang, and Julliard (2024). Note that we estimate the means of the latent factors in our Bayesian approach and fully account for estimation uncertainty in the second (centred) moments.

N Frequentist Fama-MacBeth Estimation

To make our findings on the cross-sectional fit comparable to the existing literature, we repeat the same analysis using the traditional Fama-MacBeth (FMB) regressions. We first extract the posterior mean of the consumption mean shock, denoted as $\hat{f}_t$. Next, we regress portfolio returns on $\hat{f}_t$ to obtain the asset return residuals and extract from them four principal components, denoted $\hat{\mathbf{g}}_t$, which are used as a control for omitted factors.¹⁶ Note that both $\hat{f}_t$ and $\hat{\mathbf{g}}_t$ are normalized to have unit variances, and they are uncorrelated by construction. Therefore, the risk premia estimates based on the Fama-MacBeth regressions are also their risk prices in the SDF. Unlike the Bayesian method of Bryzgalova, Huang, and Julliard (2024) used in the main body of the paper, the frequentist approach used in this section does not account for the uncertainty in estimating f_t and g_t .

Table IA.IV reports the estimation results using frequentist Fama-MacBeth regressions with and without the common pricing error (i.e., with or without an intercept) or the control for omitted factors (i.e., with or without $\hat{\mathbf{g}}_t$). The first thing to notice is that, when an intercept is not included, the FMB estimates of consumption risk premia in Table IA.IV are statistically significant and virtually identical to the (Bayesian and frequentist) robust estimates in Table 5.

Note that the standard FMB regressions are not robust to neither omitted variable bias nor simple transformations of the test assets that preserve their span. Both types of fragility are on display in Table IA.IV. First, controlling for omitted factors ($\hat{\mathbf{g}}_t$) can drastically change the cross-sectional fit of the model. Second, both point estimates and statistical significance of the risk premia can radically change when a common intercept is included in the model and when test assets are repackaged without changing their span (see Panels D and E). This is due to the fact that a truly priced factor often appears insignificant in FMB regressions if it does not have sufficient cross-sectional dispersion in betas—but this is a property of the test assets, not of the candidate source of risk. Furthermore, this property of FMB regressions is not invariant to a simple transformation of the test assets that leaves their spanning (Sharpe ratios achievable and payoffs) unchanged (see Panel B vs. C and Panel D vs. E). That is, the popular empirical distinction between a “level” and a “slope” risk factor is an artifact of FMB regressions and not an inherent property of the risk factor. That is exactly why

¹⁶To avoid collinearity in the time-series step of the Fama-MacBeth regressions, we add a small amount of noise to $\hat{\mathbf{g}}_t$.

modern methods of risk premia estimation have largely abandoned the usage of canonical Fama-MacBeth regressions in favor of more robust approaches—for detailed discussions of these issues, see Giglio, Xiu, and Zhang (2021), Giglio and Xiu (2021), and Bryzgalova, Quaini, Trojani, and Ming (2024).

Conversely, Table 5 shows (see Panel B vs. C and Panel D vs. E therein) that both Bayesian (Bryzgalova, Huang, and Julliard (2024)) and frequentist (Giglio and Xiu (2021)) robust inference methods deliver virtually identical estimates of risk premia regardless of the asset repackaging.

O Mapping to structural models

In this Appendix, we show how our state-space formulation can be used to deliver the calibration in Section IV.2 of the main text. In particular, we show how to map the reduced-form estimate of f_t 's risk price to the structural parameters governing investors' preferences. We consider two types of investor preferences.

First, in the additively separable power utility setting, the log SDF is¹⁷:

$$m_t = \log(\delta) - \gamma \Delta c_{t-1,t}, \text{ and } \Delta \mathbb{E}_t[m_t] := m_t - \mathbb{E}_{t-1}[m_t] = -\gamma \cdot \Delta \mathbb{E}_t[\Delta c_{t-1,t}], \quad (\text{IA.41})$$

where $-\log(\delta)$ is the real risk-free rate in a hypothetical economy without growth or uncertainty, and γ is the coefficient of relative risk aversion (RRA). Assuming the log normality of the consumption shock w_t^c , we obtain a closed-form solution for the real interest rate:

$$r_{f,t} = -\log(\delta) + \gamma \cdot \mathbb{E}_{t-1}[\Delta c_{t-1,t}] - \frac{\gamma^2 \text{var}_{t-1}(\Delta c_{t-1,t})}{2} = -\log(\delta) + \gamma \cdot (\mu_c + \sum_{j=1}^S \rho_j f_{t-j}) - \frac{\gamma^2 \text{var}_{t-1}(\Delta c_{t-1,t})}{2}, \quad (\text{IA.42})$$

where $\text{var}_{t-1}(\Delta c_{t-1,t}) = \sigma_{c,t-1}^2 + \rho_0^2 \sigma_{f,t-1}^2$.

Our MA representation of consumption growth implies that $\Delta \mathbb{E}_t[m_t] = -\gamma \rho_0 f_t - \gamma w_t^c$. As seen in the previous analysis, the contemporaneous covariance between consumption growth and f_t , ρ_0 , is small, which implies an unreasonably high RRA – the canonical equity premium puzzle. Parker and Julliard (2005) show that, iterating forward the intertemporal Euler equation, a valid log SDF can be constructed as

$$m_t^S = r_{f,t,t+S} - \gamma_S \Delta c_{t-1,t+S}. \quad (\text{IA.43})$$

¹⁷Note that only the shocks to the time- t SDF matter for the pricing of time- t *excess* returns.

Table IA.IV: Cross-sectional pricing ability of the shock to conditional consumption mean: Frequentist Fama-MacBeth regressions

		No intercept		With intercept		
		b_f	R^2	intercept	b_f	R^2
Panel A. 37 value-size-industry stock and nine bond portfolios						
No control for omitted factors:	coefficient	0.226	0.555	0.006	0.168	0.654
	t -stat	(3.355)		(2.679)	(2.314)	
Control for omitted factors:	coefficient	0.251	0.794	0.002	0.226	0.798
	t -stat	(3.751)		(1.659)	(3.319)	
Panel B. 74 Kozak, Nagel, and Santosh (2020) decile 1 and decile 10 anomaly portfolios						
No control for omitted factors:	coefficient	0.202	-0.316	0.020	-0.010	0.001
	t -stat	(2.746)		(3.647)	(-0.115)	
Control for omitted factors:	coefficient	0.247	0.377	0.012	0.127	0.400
	t -stat	(3.702)		(1.995)	(1.400)	
Panel C. 37 Kozak, Nagel, and Santosh (2020) decile 10 portfolios plus 37 long-short portfolios						
No control for omitted factors:	coefficient	0.231	0.542	-0.008	0.316	0.625
	t -stat	(3.265)		(-3.436)	(4.213)	
Control for omitted factors:	coefficient	0.233	0.803	-0.008	0.312	0.862
	t -stat	(3.480)		(-6.638)	(4.567)	
Panel D. 37 value-size-industry stock portfolios						
No control for omitted factors:	coefficient	0.238	-0.218	0.013	0.092	0.129
	t -stat	(3.340)		(2.372)	(0.993)	
Control for omitted factors:	coefficient	0.227	0.359	0.027	-0.025	0.587
	t -stat	(3.359)		(4.300)	(-0.258)	
Panel E. Transformed 37 value-size-industry stock portfolios						
No control for omitted factors:	coefficient	0.214	0.515	0.006	0.178	0.623
	t -stat	(2.883)		(3.248)	(2.290)	
Control for omitted factors:	coefficient	0.283	0.882	0.004	0.217	0.892
	t -stat	(4.136)		(2.346)	(2.900)	

Frequentist Fama-MacBeth estimation results for five cross-sections of excess returns: 37 value-size-industry stock and nine bond portfolios (Panel A), Kozak, Nagel, and Santosh (2020) 74 decile 1 and 10 characteristic-sorted portfolios (Panel B), Kozak, Nagel, and Santosh (2020) 37 decile 10 portfolios plus 37 long-short (decile 1 minus decile 10) portfolios (Panel C), 37 value-size-industry stock portfolios (Panel D), and transformed 37 value-size-industry stock portfolios (Panel E). In the transformed cross-section of 37 value-size-industry stock portfolios (Panel E), we long portfolios 1–19; for the remaining 18 portfolios, we construct the long-short portfolios by buying portfolios 20–37 but short-selling portfolios 1–18. We report: (1) the risk price of the shock to the conditional consumption mean f_t (b_f); (2) the cross-sectional R^2 ; and (3) the common pricing error (see column intercept) if it is included in the cross-sectional regression. The t -statistics are based on the Shanken (1992) corrected standard errors. In each cross-section of test assets, we report the results for the specification (1) with and without a common intercept in the cross-sectional step and (2) with and without the control for omitted factors.

Table IA.V: Individual pricing errors: 37 stock and nine bond portfolios

	No control for omitted factors		Control for omitted factors	
	No intercept	With intercept	No intercept	With intercept
0.5Y	0.001 (2.744)	-0.005 (-2.433)	0.001 (2.223)	-0.001 (-1.228)
1Y	0.002 (3.143)	-0.004 (-2.213)	0.001 (2.141)	-0.001 (-1.022)
2Y	0.002 (2.385)	-0.003 (-2.187)	0.001 (0.919)	-0.001 (-1.562)
3Y	0.003 (2.212)	-0.002 (-1.908)	0.000 (0.496)	-0.001 (-2.128)
4Y	0.004 (2.142)	-0.001 (-1.328)	0.000 (0.245)	-0.001 (-2.010)
5Y	0.005 (2.082)	-0.001 (-0.651)	0.000 (-0.001)	-0.001 (-1.418)
6Y	0.005 (2.009)	0.000 (-0.122)	0.000 (-0.285)	-0.001 (-1.072)
7Y	0.006 (1.923)	0.000 (0.209)	-0.001 (-0.593)	-0.001 (-0.929)
10Y	0.006 (1.644)	0.001 (0.552)	-0.002 (-1.281)	-0.001 (-0.849)
SizeValue11	-0.020 (-4.502)	-0.017 (-4.443)	-0.011 (-5.519)	-0.011 (-5.815)
SizeValue12	-0.002 (-0.658)	0.000 (-0.175)	0.002 (1.443)	0.002 (1.314)
SizeValue13	-0.001 (-0.207)	0.000 (0.150)	0.000 (0.230)	0.000 (0.082)
SizeValue14	0.007 (2.704)	0.008 (2.973)	0.007 (4.525)	0.006 (4.446)
SizeValue15	0.008 (2.558)	0.009 (2.974)	0.006 (3.837)	0.006 (3.810)
SizeValue21	-0.011 (-3.381)	-0.009 (-3.038)	-0.004 (-2.634)	-0.004 (-2.557)
SizeValue22	-0.001 (-0.373)	0.000 (0.116)	0.001 (0.473)	0.001 (0.566)
SizeValue23	0.004 (1.989)	0.004 (2.244)	0.002 (1.216)	0.002 (1.364)
SizeValue24	0.006 (2.604)	0.006 (2.635)	0.002 (1.522)	0.002 (1.593)
SizeValue25	0.006 (2.246)	0.007 (2.516)	0.003 (1.647)	0.003 (1.618)
SizeValue31	-0.008 (-2.656)	-0.007 (-2.372)	-0.002 (-1.236)	-0.002 (-1.132)
SizeValue32	0.001 (0.799)	0.002 (1.033)	0.001 (0.635)	0.001 (0.847)
SizeValue33	0.002 (1.393)	0.002 (1.291)	0.000 (0.009)	0.000 (0.101)
SizeValue34	0.006 (2.829)	0.006 (2.773)	0.002 (1.132)	0.002 (1.333)
SizeValue35	0.008 (2.689)	0.008 (2.741)	0.003 (1.346)	0.003 (1.466)
SizeValue41	-0.003 (-0.857)	-0.002 (-0.750)	0.003 (1.643)	0.003 (1.779)
SizeValue42	-0.002 (-0.884)	-0.002 (-0.942)	-0.003 (-1.585)	-0.002 (-1.464)
SizeValue43	0.002 (0.890)	0.001 (0.697)	-0.002 (-1.060)	-0.001 (-0.908)
SizeValue44	0.005 (2.502)	0.005 (2.379)	0.001 (0.732)	0.002 (0.985)
SizeValue45	0.005 (1.588)	0.005 (1.648)	0.000 (-0.053)	0.000 (0.035)
SizeValue51	0.000 (-0.058)	-0.001 (-0.464)	0.003 (2.628)	0.003 (2.569)
SizeValue52	0.000 (-0.061)	-0.002 (-0.699)	-0.001 (-0.596)	-0.001 (-0.619)
SizeValue53	0.003 (1.048)	0.001 (0.425)	0.000 (0.118)	0.000 (0.075)
SizeValue54	-0.001 (-0.338)	-0.002 (-0.763)	-0.005 (-2.631)	-0.005 (-2.714)
SizeValue55	0.002 (0.558)	0.001 (0.291)	0.000 (0.108)	0.000 (-0.063)
NonDurables	0.005 (1.597)	0.003 (1.224)	0.002 (0.901)	0.002 (1.081)
Durables	-0.008 (-1.940)	-0.008 (-1.893)	-0.008 (-2.527)	-0.009 (-2.889)
Manufact	-0.001 (-0.562)	-0.002 (-0.693)	-0.001 (-0.584)	-0.001 (-0.646)
Energy	0.005 (0.973)	0.002 (0.517)	0.004 (1.644)	0.003 (1.462)
Chemicals	0.001 (0.232)	-0.001 (-0.239)	-0.001 (-0.207)	-0.001 (-0.272)
BusEq	-0.001 (-0.184)	-0.001 (-0.184)	0.008 (2.555)	0.008 (2.508)
Telecom	0.002 (0.479)	0.000 (-0.062)	0.002 (0.503)	0.001 (0.413)
Utilities	0.003 (0.834)	0.001 (0.155)	-0.004 (-1.455)	-0.003 (-1.278)
Shops	0.001 (0.339)	0.001 (0.204)	0.001 (0.544)	0.001 (0.602)
Health	0.007 (1.649)	0.005 (1.276)	0.009 (2.860)	0.009 (2.896)
Money	-0.002 (-0.631)	-0.002 (-0.669)	-0.006 (-2.245)	-0.005 (-2.084)
Other	-0.007 (-3.791)	-0.007 (-3.726)	-0.007 (-3.810)	-0.007 (-3.758)
# of significance at 5% level	19	15	13	13
# of significance at 10% level	22	18	14	14

This table displays the individual pricing errors (with the t -statistics given in the parentheses) of 37 stock and nine bond portfolios based on the frequentist Fama-MacBeth regressions. The final two rows report the number of significant pricing errors at the 5% or 10% significance level. We report the results for the specification (1) with and without a common intercept in the cross-sectional step and (2) with and without the control for omitted factors.

Table IA.VI: Individual pricing errors: 74 KNS anomaly portfolios

	No control for omitted factors		Control for omitted factors	
	No intercept	With intercept	No intercept	With intercept
accruals_p1	-0.009 (-3.462)	-0.007 (-2.832)	-0.006 (-2.487)	-0.006 (-2.246)
accruals_p10	0.005 (1.766)	0.003 (1.270)	0.008 (3.088)	0.006 (2.688)
aturnover_p1	-0.003 (-1.031)	-0.009 (-3.099)	0.000 (0.000)	-0.003 (-1.232)
aturnover_p10	0.005 (1.845)	0.003 (1.186)	-0.001 (-0.207)	0.000 (0.028)
betaarb_p1	-0.009 (-2.858)	-0.003 (-0.897)	-0.004 (-1.310)	-0.001 (-0.567)
betaarb_p10	0.007 (2.408)	-0.002 (-0.556)	0.001 (0.347)	-0.002 (-1.059)
cfp_p1	-0.002 (-0.650)	-0.003 (-1.333)	-0.003 (-2.032)	-0.002 (-1.512)
cfp_p10	0.008 (2.295)	0.006 (1.921)	0.001 (0.537)	0.001 (0.590)
ciss_p1	-0.009 (-4.150)	-0.009 (-4.240)	-0.009 (-3.997)	-0.008 (-4.025)
ciss_p10	0.011 (4.294)	0.005 (2.131)	0.006 (2.772)	0.005 (2.502)
nissa_p1	-0.010 (-4.960)	-0.011 (-5.729)	-0.009 (-4.901)	-0.008 (-4.711)
nissa_p10	0.013 (5.186)	0.008 (3.616)	0.006 (2.837)	0.005 (2.739)
divp_p1	-0.001 (-0.565)	-0.003 (-1.209)	-0.004 (-1.771)	-0.002 (-1.418)
divp_p10	0.005 (1.025)	0.001 (0.242)	-0.004 (-1.081)	-0.004 (-1.096)
dur_p1	-0.001 (-0.374)	-0.004 (-1.509)	-0.001 (-1.255)	-0.001 (-1.127)
dur_p10	0.007 (1.698)	0.007 (1.609)	0.003 (0.845)	0.002 (0.552)
ep_p1	-0.011 (-3.165)	-0.006 (-1.832)	-0.007 (-2.204)	-0.006 (-2.118)
ep_p10	0.010 (2.852)	0.008 (2.361)	0.000 (0.060)	0.001 (0.506)
gmargins_p1	0.000 (-0.222)	-0.002 (-0.930)	-0.003 (-1.406)	-0.003 (-1.489)
gmargins_p10	0.004 (1.484)	-0.001 (-0.359)	0.003 (1.915)	0.001 (1.128)
growth_p1	-0.004 (-1.574)	-0.004 (-1.686)	0.000 (-0.039)	0.001 (0.280)
growth_p10	0.003 (1.249)	0.003 (1.141)	0.000 (-0.003)	0.000 (-0.035)
igrowth_p1	-0.007 (-2.632)	-0.006 (-2.426)	-0.003 (-1.411)	-0.003 (-1.424)
igrowth_p10	0.002 (1.093)	0.003 (1.665)	0.001 (0.379)	0.001 (0.426)
indmom_p1	-0.008 (-1.931)	-0.005 (-1.308)	0.003 (0.868)	0.003 (0.918)
indmom_p10	0.013 (2.963)	0.006 (1.690)	-0.001 (-0.211)	-0.001 (-0.240)
indrrev_p1	-0.013 (-4.879)	-0.015 (-5.836)	-0.016 (-6.553)	-0.016 (-7.006)
indrrev_p10	0.009 (3.560)	0.014 (6.200)	0.015 (6.165)	0.016 (6.926)
inv_p1	-0.004 (-1.485)	-0.005 (-1.937)	-0.003 (-1.090)	-0.002 (-0.666)
inv_p10	0.009 (3.837)	0.007 (3.065)	0.005 (2.189)	0.005 (2.079)
invcap_p1	-0.001 (-0.174)	0.001 (0.297)	0.005 (2.118)	0.004 (1.907)
invcap_p10	0.007 (2.324)	0.002 (0.571)	0.003 (1.475)	0.003 (1.244)
lev_p1	0.001 (0.188)	-0.003 (-1.092)	0.000 (-0.650)	-0.001 (-1.226)
lev_p10	0.003 (0.687)	0.004 (0.922)	-0.007 (-1.824)	-0.003 (-1.025)
lrrev_p1	-0.001 (-0.193)	-0.002 (-0.701)	0.000 (0.071)	0.002 (0.780)
lrrev_p10	0.002 (0.469)	0.008 (2.000)	-0.003 (-0.929)	-0.003 (-0.947)
mom_p1	-0.005 (-1.218)	0.000 (0.145)	0.008 (3.064)	0.008 (3.383)
mom_p10	0.015 (3.796)	0.011 (3.189)	0.006 (3.521)	0.007 (3.842)
mom12_p1	-0.029 (-6.420)	-0.021 (-6.021)	-0.014 (-4.571)	-0.013 (-4.603)
mom12_p10	0.018 (4.940)	0.014 (4.414)	0.012 (5.738)	0.012 (5.710)
momrev_p1	-0.002 (-0.721)	-0.005 (-1.436)	0.000 (0.144)	0.000 (0.017)
momrev_p10	0.004 (1.091)	0.008 (2.368)	0.001 (0.270)	0.002 (0.604)
noa_p1	-0.007 (-2.431)	-0.009 (-3.440)	-0.002 (-0.832)	-0.002 (-1.097)
noa_p10	0.004 (1.582)	0.003 (1.185)	0.004 (1.863)	0.004 (1.732)
price_p1	-0.013 (-2.368)	-0.001 (-0.256)	-0.003 (-1.096)	-0.004 (-1.666)
price_p10	0.003 (1.452)	-0.003 (-2.332)	0.000 (-0.044)	-0.001 (-1.131)

Table IA.VI: Individual pricing errors: 74 KNS anomaly portfolios

	No control for omitted factors		Control for omitted factors	
	No intercept	With intercept	No intercept	With intercept
prof_p1	-0.001 (-0.396)	-0.007 (-2.629)	-0.001 (-0.531)	-0.004 (-1.758)
prof_p10	0.008 (2.790)	0.005 (1.885)	0.002 (1.476)	0.003 (1.720)
roaa_p1	-0.009 (-2.565)	-0.005 (-1.484)	-0.004 (-1.393)	-0.005 (-1.814)
roaa_p10	0.004 (1.606)	0.000 (-0.035)	0.001 (0.839)	0.000 (0.200)
roea_p1	-0.006 (-1.718)	0.000 (-0.159)	-0.001 (-0.357)	-0.002 (-0.646)
roea_p10	0.004 (1.727)	0.001 (0.368)	0.000 (0.086)	0.001 (0.419)
season_p1	-0.011 (-3.763)	-0.011 (-3.801)	-0.015 (-4.990)	-0.014 (-5.051)
season_p10	0.013 (4.970)	0.012 (4.824)	0.011 (4.588)	0.012 (5.017)
sgrowth_p1	0.002 (0.745)	0.001 (0.458)	0.006 (3.270)	0.006 (3.312)
sgrowth_p10	0.000 (0.032)	-0.002 (-0.686)	-0.001 (-0.412)	-0.002 (-0.980)
shvol_p1	-0.006 (-1.644)	-0.001 (-0.297)	0.001 (0.615)	0.002 (1.079)
shvol_p10	0.004 (1.327)	-0.004 (-1.595)	-0.001 (-0.688)	-0.004 (-1.690)
size_p1	0.002 (0.869)	-0.004 (-3.096)	-0.001 (-1.089)	-0.002 (-2.221)
size_p10	0.001 (0.193)	0.005 (1.359)	-0.002 (-0.840)	-0.003 (-1.023)
sp_p1	-0.001 (-0.427)	-0.004 (-2.053)	-0.001 (-1.179)	-0.002 (-1.938)
sp_p10	0.007 (1.762)	0.010 (2.605)	-0.001 (-0.254)	0.001 (0.473)
strev_p1	-0.007 (-2.364)	-0.010 (-3.915)	-0.011 (-4.064)	-0.012 (-4.685)
strev_p10	-0.002 (-0.656)	0.003 (0.849)	0.006 (1.998)	0.007 (2.383)
valmom_p1	-0.005 (-1.622)	-0.005 (-1.667)	0.002 (1.050)	0.002 (1.106)
valmom_p10	0.012 (3.671)	0.008 (2.468)	0.002 (0.819)	0.001 (0.605)
valmomprof_p1	-0.007 (-2.413)	-0.007 (-2.487)	0.002 (0.972)	0.002 (0.922)
valmomprof_p10	0.018 (5.901)	0.016 (5.275)	0.007 (3.397)	0.007 (3.327)
valprof_p1	-0.006 (-2.137)	-0.009 (-3.855)	-0.002 (-1.144)	-0.002 (-1.321)
valprof_p10	0.011 (3.019)	0.012 (3.377)	0.003 (1.360)	0.004 (1.541)
value_p1	0.000 (-0.005)	-0.003 (-1.291)	-0.001 (-0.770)	-0.001 (-0.782)
value_p10	0.008 (2.040)	0.007 (1.959)	0.002 (0.679)	0.001 (0.535)
valuem_p1	0.002 (0.763)	-0.001 (-0.579)	0.000 (-0.340)	0.000 (-0.202)
valuem_p10	0.004 (0.828)	0.009 (1.956)	0.013 (4.010)	0.011 (3.918)
# of significance at 5% level	32	30	23	22
# of significance at 10% level	39	40	27	30

This table displays the individual pricing errors (with the t -statistics given in the parentheses) of 74 Kozak, Nagel, and Santos (2020) anomaly portfolios based on the frequentist Fama-MacBeth regressions. The final two rows report the number of significant pricing errors at the 5% or 10% significance level. We report the results for the specification (1) with and without a common intercept in the cross-sectional step and (2) with and without the control for omitted factors.

When pricing the time- t *excess* returns, only the time- t shocks to the SDF matters, as follows:

$$\begin{aligned}
\Delta \mathbb{E}_t[m_t^S] &= (r_{f,t,t+S} - \mathbb{E}_{t-1}[r_{f,t,t+S}]) - \gamma_S (\mathbb{E}_t[\Delta c_{t-1,t+S}] - \mathbb{E}_{t-1}[\Delta c_{t-1,t+S}]) \\
&= (r_{f,t,t+S} - \mathbb{E}_{t-1}[r_{f,t,t+S}]) - \gamma_S \left(\sum_{j=0}^S \rho_j \right) f_t - \gamma_S w_t^c.
\end{aligned}$$

The first term in the above equation is the shock to the long-term risk-free rate, which is relatively small compared to the f_t shock. w_t^c is orthogonal to excess returns and, therefore, unpriced. Hence, the SDF representation in equation (20) implies that we can elicit the

RRA coefficient as follows:

$$\gamma_S \approx \frac{b_f}{\sum_{j=0}^S \rho_j}, \quad (\text{IA.44})$$

where b_f is the risk price of the f shock. An extremely high risk price, or a low cumulative impulse response in consumption growth, lead to a extremely high RRA. Moreover, a high RRA implies high mean and volatility of the risk-free rate in equation (IA.42).

The second setting that we consider is the Epstein-Zin recursive utility,

$$V_t = \left[(1 - \delta) C_t^{\frac{1-\gamma}{\theta}} + \delta \left(\mathbb{E}_t [V_{t+1}^{1-\gamma}] \right)^{\frac{1}{\theta}} \right]^{\frac{\theta}{1-\gamma}}, \quad (\text{IA.45})$$

where $0 < \delta < 1$ is the time preference parameter, ψ is the elasticity of intertemporal substitution (IES), and $\theta = \frac{1-\gamma}{1-\frac{1}{\psi}}$. The intertemporal budget constraint of the representative agent is

$$W_{t+1} = (1 + R_{w,t+1})(W_t - C_t), \quad (\text{IA.46})$$

where W_{t+1} denotes the total wealth and $R_{w,t+1}$ is the gross return on the wealth portfolio. The log SDF is then

$$m_t = \theta \log \delta - \frac{\theta}{\psi} \Delta c_{t-1,t} - (1 - \theta) r_{w,t}, \quad \text{where } r_{w,t} = \log(1 + R_{w,t}). \quad (\text{IA.47})$$

Assuming jointly lognormal $r_{w,t}$ and $\Delta c_{t-1,t}$, we obtain the risk-free rate as

$$r_{f,t} = \begin{cases} -\log \delta + \frac{1}{\psi} \mathbb{E}_{t-1} [\Delta c_{t-1,t}] - \frac{\theta}{2\psi^2} \text{var}_{t-1}(\Delta c_{t-1,t}) + \frac{\theta-1}{2} \text{var}_{t-1}(r_{w,t}), & \text{if } \psi \neq 1, \\ -\log \delta + \mathbb{E}_{t-1} [\Delta c_{t-1,t}] - (\gamma - \frac{1}{2}) \text{var}_{t-1}(\Delta c_{t-1,t}), & \text{if } \psi = 1, \end{cases} \quad (\text{IA.48})$$

where $\text{var}_{t-1}(r_{w,t})$ is the variance of the shock to $r_{w,t}$. Note that $\text{var}_{t-1}(r_{w,t})$ and $\text{var}_{t-1}(\Delta c_{t-1,t})$ are closely related because of the budget constraint in equation (IA.46). Using the log-linear return approximation of Campbell (1993) and the state-space formulation, we have

$$\Delta \mathbb{E}_t [r_{w,t}] = \Delta \mathbb{E}_t [\Delta c_{t-1,t}] + \left(1 - \frac{1}{\psi}\right) \Delta \mathbb{E}_t \sum_{j=1}^{\infty} \kappa_z^j \Delta c_{t+j-1,t+j} \quad (\text{IA.49})$$

$$= \left[\rho_0 + \left(1 - \frac{1}{\psi}\right) \sum_{j=1}^{\bar{S}} \kappa_z^j \rho_j \right] f_t + w_t^c, \quad (\text{IA.50})$$

which implies

$$\text{var}_{t-1}(r_{w,t}) = \left[\rho_0 + \left(1 - \frac{1}{\psi}\right) \sum_{j=1}^{\bar{S}} \kappa_z^j \rho_j \right]^2 \sigma_{f,t-1}^2 + \sigma_{c,t-1}^2, \quad (\text{IA.51})$$

where κ_z is a constant determined by the steady-state consumption-to-wealth ratio. In Bansal and Yaron (2004), $\kappa_z \cong 0.997$.

Using the same logic, we can express the shock to the log SDF as follows:

$$\Delta \mathbb{E}_t[m_t] = m_t - \mathbb{E}_{t-1}[m_t] = -\gamma \cdot \Delta \mathbb{E}_t[\Delta c_{t-1,t}] - \left(\gamma - \frac{1}{\psi}\right) \cdot \Delta \mathbb{E}_t \sum_{j=1}^{\infty} \kappa_z^j \Delta c_{t+j-1,t+j}. \quad (\text{IA.52})$$

When the inverse IES equals RRA, equation (IA.52) can be simplified to the CRRA preference in equation (IA.41). Replacing the MA representation of $\Delta c_{t-1,t}$ in equation (IA.52), and ignoring the unpriced component $-\gamma w_t^c$, yields

$$\Delta \mathbb{E}_t[m_t] = -\gamma \rho_0 f_t - \left(\gamma - \frac{1}{\psi}\right) \sum_{j=1}^{\bar{S}} \kappa_z^j \rho_j f_t. \quad (\text{IA.53})$$

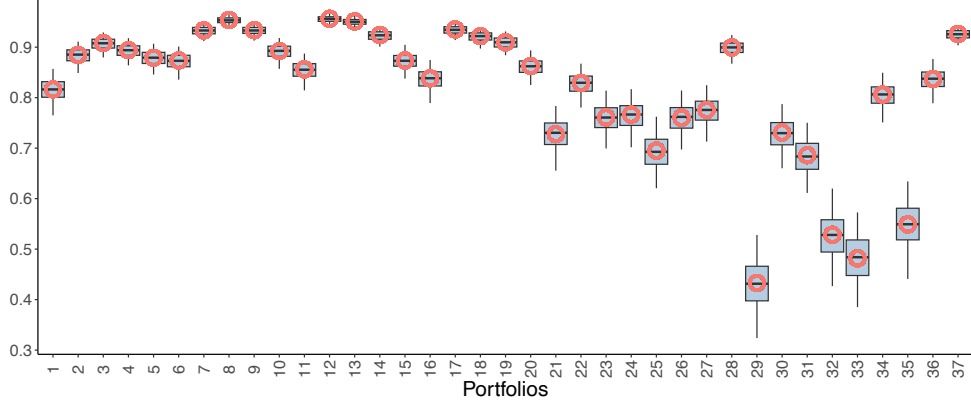
Finally, fixing the value of IES, and matching the above to the risk price of f_t in equation (20), we can reverse engineer γ as

$$\gamma \rho_0 + \left(\gamma - \frac{1}{\psi}\right) \sum_{j=1}^{\bar{S}} \kappa_z^j \rho_j = b_f \implies \gamma = \frac{b_f + \frac{1}{\psi} \sum_{j=1}^{\bar{S}} \kappa_z^j \rho_j}{\rho_0 + \sum_{j=1}^{\bar{S}} \kappa_z^j \rho_j}. \quad (\text{IA.54})$$

Using the estimated impulse responses $\{\rho_j\}_{j=0}^{\bar{S}}$, we can infer the risk price of the conditional consumption mean shock given the values of γ and ψ . We can further use these calibrated risk prices to infer the equity risk premium, which equals $\text{cov}(b_f f_t, r_t^{mkt})$. For example, assuming that $\gamma = 10$, $\psi = 1.5$, and there is no stochastic volatility in consumption, the implied annualized equity risk premium is about 2.6% and 4.1%, respectively, in the baseline cross-section of 46 assets and KNS anomalies. If we consider the estimates with stochastic volatilities, the implied annualized equity risk premium is 3.2% and 4.4% in those two cross-sections.

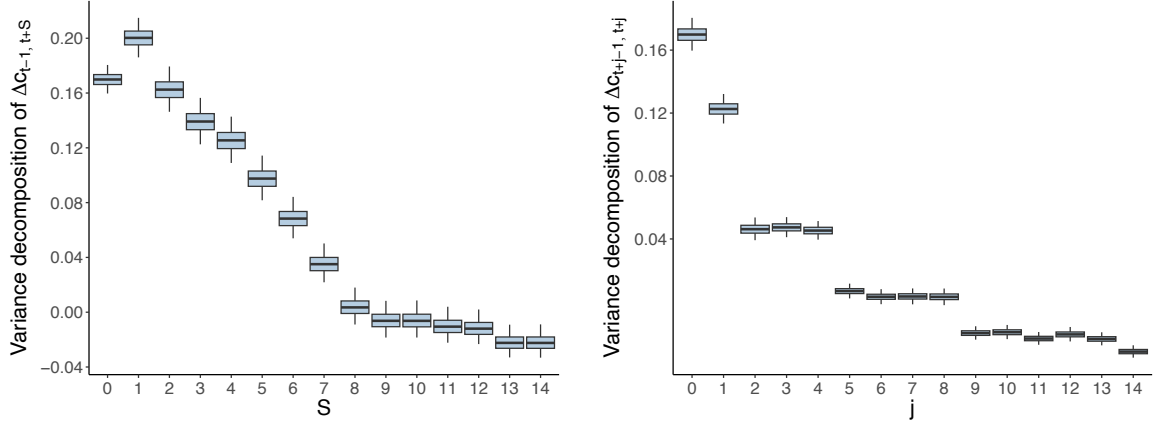
P Additional figures

Figure IA.19: Variance decomposition of asset returns (average of 1,000 simulations).



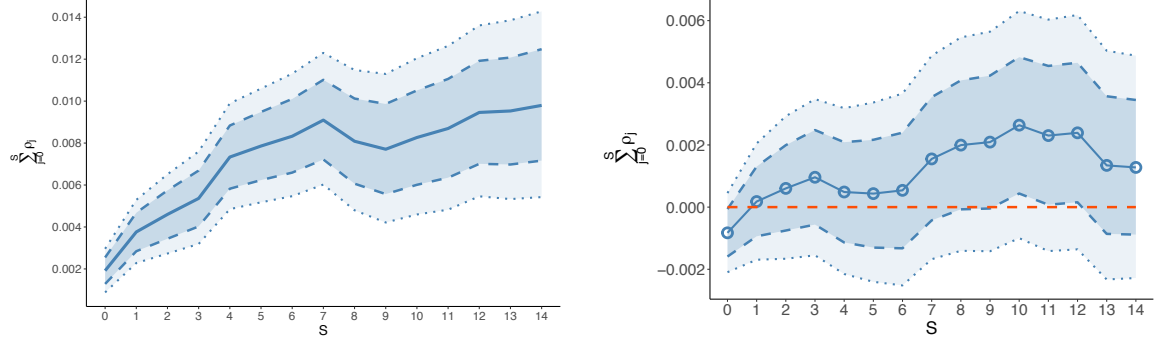
The figure shows the box plots (95% percentiles) of the percentage of time-series variances of individual stock portfolio returns explained by the f component in the one-factor model, as estimated by the state-space model. Red circles denote hypothetical true calibrated values.

Figure IA.20: Share of consumption growth variance driven by its moving average component.



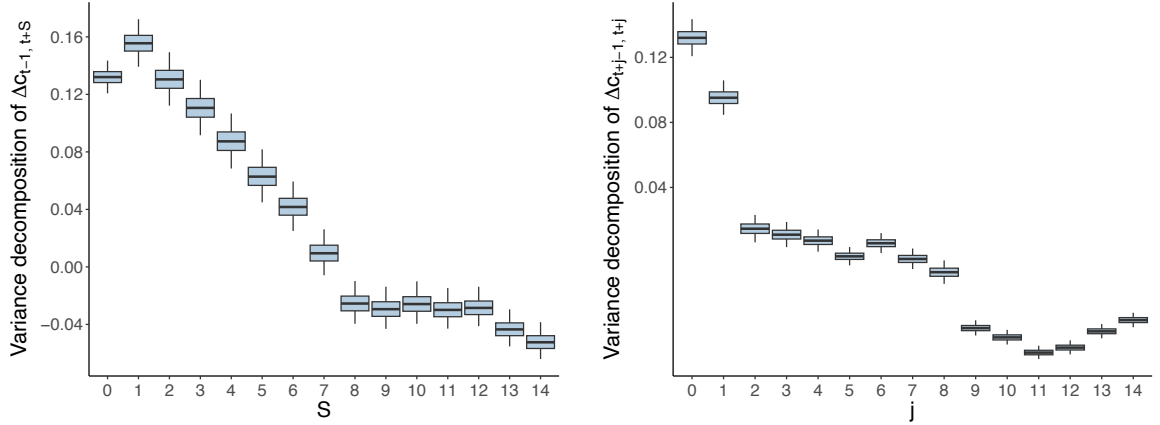
The figure shows the box plots (posterior 95% coverage area) of the percentage of time-series variances of consumption growth explained by the MA component. These plots report the adjusted R-squared, $R_{adj}^2 = 1 - \frac{T-1}{T-1-df}(1 - R^2)$, where df denotes the degree of freedom, which equals 15 in this model. Left panel: Cumulated consumption growth $\Delta c_{t-1,t+S}$. Right panel: One-period consumption growth $\Delta c_{t-1,t+j}$. We study a single-factor model in equations (2) and (3), with $\bar{S} = 14$. The cross-section of test assets includes 25 size-and-value-sorted portfolios, 12 industry portfolios, and nine bond portfolios.

Figure IA.21: Consumption growth response to the latent factors f_t and b_t shocks.



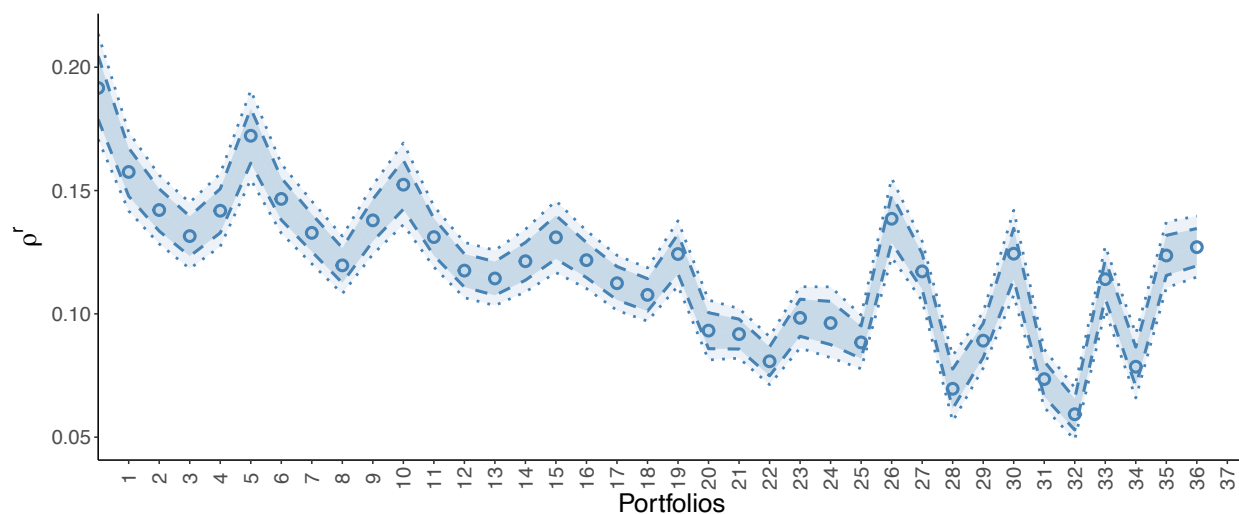
The figure shows the posterior means (continuous solid line) and centered posterior 90% (dashed line) and 68% (dotted line) coverage regions. The estimation is based on the two-factor model in equations (5) and (6). Left panel: Cumulated consumption response to common factor (f_t) shocks. Right panel: Cumulated consumption response to bond factor (b_t) shocks.

Figure IA.22: Variance of consumption growth explained by the MA components f and b .



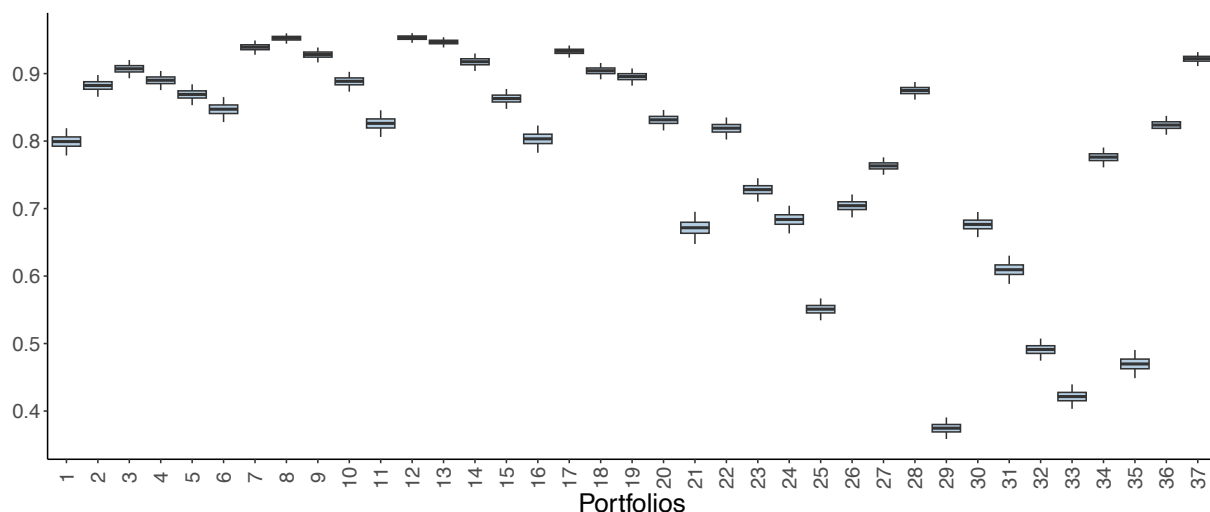
The figure shows the box plots (posterior 95% coverage area) of the percentage of time-series variances of consumption growth explained by the MA components f and b . These plots report the adjusted R-squared, $R_{adj}^2 = 1 - \frac{T-1}{T-1-df}(1-R^2)$, where df denotes the degree of freedom, which equals 30 in this model. Left panel: Cumulated consumption growth $\Delta c_{t-1,t+s}$. Right panel: One-period consumption growth $\Delta c_{t+j-1,t+j}$.

Figure IA.23: Common factor loadings (ρ^r) of the stock portfolios in the two-factor model.



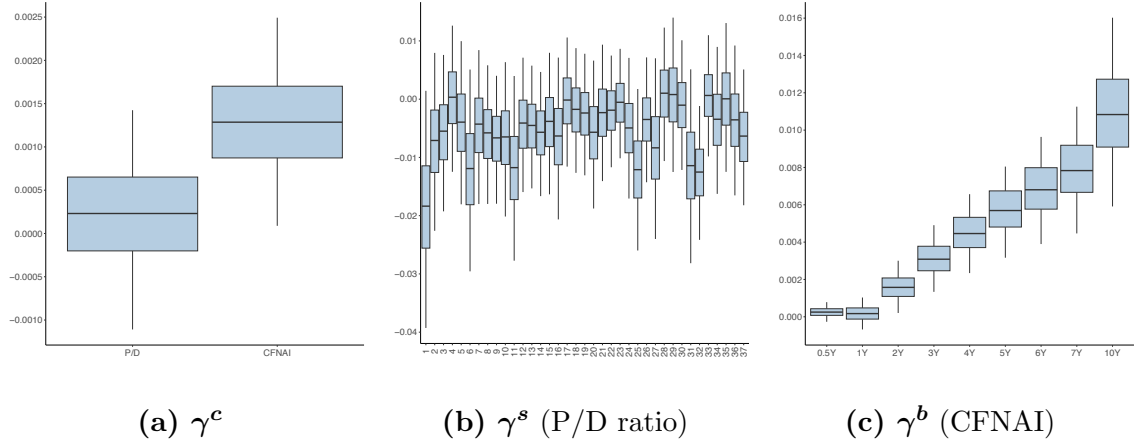
The graph presents posterior means of the stocks factor loadings on f_t (circles) and centered posterior 90% (dashed line) and 68% (dotted line) coverage regions in the two-factor model. Ordering of portfolios: 25 Fama and French (1992) size- and book-to-market-sorted portfolios (e.g., portfolio 2 is the smallest decile of size and the second smaller decile of book-to-market ratio) and 12 industry portfolios.

Figure IA.24: Share of stock portfolios' return variance explained by the f component in the two-factor model.



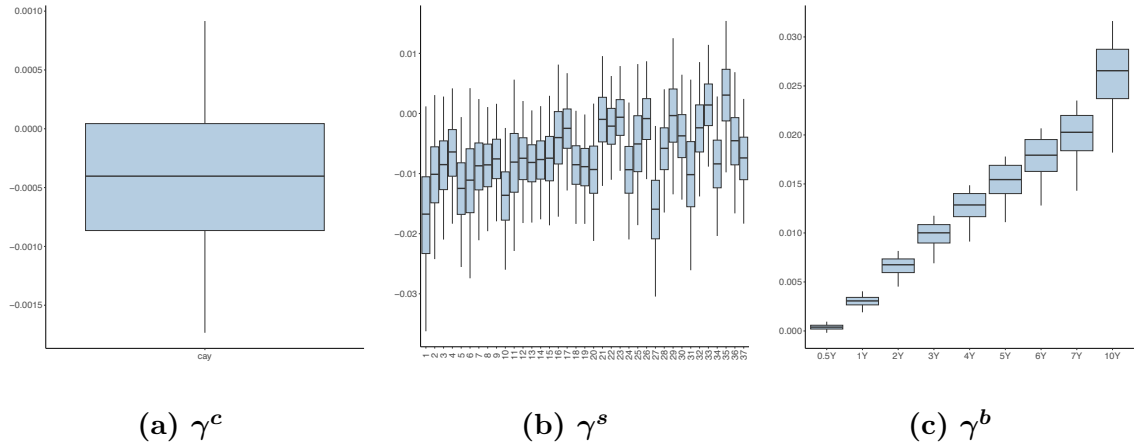
The figure shows the box plots (posterior 95% coverage area) of the percentage of time-series variances of individual stock portfolio returns explained by the f component in the two-factor model. Ordering of portfolios: 25 Fama and French (1992) size- and book-to-market-sorted portfolios (e.g., portfolio 2 is the smallest decile of size and the second smaller decile of book-to-market ratio) and 12 industry portfolios.

Figure IA.25: Box plots of γ^c , γ^s , and γ^b in equations (13)–(15).



This figure plots the distribution of the predictors' coefficient estimates in equations (13)–(15).

Figure IA.26: Box plots of *cay*.



The two-factor model is as follows:

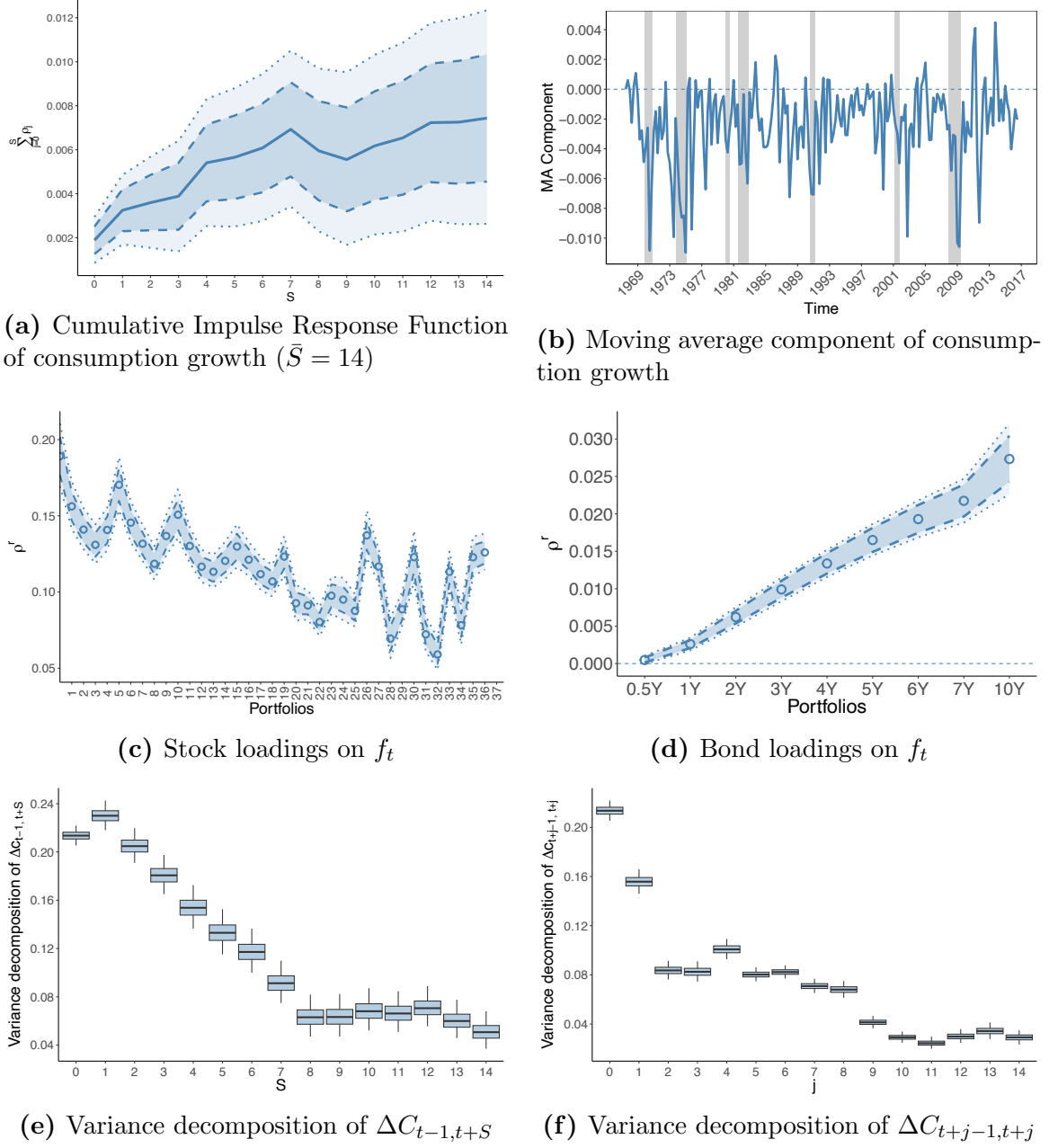
$$\Delta c_{t-1,t} = \mu_c + \sum_{j=0}^{\bar{S}} \rho_j f_{t-j} + \sum_{j=0}^{\bar{S}} \theta_j b_{t-j} + \gamma^c \text{cay}_{t-1} + w_t^c,$$

Stocks: $r_{st}^e = \mu_{sr} + \rho^{sr} f_t + \gamma^s \text{cay}_{t-1} + w_t^{sr}$, and

Bonds: $r_{bt}^e = \mu_{br} + \rho^{br} f_t + \theta^b b_t + \gamma^b \text{cay}_{t-1} + w_t^{br}$,

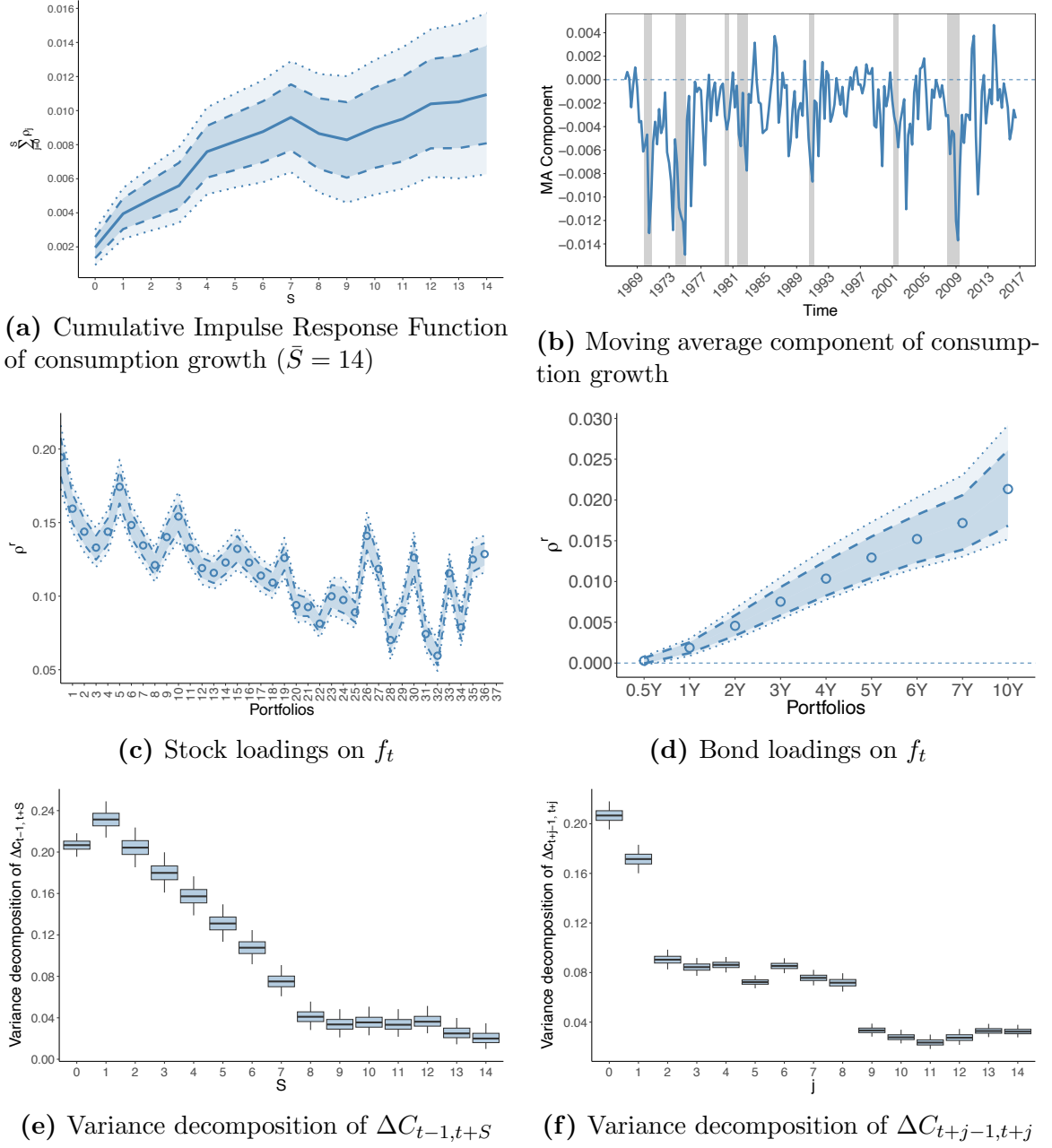
where cay_{t-1} comes from Lettau and Ludvigson (2001).

Figure IA.27: Plots of the two-factor model with P/D ratio and CFNAI in Section III.4.



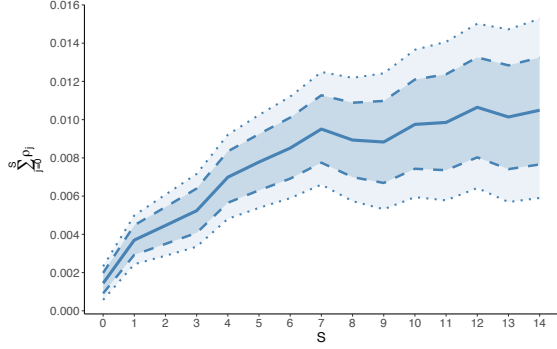
The figure plots (a) cumulative impulse response function of consumption growth ($\bar{S} = 14$), (b) moving average component of consumption growth, (c) stock loadings on f_t , (d) bond loadings on f_t , (e) variance decomposition of $\Delta C_{t-1,t+S}$, and (f) variance decomposition of $\Delta C_{t+j-1,t+j}$. We consider a two-factor model for asset returns, including the market's P/D ratio and the CFNAI index as the predictors for stock and bond portfolios, respectively. The cross-section of test assets includes 25 size-and-value-sorted portfolios, 12 industry portfolios, and nine bond portfolios. Details can be found in Section III.4.

Figure IA.28: Plots of the two-factor model with *cay* in Section III.4.

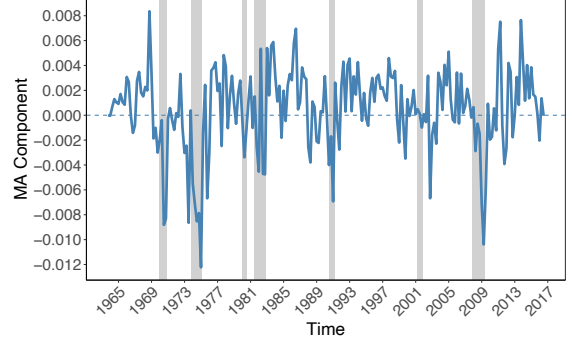


The figure plots (a) cumulative impulse response function of consumption growth ($\bar{S} = 14$), (b) moving average component of consumption growth, (c) stock loadings on f_t , (d) bond loadings on f_t , (e) variance decomposition of $\Delta C_{t-1,t+S}$, and (f) variance decomposition of $\Delta C_{t+j-1,t+j}$. We consider a two-factor model for asset returns, including *cay* in Lettau and Ludvigson (2001) as the predictors for stock and bond portfolios, respectively. The cross-section of test assets includes 25 size-and-value-sorted portfolios, 12 industry portfolios, and nine bond portfolios. Details can be found in Section III.4.

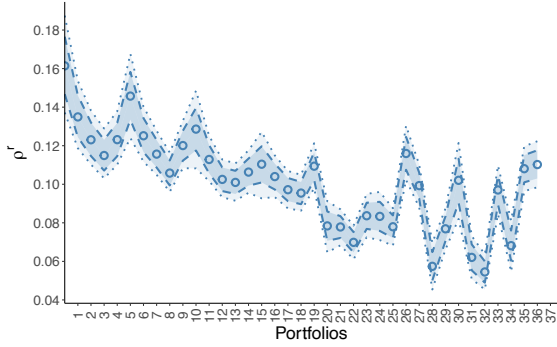
Figure IA.29: Plots of a five-factor model in Section III.5.



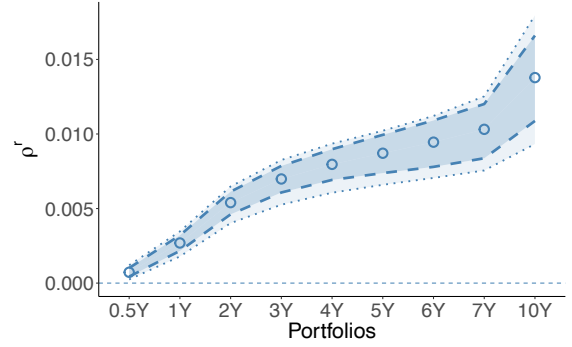
(a) Cumulative Impulse Response Function of consumption growth ($\bar{S} = 14$)



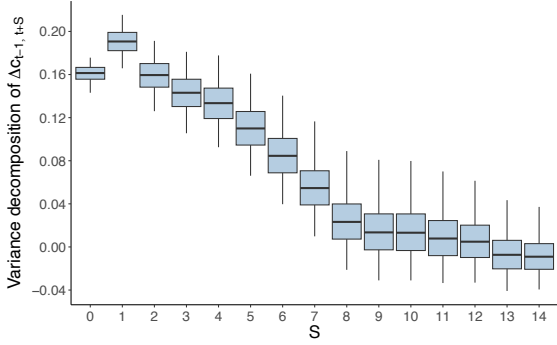
(b) Moving average component of consumption growth



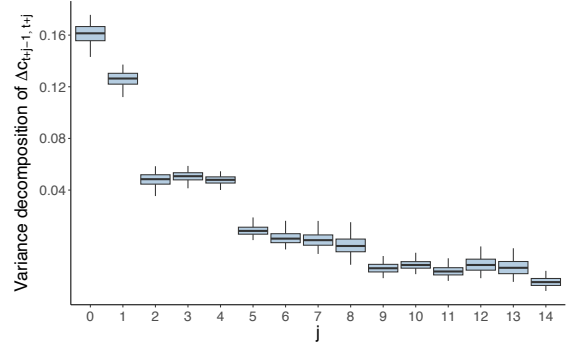
(c) Stock loadings on f_t



(d) Bond loadings on f_t



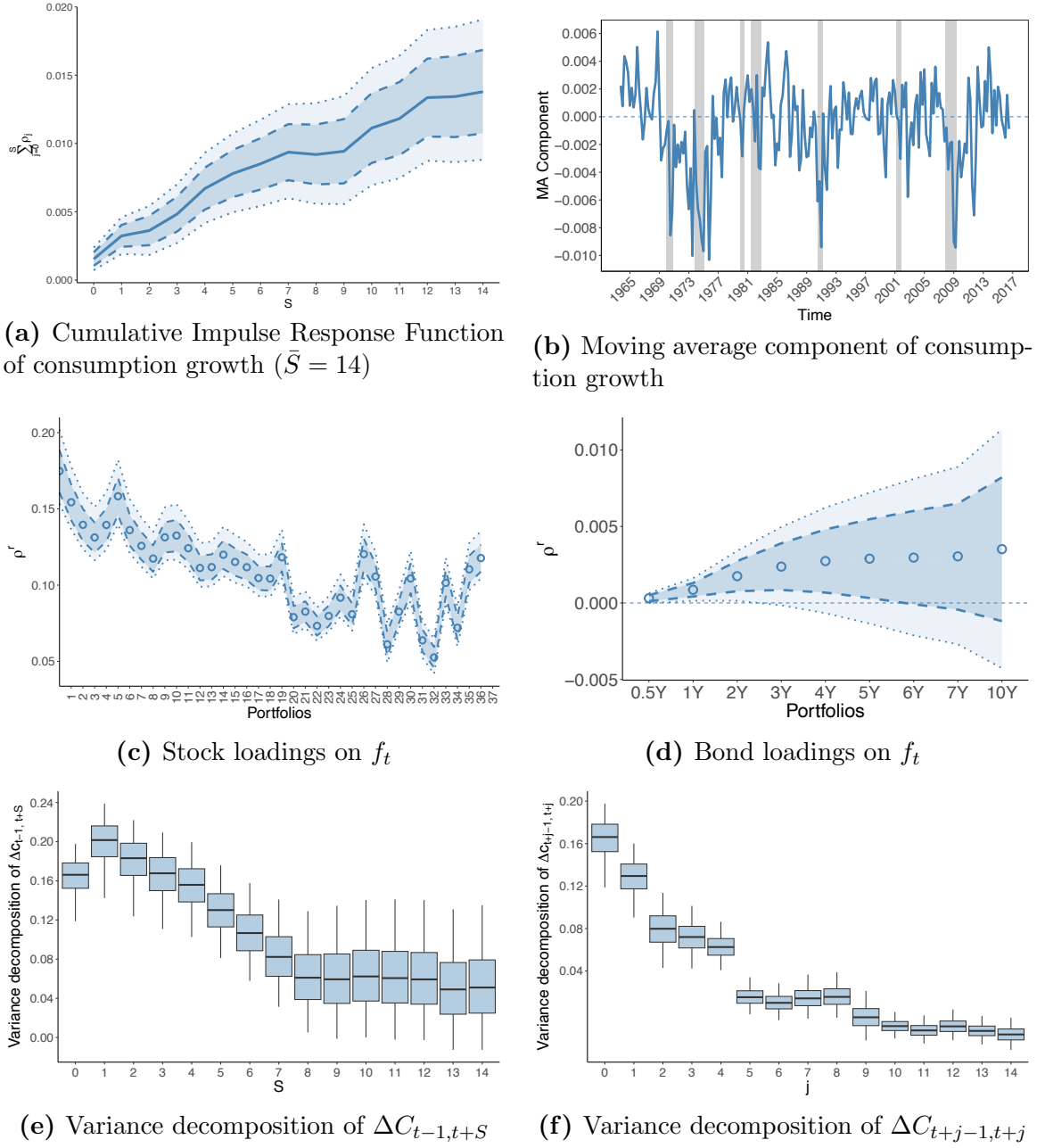
(e) Variance decomposition of $\Delta C_{t-1,t+S}$



(f) Variance decomposition of $\Delta C_{t+j-1,t+j}$

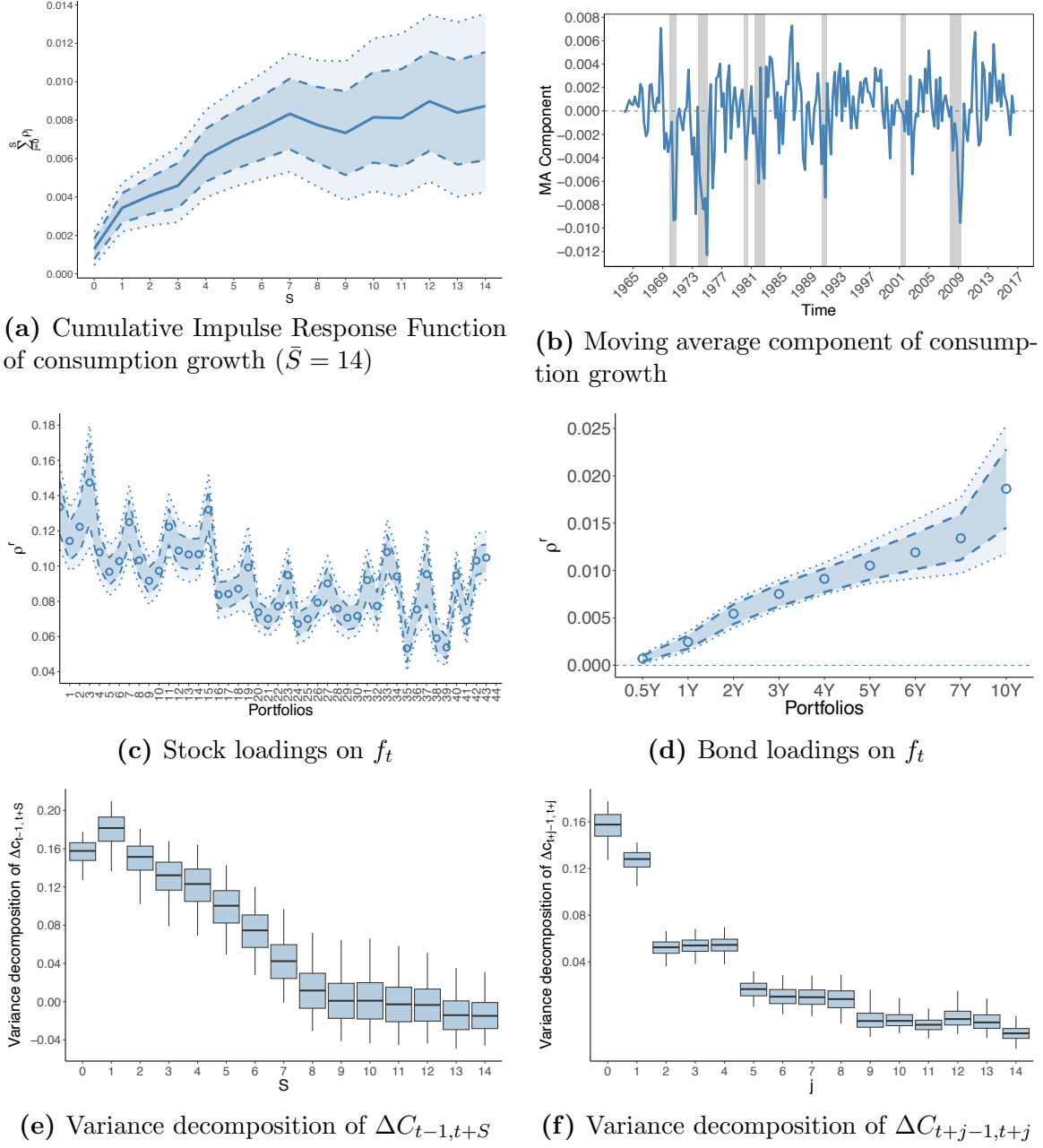
The figure plots (a) cumulative impulse response function of consumption growth ($\bar{S} = 14$), (b) moving average component of consumption growth, (c) stock loadings on f_t , (d) bond loadings on f_t , (e) variance decomposition of $\Delta C_{t-1,t+S}$, and (f) variance decomposition of $\Delta C_{t+j-1,t+j}$. We consider a five-factor model for asset returns. The cross-section of test assets includes 25 size-and-value-sorted portfolios, 12 industry portfolios, and nine bond portfolios. Details can be found in Section III.5.

Figure IA.30: Plots under the stochastic volatility assumption in Section III.6.



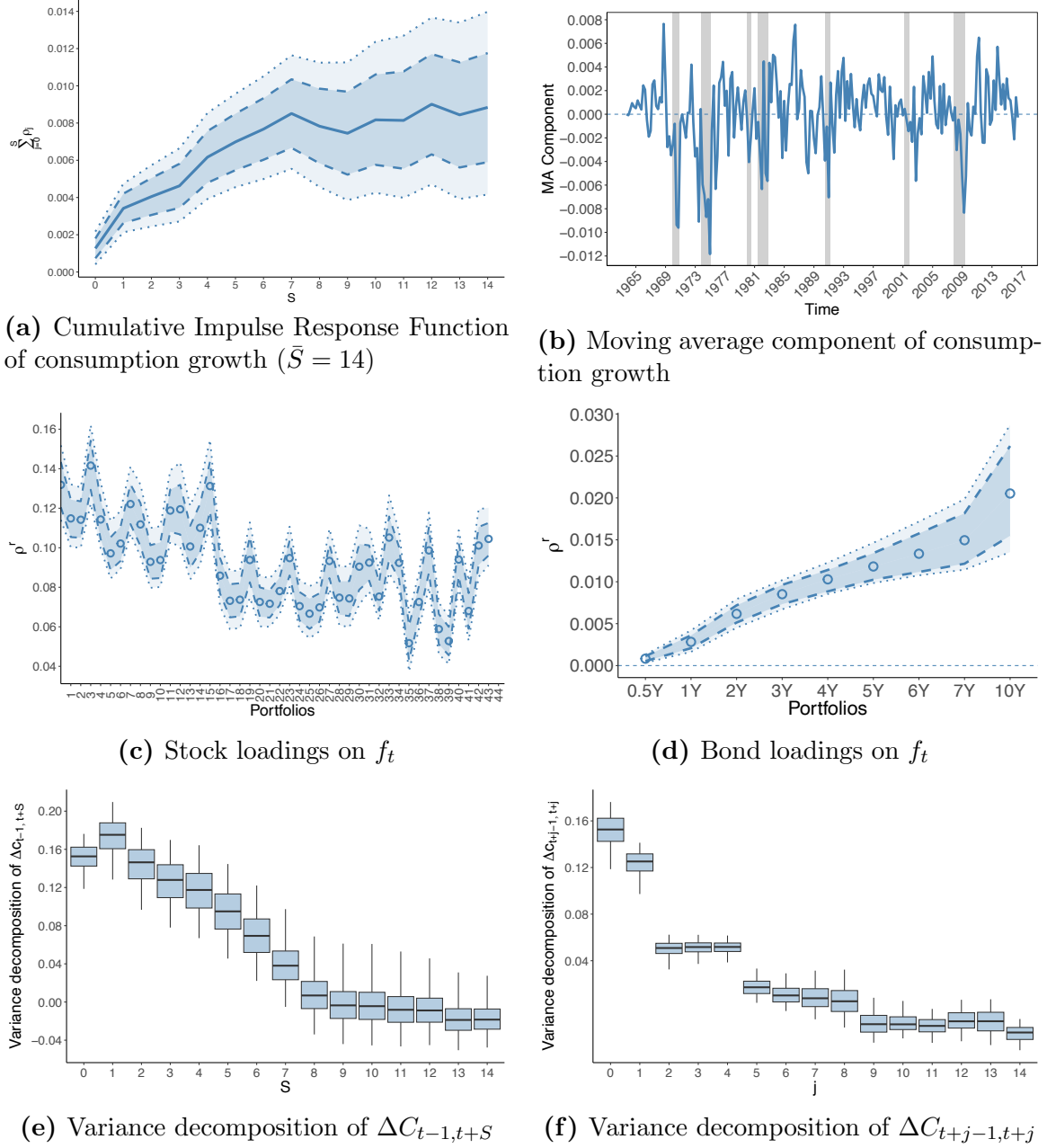
The figure plots (a) cumulative impulse response function of consumption growth ($\bar{S} = 14$), (b) moving average component of consumption growth, (c) stock loadings on f_t , (d) bond loadings on f_t , (e) variance decomposition of $\Delta C_{t-1,t+S}$, and (f) variance decomposition of $\Delta C_{t+j-1,t+j}$. We assume different stochastic volatility processes for f_t , w_t^c , and w_t^r . The cross-section of test assets includes 25 size-and-value-sorted portfolios, 12 industry portfolios, and nine bond portfolios. Details can be found in Section III.6.

Figure IA.31: Robustness check: 32 size-profitability-investment-sorted portfolios, 12 industry portfolios, and nine bond portfolios in a five-factor model.



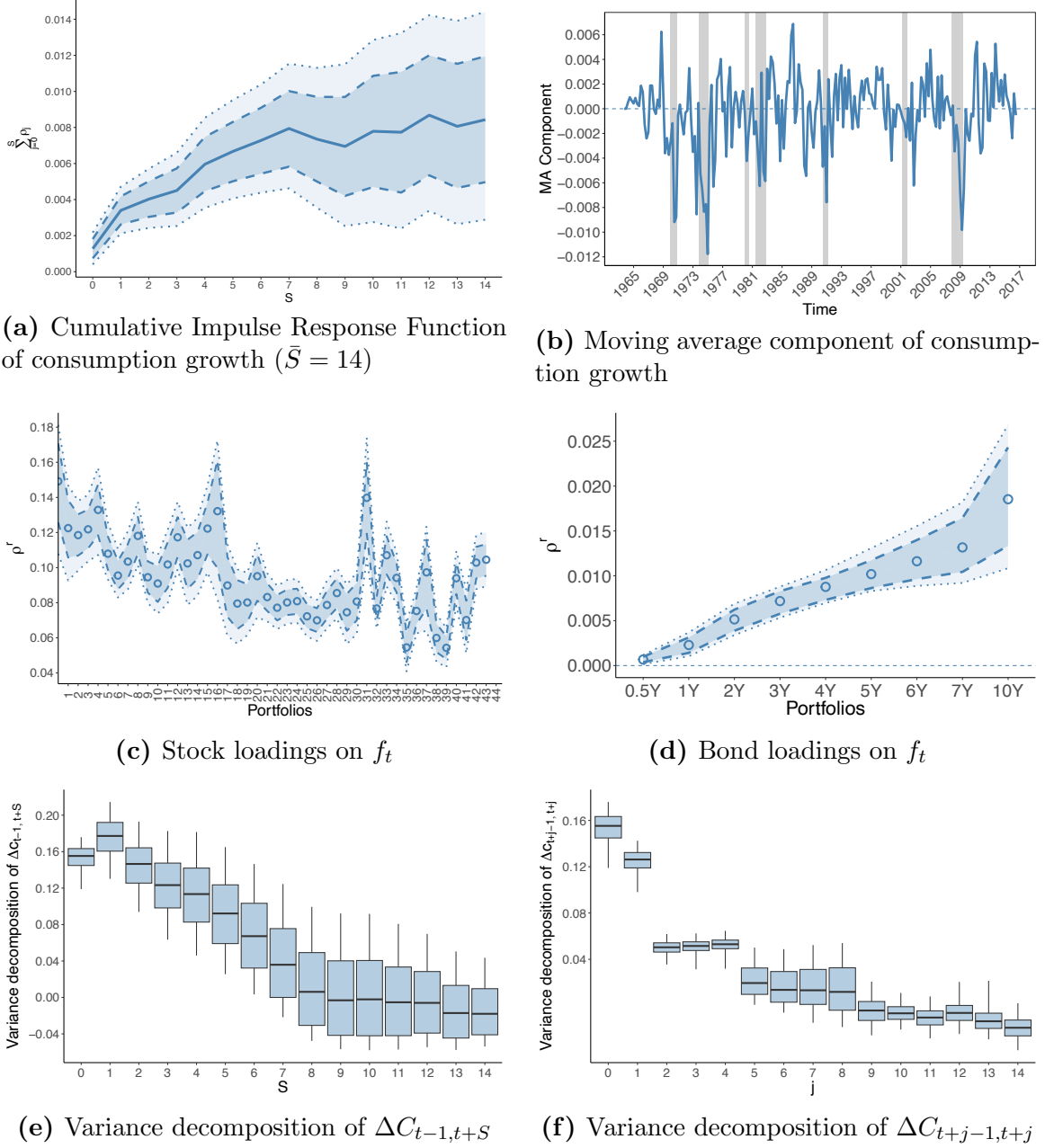
The figure plots (a) cumulative impulse response function of consumption growth ($\bar{S} = 14$), (b) moving average component of consumption growth, (c) stock loadings on f_t , (d) bond loadings on f_t , (e) variance decomposition of $\Delta C_{t-1,t+S}$, and (f) variance decomposition of $\Delta C_{t+j-1,t+j}$. We study the cross-section of 32 size-profitability-investment-sorted portfolios, 12 industry portfolios, and nine bond portfolios. Moreover, asset returns are modeled using a five-factor model, with $\bar{S} = 14$.

Figure IA.32: Robustness check: 32 size-value-investment-sorted portfolios, 12 industry portfolios, and nine bond portfolios in a five-factor model.



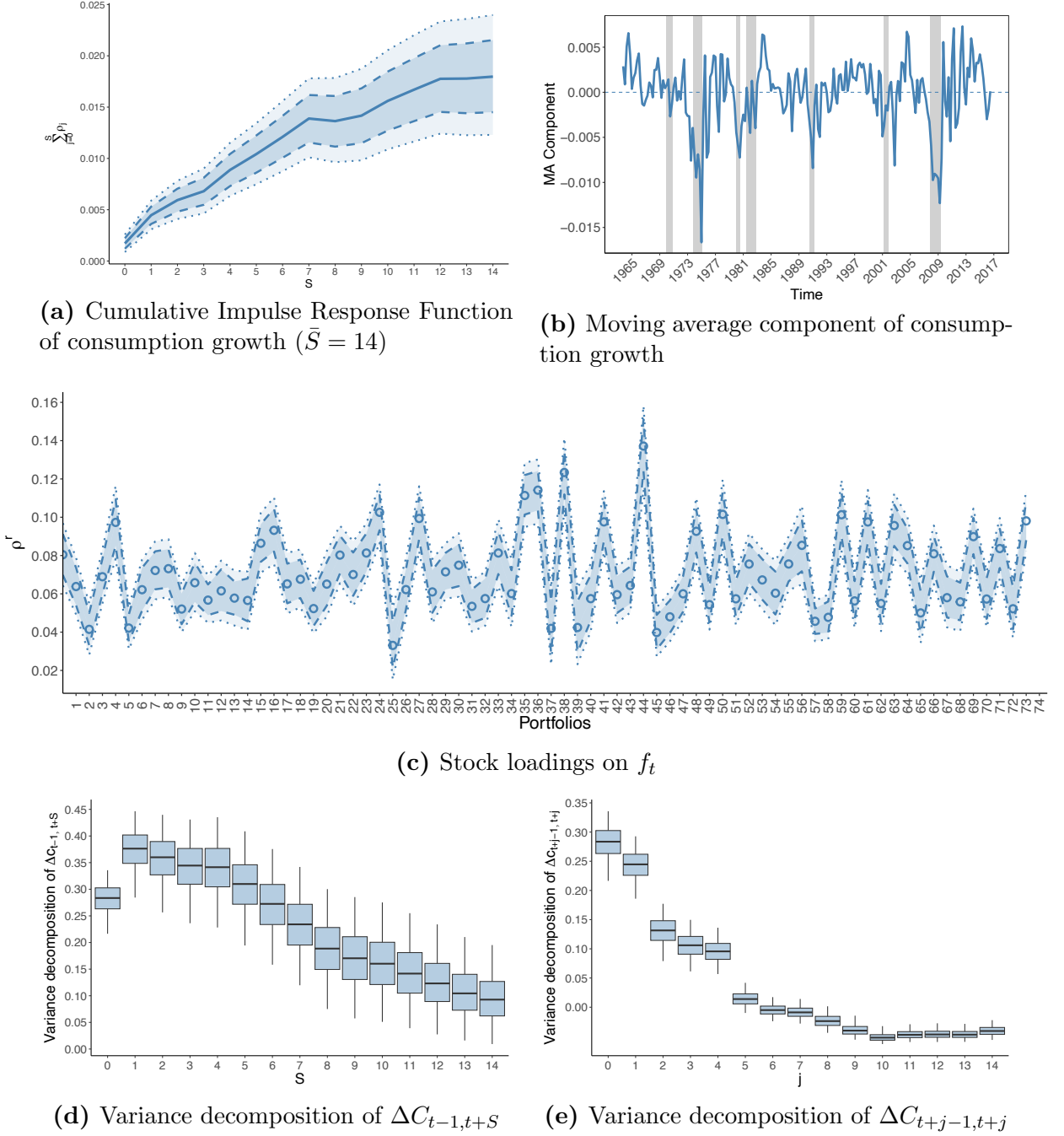
The figure plots (a) cumulative impulse response function of consumption growth ($\bar{S} = 14$), (b) moving average component of consumption growth, (c) stock loadings on f_t , (d) bond loadings on f_t , (e) variance decomposition of $\Delta C_{t-1,t+S}$, and (f) variance decomposition of $\Delta C_{t+j-1,t+j}$. We study the cross-section of 32 size-value-investment-sorted portfolios, 12 industry portfolios, and nine bond portfolios. Moreover, asset returns are modeled using a five-factor model, with $\bar{S} = 14$.

Figure IA.33: Robustness check: 32 size-value-profitability-sorted portfolios, 12 industry portfolios, and nine bond portfolios in a five-factor model.



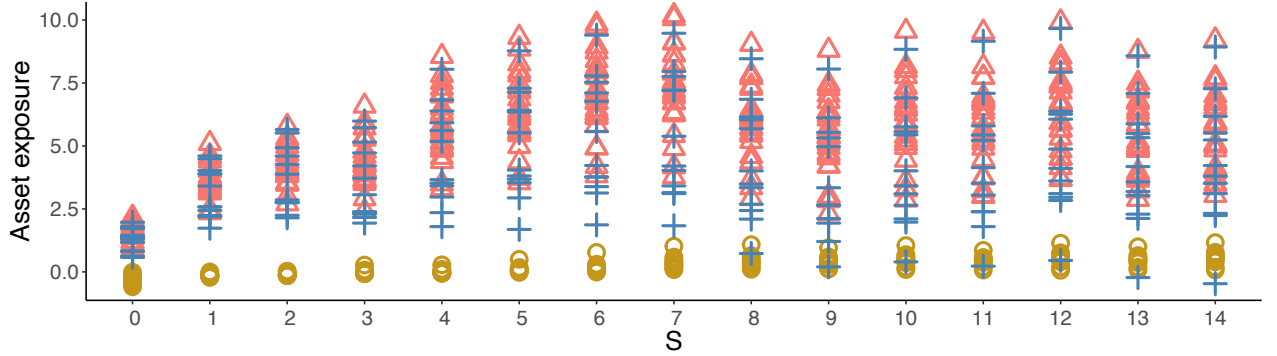
The figure plots (a) cumulative impulse response function of consumption growth ($\bar{S} = 14$), (b) moving average component of consumption growth, (c) stock loadings on f_t , (d) bond loadings on f_t , (e) variance decomposition of $\Delta C_{t-1,t+S}$, and (f) variance decomposition of $\Delta C_{t+j-1,t+j}$. We study the cross-section of 32 size-value-profitability-sorted portfolios, 12 industry portfolios, and nine bond portfolios. Moreover, asset returns are modeled using a five-factor model, with $\bar{S} = 14$.

Figure IA.34: Robustness check: Kozak, Nagel, and Santosh (2020) 74 portfolios in a five-factor model.



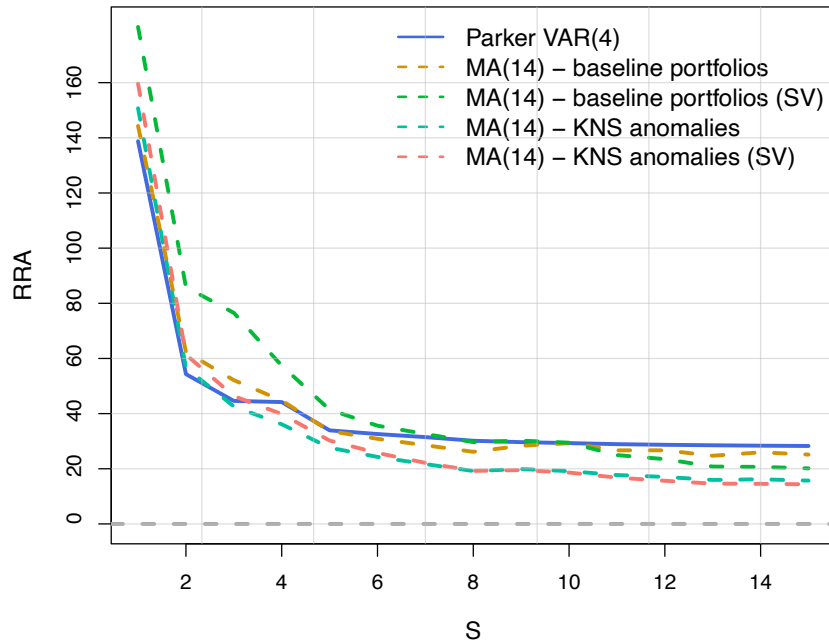
The figure plots (a) cumulative impulse response function of consumption growth ($\bar{S} = 14$), (b) moving average component of consumption growth, (c) stock loadings on f_t , (d) variance decomposition of $\Delta C_{t-1,t+S}$, and (e) variance decomposition of $\Delta C_{t+j-1,t+j}$. We study the cross-section of 74 characteristic-sorted portfolios used in Kozak, Nagel, and Santosh (2020). Specifically, they create value-weighted decile portfolios sorted by firm characteristics, and we use the long and short legs (those in deciles 1 and 10) that have data since July 1963. Hence, we end up with 74 portfolios (both long and short legs of 37 characteristics). Moreover, asset returns are modeled using a five-factor model, with $\bar{S} = 14$.

Figure IA.35: Term structure of consumption exposure.



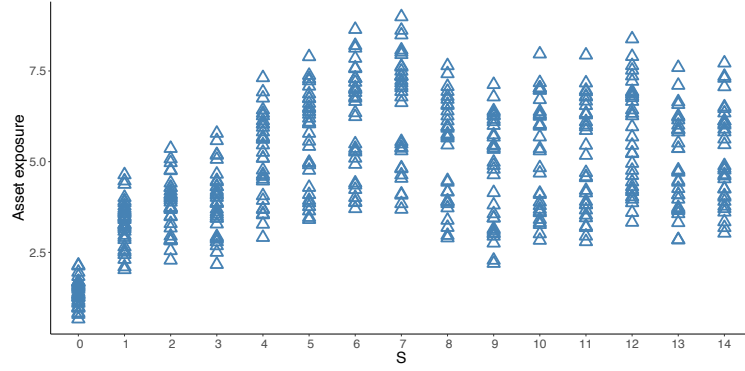
The figure shows the spreads of consumption betas measured as $Cov(\Delta c_{t,t+1+S}, r_{j,t+1}^e)$ for different horizon S (0-14) and asset j : nine bonds (circle), 25 Fama-French size and book-to-market (triangle), and 12 industry (cross) portfolios. The asset exposures are in basis point units.

Figure IA.36: Implied coefficients of relative risk aversion.

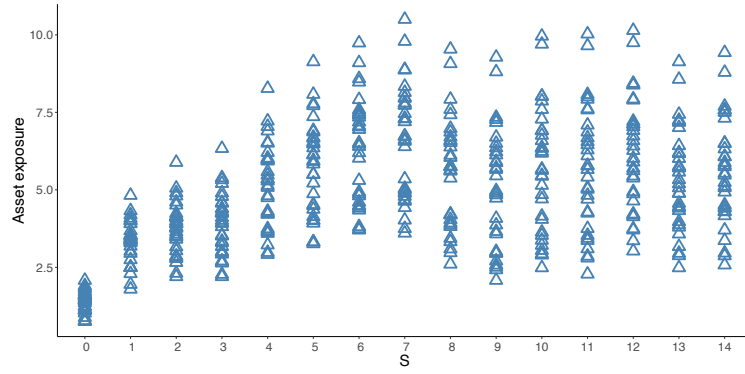


The figure reports the point estimates of the coefficients of relative risk aversion (RRA) implied by our MA(14) state-space formulation and the VAR(4) in Parker (2001). We consider the additively separable CRRA preferences. Details can be found in the footnote of Table 7 and Internet Appendices L and O.

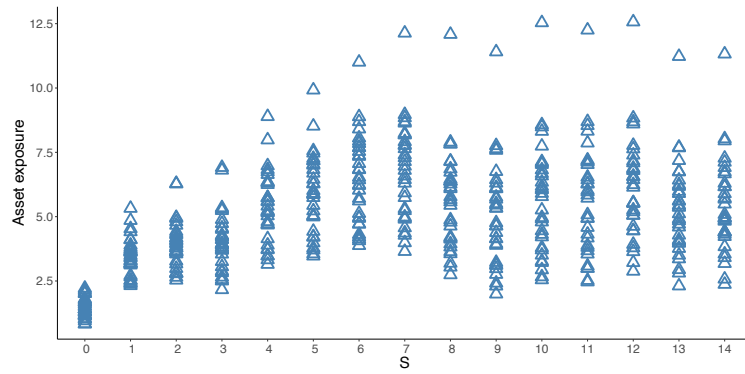
Figure IA.37: Cross-sectional spread of exposure to slow consumption adjustment risk.



(a) 32 size-profitability-investment-sorted portfolios



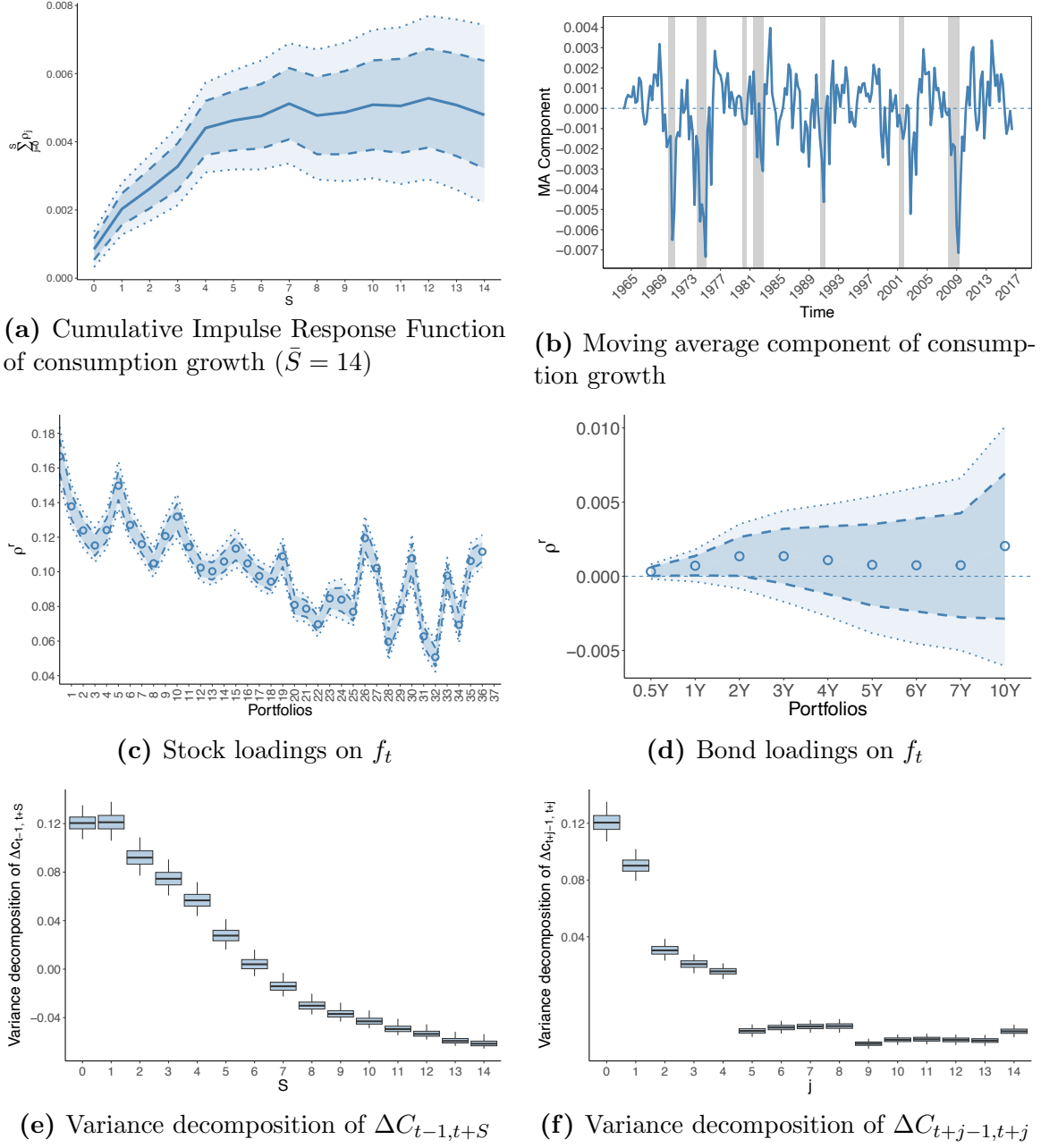
(b) 32 size-value-investment-sorted portfolios



(c) 32 size-value-profitability-sorted portfolios

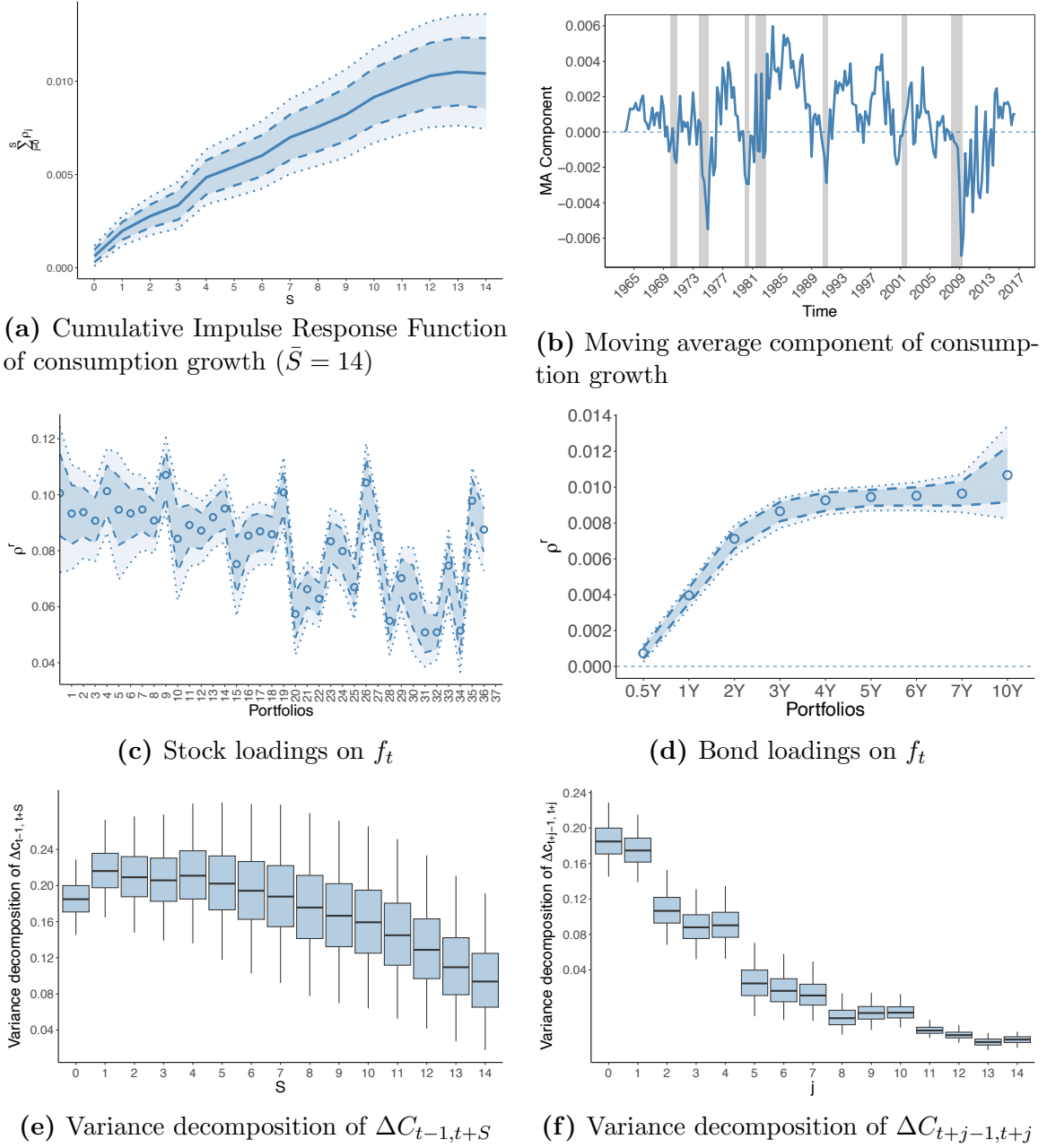
The figure presents the spread of consumption betas measured as $Cov(\Delta c_{t,t+1+S}, r_{j,t+1}^e)$ for different horizon S (0-14) and asset j : (a) 32 size-profitability-investment-sorted portfolios, (b) 32 size-value-investment-sorted portfolios, and (c) 32 size-value-profitability-sorted portfolios. The asset exposures are in basis point units.

Figure IA.38: Robustness check: Modeling the growth rate in nondurable consumption goods plus service in a single-factor model of asset returns.



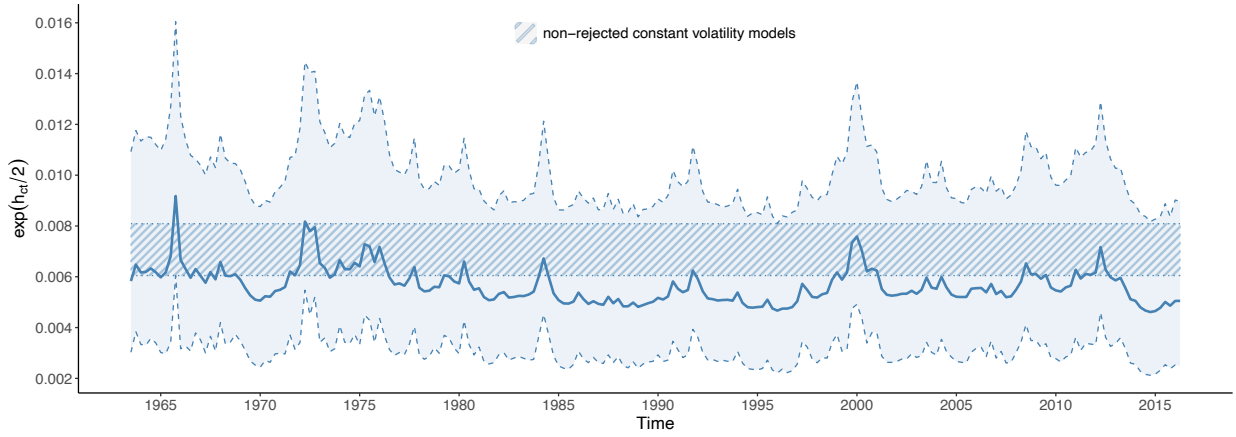
The figure plots (a) cumulative impulse response function of consumption growth ($\bar{S} = 14$), (b) moving average component of consumption growth, (c) stock loadings on f_t , (d) bond loadings on f_t , (e) variance decomposition of $\Delta C_{t-1,t+S}$, and (f) variance decomposition of $\Delta C_{t+j-1,t+j}$. We consider a single-factor model in equations (2) and (3), but $\Delta C_{t-1,t}$ is the quarterly growth rate in nondurable consumption goods and service. Moreover, asset returns are modeled using a single-factor model, with $\bar{S} = 14$. The cross-section of test assets includes 25 size-and-value-sorted portfolios, 12 industry portfolios, and nine bond portfolios.

Figure IA.39: Robustness check: Modeling the growth rate in nondurable consumption goods plus service in a five-factor model of asset returns.

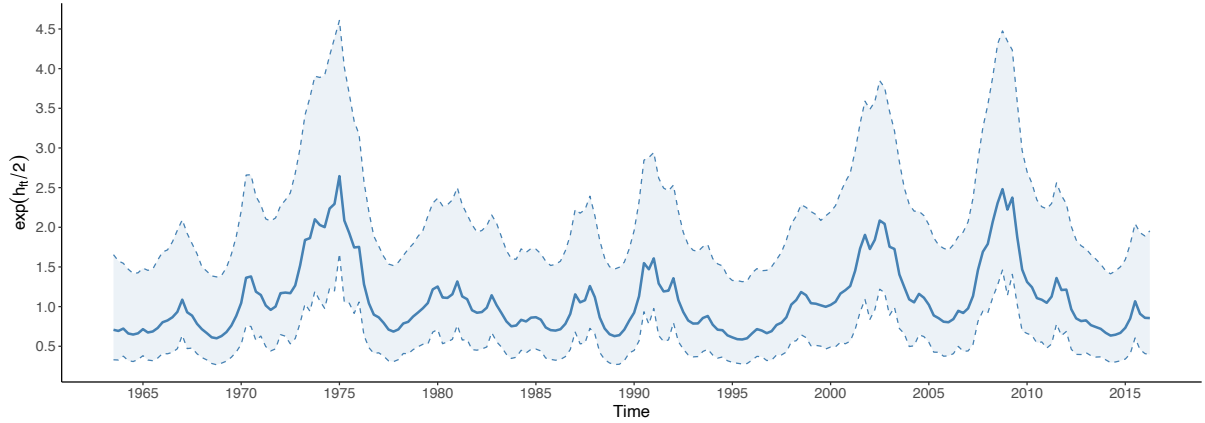


The figure plots (a) cumulative impulse response function of consumption growth ($\bar{S} = 14$), (b) moving average component of consumption growth, (c) stock loadings on f_t , (d) bond loadings on f_t , (e) variance decomposition of $\Delta C_{t-1,t+S}$, and (f) variance decomposition of $\Delta C_{t+j-1,t+j}$. We consider a single-factor model in equations (2) and (3), but $\Delta C_{t-1,t}$ is the quarterly growth rate in nondurable consumption goods and service. Moreover, asset returns are modeled using a five-factor model, with $\bar{S} = 14$. The cross-section of test assets includes 25 size-and-value-sorted portfolios, 12 industry portfolios, and nine bond portfolios.

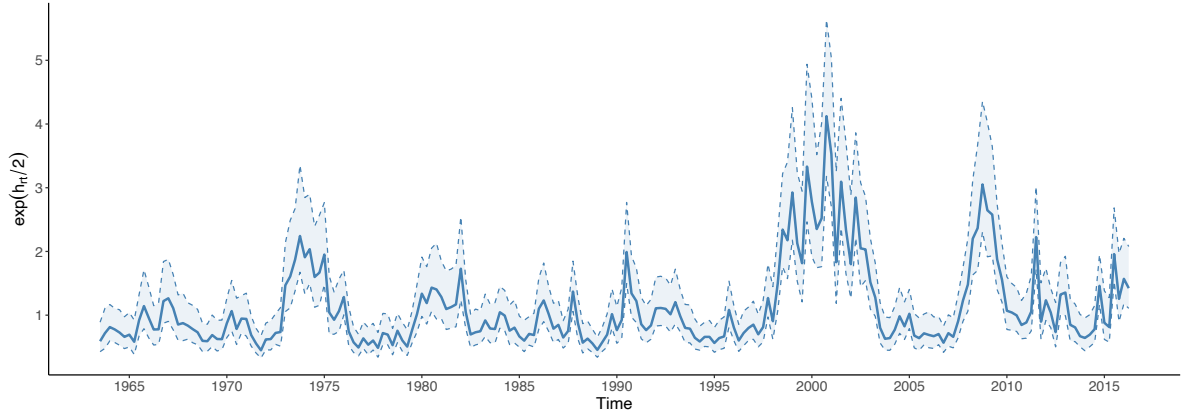
Figure IA.40: Filtered SVs with Kozak, Nagel, and Santosh (2020) 74 portfolios.



Panel A: Log volatility of the idiosyncratic shock to consumption (w_t^e) of equation (2)



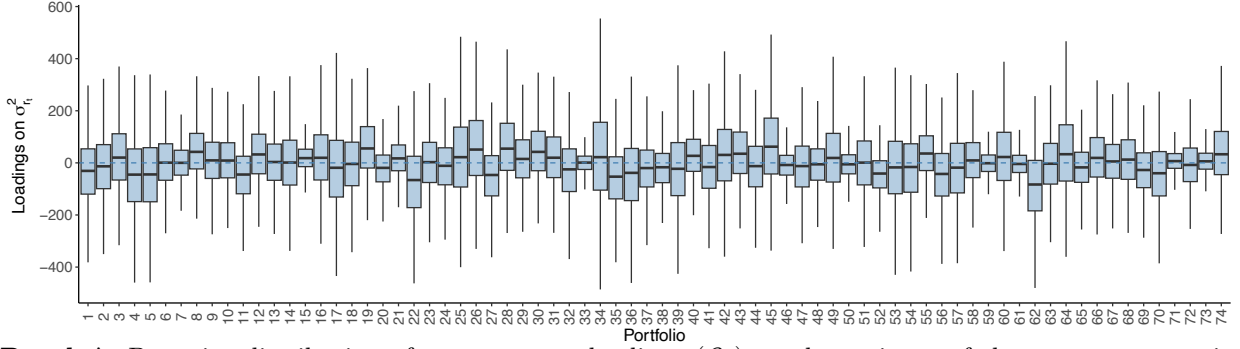
Panel B: Log volatility of the shock to the conditional mean of consumption growth (f_t).



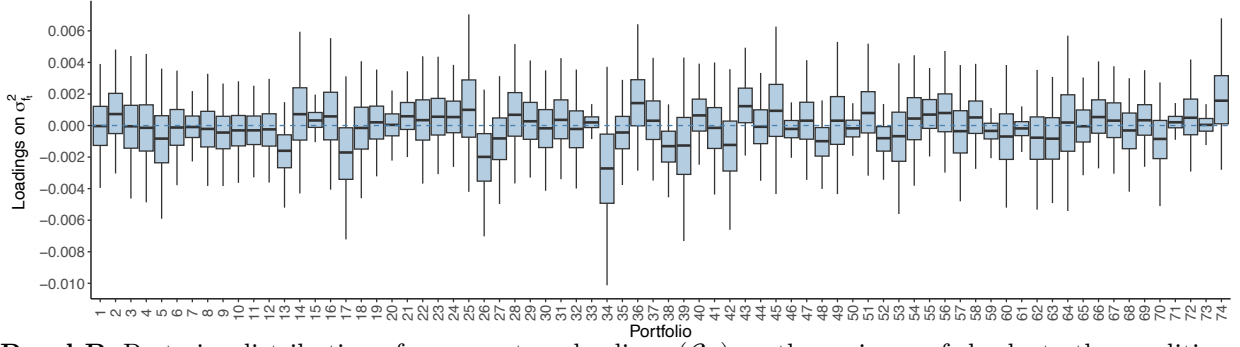
Panel C: Common log volatility of asset return (h_{rt} of equation (19)).

The figure shows the estimated stochastic volatilities of the model in equations (2) and (16)–(19), and Section B.2 under a diffuse prior for the autoregressive volatility coefficients. Solid blue lines depict the posterior median of the log volatility, while dotted red lines denote 2.5% and 97.5% credible intervals. Shaded (patterned) areas reflect constant volatility levels that would not be rejected given the credible intervals. We study the cross-section of 74 characteristic-sorted portfolios used in Kozak, Nagel, and Santosh (2020). Specifically, they create value-weighted decile portfolios sorted by firm characteristics, and we use the long and short legs (those in deciles 1 and 10) that have data since July 1963. Hence, we end up with 74 portfolios (both long and short legs of 37 characteristics).

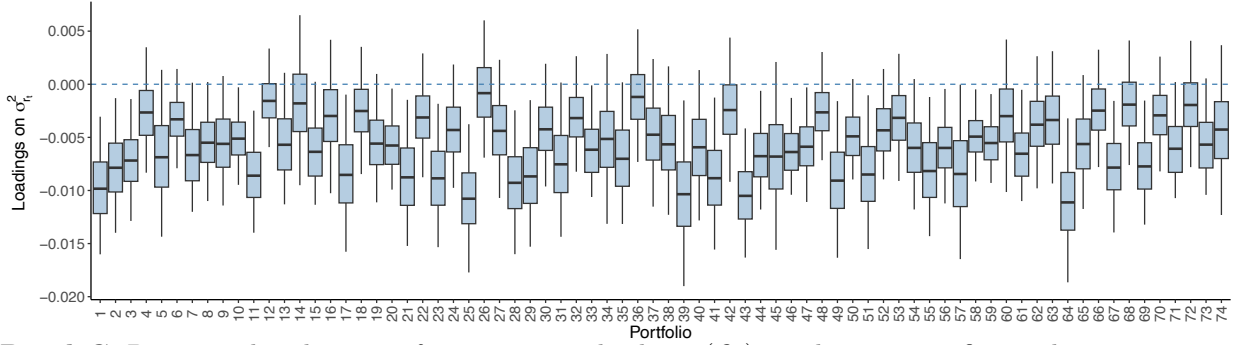
Figure IA.41: Loadings of excess returns on consumption and returns volatilities in Kozak, Nagel, and Santosh (2020) 74 portfolios.



Panel A: Posterior distribution of excess return loadings (β_c) on the variance of short-run consumption shocks ($\sigma_{c,t-1}^2$) in equation (16).



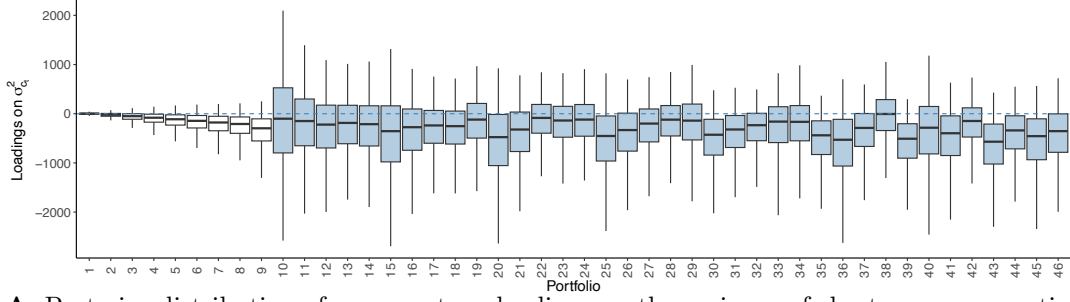
Panel B: Posterior distribution of excess return loadings (β_f) on the variance of shocks to the conditional consumption growth mean ($\sigma_{f,t-1}^2$) in equation (16).



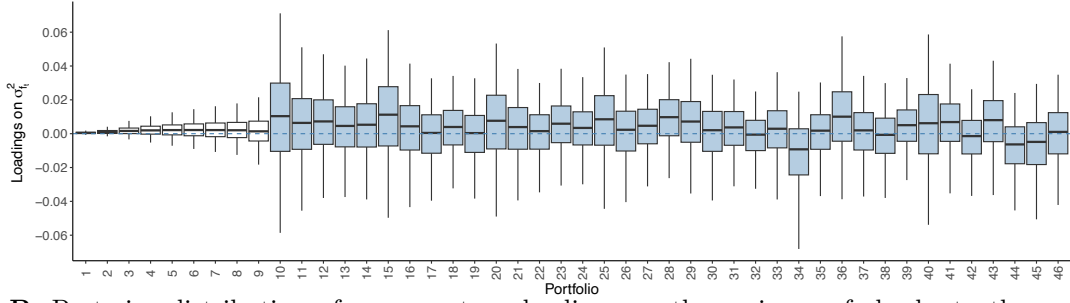
Panel C: Posterior distributions of excess return loadings (β_r) on the common financial return variance ($\sigma_{r,t-1}^2$) in equation (16).

The figure shows the box plots of the posterior distributions of the loadings of portfolio excess returns on the variance of short-run consumption shocks ($\sigma_{c,t-1}^2$), the variance of shocks to the conditional consumption growth ($\sigma_{f,t-1}^2$), and the common financial returns variance ($\sigma_{r,t-1}^2$). We study the cross-section of 74 characteristic-sorted portfolios used in Kozak, Nagel, and Santosh (2020). Specifically, they create value-weighted decile portfolios sorted by firm characteristics, and we use the long and short legs (those in deciles 1 and 10) that have data since July 1963. Hence, we end up with 74 portfolios (both long and short legs of 37 characteristics).

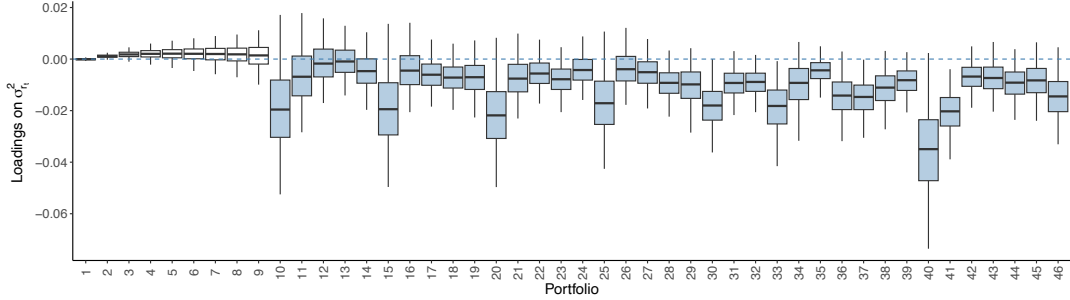
Figure IA.42: Loadings of excess returns on consumption and returns volatility: Predicting cumulative excess returns over the next *four quarters*.



Panel A: Posterior distribution of excess return loadings on the variance of short-run consumption shocks ($\sigma_{c,t-1}^2$).



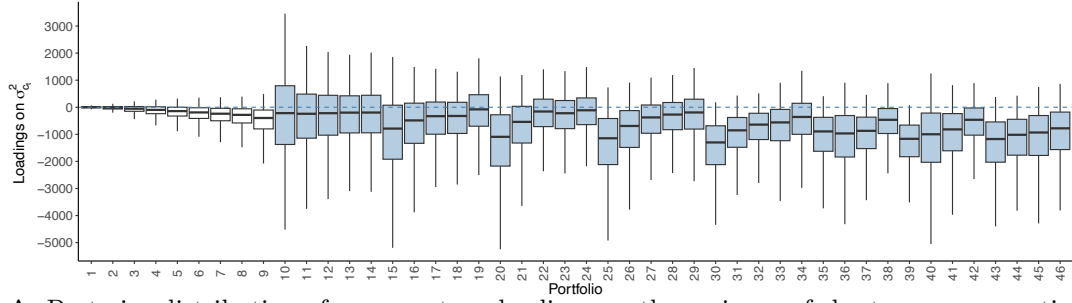
Panel B: Posterior distribution of excess return loadings on the variance of shocks to the conditional consumption growth mean ($\sigma_{f,t-1}^2$).



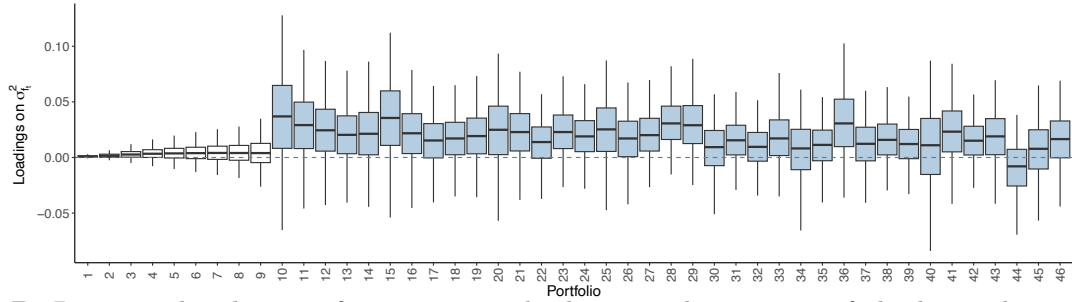
Panel C: Posterior distributions of excess return loadings on the common financial return variance ($\sigma_{r,t-1}^2$).

The figure shows the box plots of the posterior distributions of the loadings of portfolio excess returns on the variance of short-run consumption shocks ($\sigma_{c,t-1}^2$), the variance of shocks to the conditional consumption growth ($\sigma_{f,t-1}^2$), and the common financial returns variance ($\sigma_{r,t-1}^2$). Portfolios are ordered with bonds first (1–9), Fama-French 25 size and book-to-market second (10 – 34), and industry portfolios last. f_t follows the stochastic volatility process with the leverage effect (see equations (IA.25)–(IA.26)).

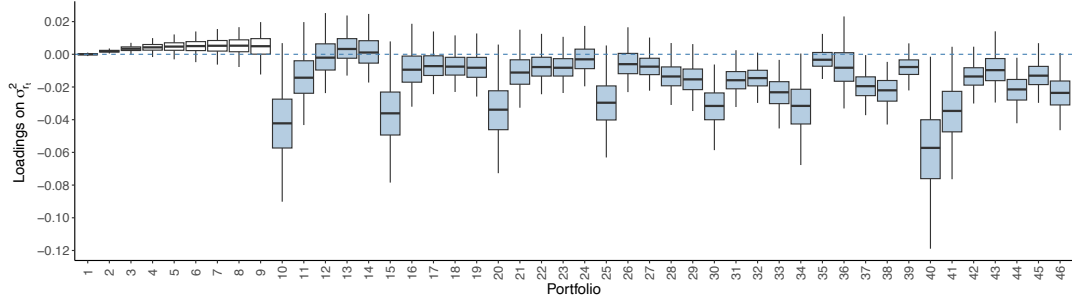
Figure IA.43: Loadings of excess returns on consumption and returns volatility: Predicting cumulative excess returns over the next *eight quarters*.



Panel A: Posterior distribution of excess return loadings on the variance of short-run consumption shocks ($\sigma_{c,t-1}^2$).



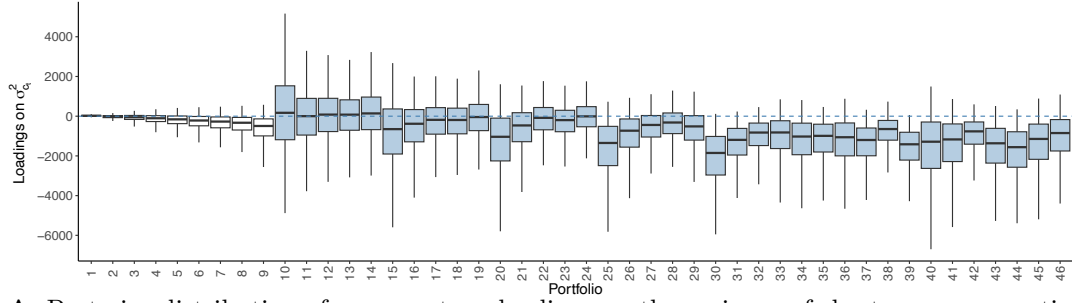
Panel B: Posterior distribution of excess return loadings on the variance of shocks to the conditional consumption growth mean ($\sigma_{f,t-1}^2$).



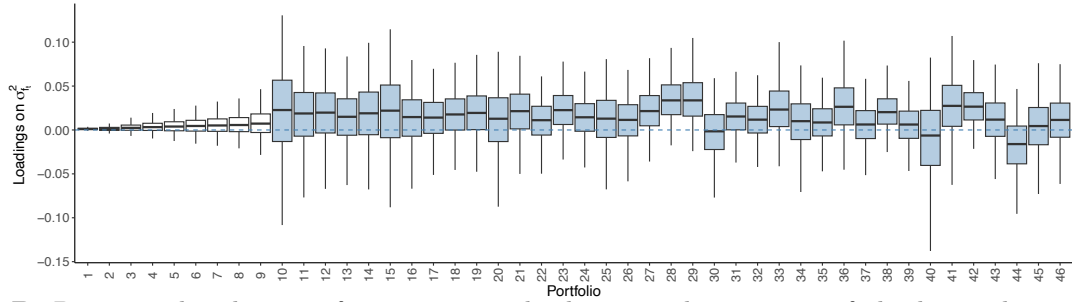
Panel C: Posterior distributions of excess return loadings on the common financial returns variance ($\sigma_{r,t-1}^2$).

This figure shows the box plots of the posterior distributions of the loadings of portfolio excess returns on the variance of short-run consumption shocks ($\sigma_{c,t-1}^2$), the variance of shocks to the conditional consumption growth ($\sigma_{f,t-1}^2$), and the common financial returns variance ($\sigma_{r,t-1}^2$). Portfolios are ordered with bonds first (1–9), Fama-French 25 size and book-to-market second (10–34), and industry portfolios last. f_t follows the stochastic volatility process with the leverage effect (see equations (IA.25)–(IA.26)).

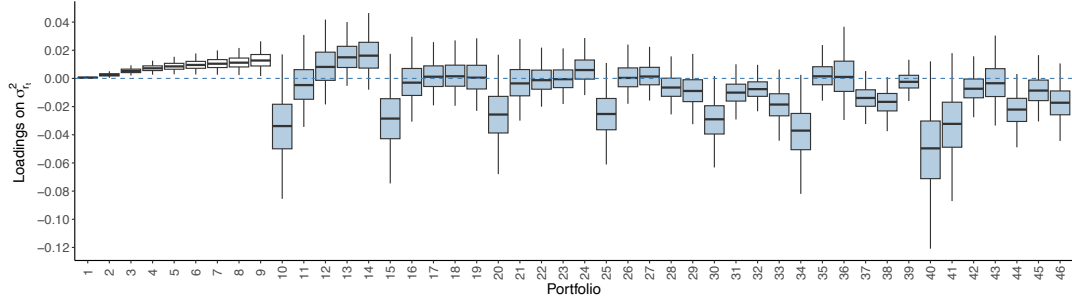
Figure IA.44: Loadings of excess returns on consumption and returns volatility: Predicting cumulative excess returns over the next *12 quarters*.



Panel A: Posterior distribution of excess return loadings on the variance of short-run consumption shocks ($\sigma_{c,t-1}^2$).



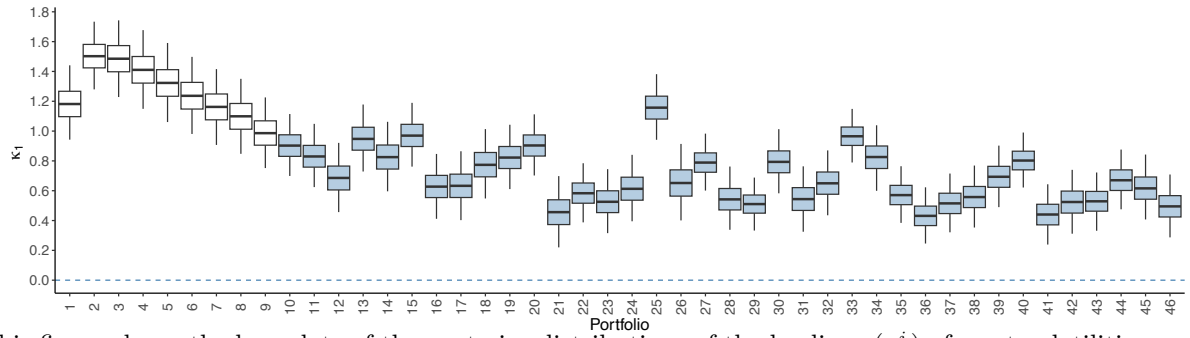
Panel B: Posterior distribution of excess return loadings on the variance of shocks to the conditional consumption growth mean ($\sigma_{f,t-1}^2$).



Panel C: Posterior distributions of excess return loadings on the common financial return variance ($\sigma_{r,t-1}^2$).

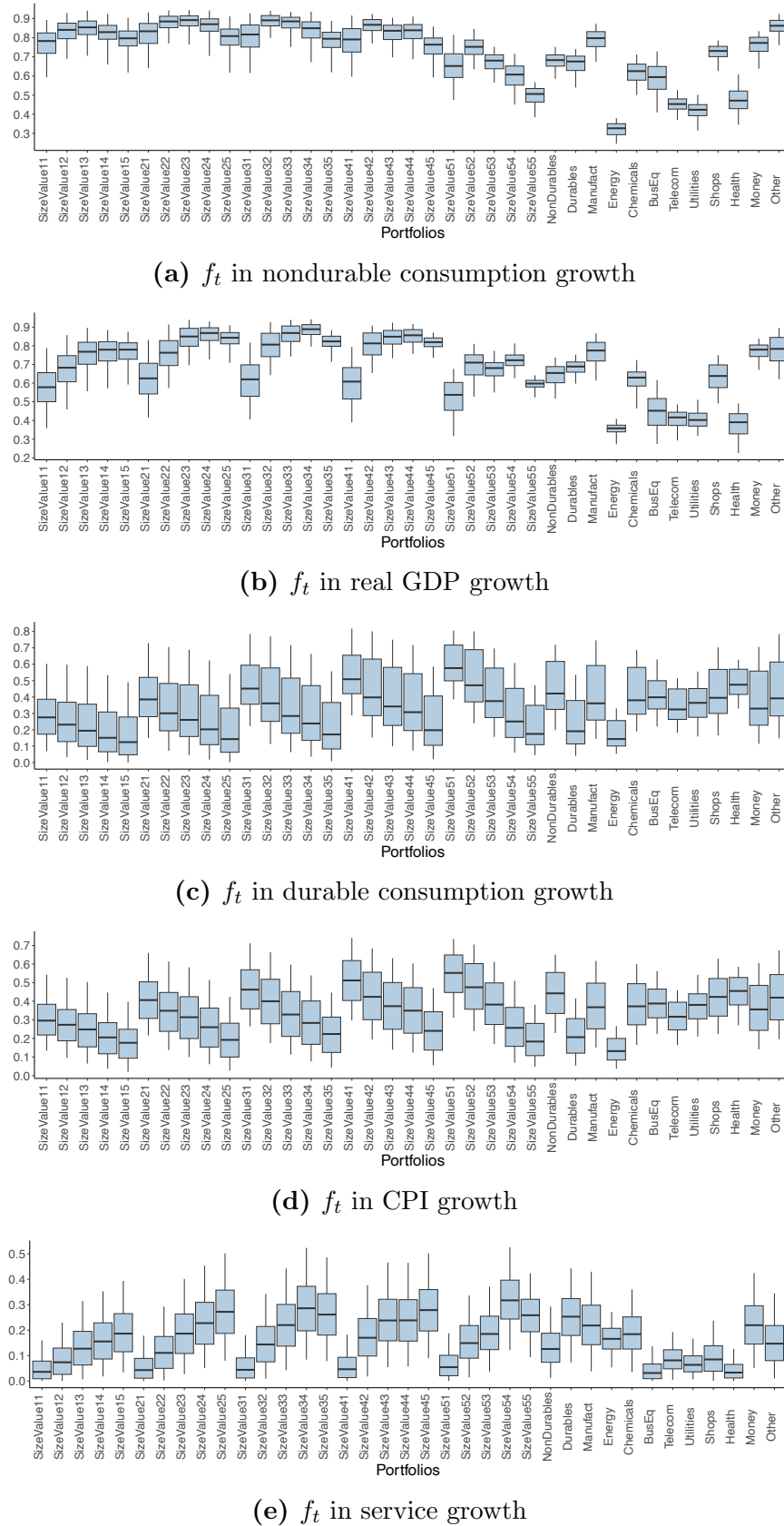
The figure shows the box plots of the posterior distributions of the loadings of portfolio excess returns on the variance of short-run consumption shocks ($\sigma_{c,t-1}^2$), the variance of shocks to the conditional consumption growth ($\sigma_{f,t-1}^2$), and the common financial returns variance ($\sigma_{r,t-1}^2$). Portfolios are ordered with bonds first (1–9), Fama-French 25 size and book-to-market second (10 – 34), and industry portfolios last. f_t follows the stochastic volatility process with the leverage effect (see equations (IA.25)–(IA.26)).

Figure IA.45: Loadings on the common financial market volatility process.



This figure shows the box plots of the posterior distributions of the loadings (κ_1^i) of asset volatilities on the common financial market volatility process. Portfolios are ordered with bonds first (1–9), Fama-French 25 size and book-to-market second (10–34), and industry portfolios last.

Figure IA.46: Share of stock portfolios' return variance explained by the f component in five-factor models.



Box-plots (posterior 95% coverage area) of the percentage of time-series variances of individual stock portfolio returns explained by the f component in the five-factor model for alternative macro variables.

Q Additional tables

Table IA.VII: Predicting nondurable consumption growth using SPF forecasts

$j =$	1	2	3	4
Panel A. Only SPF Forecast				
$\mathbb{E}_t^{SPF}[\Delta c_{t+j-1,t+j}]$	0.957	1.305	0.465	1.029
s.e. (OLS)	(0.224)	(0.315)	(0.373)	(0.425)
s.e. (NW, lag=12)	(0.221)	(0.294)	(0.472)	(0.397)
Predictive R^2	0.117	0.111	0.011	0.041
Panel B. MA Model vs. SPF Forecast				
$\mathbb{E}_t[\Delta c_{t+j-1,t+j}]$	0.580	0.622	0.856	0.829
s.e. (OLS)	(0.165)	(0.213)	(0.218)	(0.214)
s.e. (NW, lag=12)	(0.140)	(0.192)	(0.194)	(0.209)
$\mathbb{E}_t^{SPF}[\Delta c_{t+j-1,t+j}]$	0.718	0.980	0.297	0.935
s.e. (OLS)	(0.225)	(0.326)	(0.357)	(0.406)
s.e. (NW, lag=12)	(0.157)	(0.277)	(0.403)	(0.366)
Predictive R^2	0.190	0.163	0.112	0.135

The table summarizes the regressions in which future realized growth rates of nondurable consumption are forecasted by several predictors. In Panel A, we regress the future one-period nondurable consumption growth on the mean SPF expected consumption growth $\mathbb{E}_t^{SPF}[\Delta c_{t+j-1,t+j}]$, where j ranges from one to four quarters. In Panel B, we further include the conditional consumption mean implied by our MA model (see equation (9)). We report (1) the point estimates of the slope coefficients, (2) the OLS and Newey-West (12 lags) standard errors within the parentheses, and (3) the predictive R^2 . Sample: 1981:Q3–2019:Q4.

Table IA.VIII: Validating the predictability of the conditional consumption mean

$j =$	1	2	3	4	5	6	7	8	9	10	11	12
Panel A. Predicting nondurable consumption growth												
$\widehat{\mathbb{E}}_t[\Delta c_{t+j-1,t+j}]$	0.997	0.964	0.970	0.973	0.946	0.921	0.974	1.119	1.027	0.971	1.039	1.025
s.e. (OLS)	(0.149)	(0.200)	(0.206)	(0.206)	(0.276)	(0.291)	(0.318)	(0.358)	(0.416)	(0.431)	(0.571)	(0.572)
s.e. (NW, lag=12)	(0.213)	(0.212)	(0.211)	(0.217)	(0.203)	(0.219)	(0.185)	(0.202)	(0.314)	(0.343)	(0.446)	(0.446)
HNI_t	0.167	0.161	0.153	0.139	0.149	0.155	0.158	0.168	0.145	0.137	0.133	0.127
s.e. (OLS)	(0.070)	(0.073)	(0.072)	(0.072)	(0.074)	(0.074)	(0.074)	(0.074)	(0.071)	(0.072)	(0.072)	(0.072)
s.e. (NW, lag=12)	(0.081)	(0.094)	(0.095)	(0.101)	(0.116)	(0.117)	(0.118)	(0.119)	(0.112)	(0.108)	(0.105)	(0.103)
Predictive R^2	0.201	0.123	0.117	0.115	0.073	0.068	0.065	0.070	0.048	0.042	0.033	0.031
Panel B. Predicting nondurable plus service consumption growth												
$\widehat{\mathbb{E}}_t[\Delta c_{t+j-1,t+j}]$	1.002	0.893	0.831	0.820	0.678	0.635	0.604	0.517	0.542	0.550	0.469	0.376
s.e. (OLS)	(0.124)	(0.144)	(0.147)	(0.147)	(0.183)	(0.186)	(0.193)	(0.228)	(0.231)	(0.252)	(0.364)	(0.515)
s.e. (NW, lag=12)	(0.132)	(0.123)	(0.140)	(0.160)	(0.226)	(0.251)	(0.237)	(0.274)	(0.296)	(0.301)	(0.406)	(0.578)
HNI_t	0.216	0.208	0.202	0.197	0.210	0.219	0.225	0.231	0.220	0.219	0.221	0.223
s.e. (OLS)	(0.039)	(0.041)	(0.042)	(0.042)	(0.043)	(0.043)	(0.043)	(0.043)	(0.042)	(0.042)	(0.042)	(0.043)
s.e. (NW, lag=12)	(0.049)	(0.048)	(0.047)	(0.049)	(0.054)	(0.053)	(0.053)	(0.054)	(0.054)	(0.055)	(0.056)	(0.056)
Predictive R^2	0.341	0.256	0.230	0.223	0.165	0.165	0.163	0.149	0.147	0.142	0.130	0.125

The table summarizes the regressions in which future realized growth rates of consumption ($\Delta c_{t+j-1,t+j}$, $1 \leq j \leq 12$) are forecasted by several predictors. We regress $\Delta c_{t+j-1,t+j}$ on the conditional consumption mean ($\widehat{\mathbb{E}}_t[\Delta c_{t+j-1,t+j}]$) implied by the MA model and the HN-index in Liu and Matthies (2022) (one-sided HP filter of the news index). Panels A and B report the results for nondurable consumption and nondurable plus service consumption growth. We report (1) the point estimates of the slope coefficients, (2) the OLS and Newey-West (12 lags) standard errors within the parentheses, and (3) the predictive R^2 . Sample: 1963:Q3–2013:Q4. The sample ends at 2013:Q4 due to the availability of the HN-index data.

Table IA.IX: Predicting squared forecast errors of consumption growth

	Market P/D ratio	Financial uncertainty	Macro uncertainty	Real uncertainty
Coefficient	-0.007	0.123	0.069	0.134
s.e. (OLS)	(0.069)	(0.068)	(0.069)	(0.068)
s.e. (NW, lag=12)	(0.068)	(0.080)	(0.079)	(0.051)
R^2	0.000	0.015	0.005	0.018

We regress the squared forecast error, $(\Delta c_{t,t+1} - \hat{\mathbb{E}}_t[\Delta c_{t,t+1}])^2$, on several persistent economic variables measured at time t , including the market price-dividend ratio and financial/real/macro uncertainty measures from Jurado, Ludvigson, and Ng (2015) and Ludvigson, Ma, and Ng (2021). All variables are standardized to have unit variances in the regressions. The uncertainty measures have horizons of three months, consistent with the quarterly frequency of consumption growth.

Table IA.X: Model comparison using log Bayes factors with scaled consumption growth and asset returns: $\text{var}(r_{it}^e) = 1$ for $i = 1, \dots, N$, and $\text{var}(\Delta c_{t-1,t}) = N$

	Models									
	I	II	III	IV	V	VI	VII	VIII	IX	X
Log of Bayes Factor:	0	-156	-38	-49	-87	-233	-165	-439	-234	-551
Posterior Probability.:	100%	0%	0%	0%	0%	0%	0%	0%	0%	0%

Model I: w_t^c , f_t and w_t^r follow SV processes, and $r_t^e = \mu_r + \rho^r f_t + w_t^r$.

Model II: w_t^c , f_t and w_t^r follow SV processes, $r_t^e = \mu_r + \rho^r f_t + w_t^r$, and $h_{ft} = h_{rt}$.

Model III: w_t^c and w_t^r follow SV processes, $f_t \stackrel{\text{iid}}{\sim} \mathcal{N}(0, 1)$, and $r_t^e = \mu_r + \rho^r f_t + w_t^r$.

Model IV: f_t and w_t^r follow SV processes, $w_t^c \stackrel{\text{iid}}{\sim} \mathcal{N}(0, \sigma_c^2)$, and $r_t^e = \mu_r + \rho^r f_t + w_t^r$.

Model V: w_t^r follows SV process, $w_t^c \stackrel{\text{iid}}{\sim} \mathcal{N}(0, \sigma_c^2)$, $f_t \stackrel{\text{iid}}{\sim} \mathcal{N}(0, 1)$, and $r_t^e = \mu_r + \rho^r f_t + w_t^r$.

Model VI: w_t^c , f_t and w_t^r follow SV processes, $r_t^e = \mu_r + \rho^r f_t + \beta_f \sigma_{f,t-1}^2 + w_t^r$, and $h_{ft} = h_{rt}$.

Model VII: w_t^r follows SV process, $w_t^c \stackrel{\text{iid}}{\sim} \mathcal{N}(0, \sigma_c^2)$, $f_t \stackrel{\text{iid}}{\sim} \mathcal{N}(0, 1)$, and $r_t^e = \mu_r + \rho^r f_t + \beta_r \sigma_{r,t-1}^2 + w_t^r$.

Model VIII: w_t^c , f_t and w_t^r follow SV processes, and $r_t^e = \mu_r + \rho^r f_t + \beta_f \sigma_{f,t-1}^2 + \beta_r \sigma_{r,t-1}^2 + w_t^r$.

Model IX: f_t and w_t^r follow SV processes, $w_t^c \stackrel{\text{iid}}{\sim} \mathcal{N}(0, \sigma_c^2)$, and $r_t^e = \mu_r + \rho^r f_t + \beta_f \sigma_{f,t-1}^2 + \beta_r \sigma_{r,t-1}^2 + w_t^r$.

Model X: w_t^c , f_t and w_t^r follow SV processes, and $r_t^e = \mu_r + \rho^r f_t + \beta_c \sigma_{c,t-1}^2 + \beta_f \sigma_{f,t-1}^2 + \beta_r \sigma_{r,t-1}^2 + w_t^r$.

The table summarizes the model comparison for restricted and unrestricted versions of the specification in equations (2) and (16)–(19), and Section B.2. We approximate the Bayes factor using the Schwartz criterion. We use Model I as a benchmark and calculate the (log) odds of each model compared to Model I. A negative number implies that the chosen model is less likely than Model I conditional on the observed data. The model posterior probabilities are computed under the prior of the specifications being all equally likely. Different from Table IA.X, we rescale the variance of consumption growth and asset returns: $\text{var}(r_{it}^e) = 1$ for $i = 1, \dots, N$, and $\text{var}(\Delta c_{t-1,t}) = N$. Under such a normalization, we ensure that we put equal weights to fit consumption growth and asset returns equations.

Table IA.XI: Correlations of stochastic volatility processes and VXO: Kozak, Nagel, and Santosh (2020) 74 portfolios

	Mean	2.5%	5%	50%	95%	97.5%
Panel A: correlations of vol processes with VXO ² index						
$cor(\sigma_{ct}^2, VXO_t^2)$	0.10	-0.12	-0.09	0.09	0.31	0.35
$cor(\sigma_{ft}^2, VXO_t^2)$	0.43	0.22	0.26	0.44	0.58	0.60
$cor(\sigma_{rt}^2, VXO_t^2)$	0.53	0.41	0.44	0.53	0.63	0.65
Panel B: Pairwise correlations of vol processes						
$cor(\sigma_{ct}^2, \sigma_{ft}^2)$	0.07	-0.11	-0.08	0.05	0.27	0.31
$cor(\sigma_{ct}^2, \sigma_{rt}^2)$	0.08	-0.09	-0.07	0.06	0.26	0.31
$cor(\sigma_{ft}^2, \sigma_{rt}^2)$	0.30	0.12	0.14	0.30	0.48	0.52

The table summarises the posterior mean, 2.5%, 5%, 50%, 95%, and 97.5% quantiles of correlation among σ_{ct}^2 , σ_{ft}^2 , σ_{rt}^2 , and the VXO index under a diffuse prior for the autoregressive coefficients of the volatility processes. We study the cross-section of 74 characteristic-sorted portfolios in Kozak, Nagel, and Santosh (2020). Specifically, they create value-weighted decile portfolios sorted by firm characteristics, and we use the long and short legs (those in deciles 1 and 10) that have data since July 1963. Hence, we end up with 74 portfolios (both long and short legs of 37 characteristics).

Table IA.XII: Correlations of stochastic volatility processes and real/macro/financial uncertainty

	Posterior Mean	2.5%	5%	50%	95%	97.5%
Panel A: correlations of vol processes with real uncertainty						
$cor(\sigma_{ct}^2, real_t^2)$	0.22	0.02	0.04	0.22	0.42	0.45
$cor(\sigma_{ft}^2, real_t^2)$	0.26	0.05	0.08	0.26	0.44	0.47
$cor(\sigma_{rt}^2, real_t^2)$	0.36	0.25	0.27	0.37	0.44	0.46
Panel B: correlations of vol processes with macro uncertainty						
$cor(\sigma_{ct}^2, macro_t^2)$	0.16	-0.03	-0.01	0.15	0.36	0.41
$cor(\sigma_{ft}^2, macro_t^2)$	0.28	0.07	0.10	0.28	0.47	0.51
$cor(\sigma_{rt}^2, macro_t^2)$	0.50	0.36	0.39	0.51	0.59	0.61
Panel C: correlations of vol processes with financial uncertainty						
$cor(\sigma_{ct}^2, financial_t^2)$	0.16	-0.04	-0.01	0.15	0.34	0.38
$cor(\sigma_{ft}^2, financial_t^2)$	0.45	0.21	0.25	0.46	0.62	0.64
$cor(\sigma_{rt}^2, financial_t^2)$	0.54	0.43	0.46	0.55	0.61	0.62

The table summarizes posterior mean, 2.5%, 5%, 50%, 95%, and 97.5% quantiles of correlation among σ_{ct}^2 , σ_{ft}^2 , σ_{rt}^2 , and the *squared* financial/real/macro uncertainty measures from Jurado, Ludvigson, and Ng (2015) and Ludvigson, Ma, and Ng (2021). The uncertainty measures have horizons of three months, consistent with the quarterly frequency of consumption growth.

Table IA.XIII: Cross-sectional estimates of risk premia in six-factor models

	Estimating $\Delta\mathbb{E}_t[m_t] = b_f f_t - \mathbf{b}_g^T \mathbf{g}_t$					$\mathbb{E}[r_t^{mkt}] =$	
	b_f	$\mathbb{E}[SR_m \text{data}]$	$\mathbb{E}[SR_f \text{data}]$	$\mathbb{E}[\frac{SR_f^2}{SR_m^2} \text{data}]$	R^2	$-\text{cov}(r_t^{mkt}, m_t)$	$-\text{cov}(r_t^{mkt}, -b_f f_t)$
Panel A. 37 stock and nine bond portfolios							
Posterior median	0.264	0.969	0.529	0.325	0.784	0.068	0.073
90% CIs	[0.062, 0.460]	[0.652, 1.496]	[0.156, 0.919]	[0.022, 0.688]	[0.497, 0.905]	[0.031, 0.107]	[0.018, 0.123]
Panel B. Kozak, Nagel, and Santosh (2020) 74 anomaly portfolios							
Posterior median	0.247	0.975	0.495	0.260	0.422	0.072	0.068
90% CIs	[0.106, 0.374]	[0.754, 1.212]	[0.217, 0.749]	[0.056, 0.548]	[0.216, 0.592]	[0.034, 0.108]	[0.029, 0.108]

The table reports estimation results for two cross-sections of excess returns: (1) 37 stock and nine bond portfolios and (2) Kozak, Nagel, and Santosh (2020) 74 characteristic-sorted portfolios. We report seven statistics: (1) the risk price of the shock to the conditional consumption mean f_t (b_f), (2) the annualized Sharpe ratio of the SDF in equation (20), defined as the annualized volatility of the SDF (SR_m), (3) the annualized Sharpe ratio of $b_f f_t$ (SR_f), (4) the ratio of SR_f^2 to SR_m^2 , (5) the cross-sectional R-squared (R^2), (6) the (annualized) market risk premium implied by the SDF, $-\text{cov}(r_t^{mkt}, m_t)$, and (7) the (annualized) market risk premium implied by the covariance between market excess return and $-b_f f_t$, $-\text{cov}(r_t^{mkt}, -b_f f_t)$. We estimate the risk prices using the Bayesian approach developed in Bryzgalova, Huang, and Julliard (2024). Details are provided in Internet Appendix M. We consider six-factor models of asset returns. Both the posterior median and the 90% Bayesian credible intervals are reported.

Table IA.XIV: What drives the consumption shocks spanned by financial markets?

Panel A: Single-factor model in nondurable consumption growth					
	MKT	SMB	HML	RMW	CMA
Correlation	0.938	0.634	-0.123	-0.224	-0.297
95% CI	[0.930, 0.946]	[0.613, 0.654]	[-0.150, -0.097]	[-0.238, -0.210]	[-0.318, -0.276]
	PC1	PC2	PC3	PC4	PC5
Correlation	0.994	0.052	-0.043	-0.027	-0.001
95% CI	[0.991, 0.996]	[0.024, 0.080]	[-0.070, -0.016]	[-0.051, -0.003]	[-0.018, 0.015]
Panel B: Two-factor model in nondurable consumption growth (P/D, CFNAI as predictors)					
	MKT	SMB	HML	RMW	CMA
Correlation	0.937	0.626	-0.132	-0.231	-0.288
95% CI	[0.924, 0.946]	[0.605, 0.647]	[-0.160, -0.105]	[-0.247, -0.214]	[-0.310, -0.266]
	PC1	PC2	PC3	PC4	PC5
Correlation	0.991	0.060	-0.037	0.030	0.001
95% CI	[0.982, 0.995]	[0.031, 0.089]	[-0.064, -0.009]	[0.012, 0.047]	[-0.016, 0.017]
Panel C: Two-factor model in nondurable consumption growth (<i>cay</i> as predictor)					
	MKT	SMB	HML	RMW	CMA
Correlation	0.938	0.619	-0.136	-0.225	-0.29
95% CI	[0.929, 0.947]	[0.596, 0.641]	[-0.163, -0.109]	[-0.243, -0.207]	[-0.311, -0.269]
	PC1	PC2	PC3	PC4	PC5
Correlation	0.990	0.058	-0.030	0.032	-0.003
95% CI	[0.981, 0.995]	[0.030, 0.086]	[-0.058, -0.002]	[0.016, 0.049]	[-0.019, 0.014]

The table reports the correlation coefficients between f_t and Fama-French five factors and the first five principal components of asset returns. We also report their 95% posterior credible intervals under the coefficient estimates. Moreover, asset returns are modeled using a single-factor model in Panel A and two-factor models in Panels B and C (as in Section III.4), with $\bar{S} = 14$. The cross-section of test assets includes 25 size-and-value-sorted portfolios, 12 industry portfolios, and nine bond portfolios.